OECD Economic Surveys

Sweden

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OECD ECONOMIC SURVEYS

2000-2001

Sweden

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BASIC STATISTICS OF SWEDEN

THE LAND

Land area (1 000 sq. km)	411
Lakes (1 000 sq. km)	39
Arable area (1 000 sq. km)	30
Woodland (1 000 sq. km)	228

Inhabitants in major cities, including suburbs (31 December 1999), thousands:	
Stockholm	1 643
Göteborg	789
Malmö	519

THE PEOPLE

Population (31 December 1999), thousands	8 861
Number of inhabitants per sq. km	22
Net natural increase (average 1995-99), thousands	–3.0

Net natural increase per 1 000 inhabitants (average 1995-99)	–0.1
Net migration (average 1995-99), thousands	9.4

THE PRODUCTION

Gross domestic product in 1999 (Kr billion)	1 972.1
GDP per head, US$	26 936

Gross fixed capital formation in 1999:	
Per cent of GDP	16.6
Per head, US$	4 033
Employment, 1999:	
Total civilian, thousands	4 067
Per cent of total:	
Agriculture, forestry, fishing	2.6
Industry	25.1
Other	72.2

THE GOVERNMENT

Per cent of GDP in 1999:	
Public consumption	27.0
General government current revenue	57.9
Public gross fixed capital formation	2.5

Composition of Parliament:	Number of seats
Social democrats	131
Moderates	82
Left	43
Christian democrats	42
Centre	18
Liberals	17
Greens	16
	349

Last general election: September 1998
Next general election: September 2002

THE FOREIGN TRADE

Exports of goods and services, 1999 (per cent of GDP)	43.8
Main merchandise exports (per cent of total):	
Forestry products	13.5
Mineral products	8.8
Chemical products	9.6
Energy products	2.4
Engineering products	55.8
Other products	9.9

Imports of goods and services, 1999 (per cent of GDP)	38.2
Main merchandise imports (per cent of total):	
Forestry products	3.4
Mineral products	8.2
Chemical products	10.9
Energy products	6.3
Engineering products	52.1
Other products	19.1

THE CURRENCY

Monetary unit: Krona

Currency units per US$, average of daily figures:	
Year 2000	9.15
January 2001	9.47

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Sweden by the Economic and Development Review Committee on 11 January 2001.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 30 January 2001.

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The previous Survey of Sweden was issued in July 1999.

Assessment and recommendations

Growth has been strong and based on a solid foundation…

Economic performance has continued to be impressive, and the short-term outlook remains favourable. Growth has averaged 4 per cent in 1999 and 2000, while inflation has remained low. Real disposable incomes have recovered, and households' balance sheets have strengthened remarkably, helped by rising house prices and the stock market boom. Private consumption has picked up strongly as a result, and the savings rate has declined. Business investment has increased considerably, helped by rapid growth in other demand components and low interest rates, and housing investment has risen rapidly, albeit from very low levels. At the same time, exports have risen, benefiting from stronger growth in overseas markets.

… although the labour market is becoming tighter…

Against this backdrop, employment growth has been well above trend, and unemployment has declined steadily to around 4 per cent. As a result the labour market is becoming still tighter, with vacancy rates and labour shortages on the rise. A key question is whether current or lower unemployment rates would be consistent over the medium term with continued low inflation. OECD estimates, while subject to considerable uncertainty, suggest it might not, unless accompanied by structural reforms. In contrast, external and budget balances are in comfortable positions: the current-account surplus has weakened only slightly and remains above 2 per cent of GDP, while the general government surplus strengthened to more than 3 per cent of GDP in 2000.

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… with rising inflation the main threat to the positive outlook…

Real GDP growth is projected to remain above 3 per cent in 2001, while in 2002 – assuming broadly neutral fiscal policy for that year – it should moderate to around 2½ per cent. Thus, the economy could continue to expand at rates above potential during most of the forecast period. Consequently, unemployment is expected to fall further below the OECD's estimate of its sustainable level, thereby inducing some pickup in wage increases. The OECD's *Outlook* assumes that the Riksbank will respond by raising interest rates during the course of 2001. Despite this tightening, inflation is expected to exceed the 2 per cent target in 2002. At the same time, spreading capacity constraints are projected to result in declining export market shares and a shrinking current account surplus. The main upside risks to activity over the projection period emanate from domestic sources. The savings rate might continue to decline, in particular if the Riksbank were to increase rates only moderately, or if fiscal policies continued to stimulate demand in 2002. This would occur if further tax cuts were not offset by lower spending. On the other hand, if recent signs of global slowing and weaker equity markets intensify, they may presage lower-than-projected growth.

… even though the current inflation-targeting framework has continued to lead to low inflation outcomes

The inflation-targeting monetary policy regime remains in place, with a target rate of 2 per cent and a tolerance band of 1 per cent. The government has repeated its position that it has no current plans to enter the European exchange rate mechanism (ERM2), despite favouring eventual adoption of the euro. The current monetary framework was reinforced in 1999, and its already high credibility has strengthened further. This was illustrated, for example, by the elimination in 2000 of the interest-rate differential on long-term government bonds *vis-à-vis* Germany. Consumer price inflation remained remarkably subdued at 1¼ per cent in 2000, below the Riksbank's target for the fifth consecutive year. Although greater competition has helped to contain price increases, undershooting can also be partly attributed to temporary or non-market-determined factors – falling mortgage interest costs, suppressed increases in housing rents and declining profit shares – as well as to the krona appreciation up to mid-2000. While wage increases have been historically low in nominal terms, real wage gains have nonetheless exceeded productivity increases in recent years.

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Further interest-rate increases are needed to ensure inflation stays close to its target…

Although inflation has remained low, the tightening of monetary policy in 1999 and the first half of 2000 was appropriate. Interest rates were raised in November 1999 and again in February 2000, but, with continuing positive results on inflation, the Bank held the repo rate constant at 3.75 per cent until December 2000, when it was raised by ¼ percentage point. As interest rates were increased considerably in the euro area during 2000, the short-term interest rate differential between Sweden and the euro area turned negative in mid-2000 and widened rapidly to around 1 percentage point. With fixed-income investors able to find higher returns abroad, weakness in Sweden's equity markets, and the liberalisation of pension fund portfolios, the exchange rate depreciated considerably in the second half of 2000, thereby reversing most of the earlier appreciation. In addition, the budget for 2001 implies further stimulus to activity (see below). However, while acknowledging that the risks are mainly on the upside, the Riksbank has not clearly foreshadowed any further interest rate increases, given their assessment of the prospects for inflation two years ahead, and the fact that inflation expectations are close to the target of 2 per cent. Still, the OECD projects that growth will not slow down to its potential rate of 2-2½ per cent on current macroeconomic policies, so that further interest rate rises are likely to be needed. Taking early action would help smooth the path to a more sustainable growth rate and safeguard the Bank's credibility.

… especially given the current fiscal stimulus

Sweden has achieved a huge turn-around in its public finances, and the budget surplus was above 3 per cent of GDP in 2000. Significant tax cuts were included in both the 2000 and 2001 Budgets, amounting to more than 1½ per cent of GDP in total. These cuts are part of a multi-year income tax reform, with further reductions signalled in coming years, depending on the state of the economy. Well-targeted tax cuts would certainly strengthen the supply side of the economy, particularly in Sweden where tax rates are very high. In the short term, however, they have resulted in an easing of fiscal policy, especially in 2001, when growth rates are projected to be still well above what is sustainable in the medium term. To avoid this pro-cyclical impetus, it would have been more prudent to match the tax cuts with

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spending reductions in order to avoid the risk of eventual over-heating. Otherwise more of the burden of cooling activity may simply be shifted onto monetary policy. Furthermore, while the government takes into account the role of the conjuncture in assessing the room for tax cuts and is committed to achieving a 2 per cent surplus over the course of the economic cycle, the transparency of the process suffers from the fact that the government does not publish estimates of the output gap and the cyclically-adjusted budget surplus.

Careful planning is needed to head off expenditure pressures in 2002

The main challenge for fiscal policy in 2002 will be to address spending pressures and build room for further tax cuts without compromising the government's overall fiscal stance. The expenditure ceilings, instituted in 1997, are generally perceived to have constrained outlays, although discretionary spending has in fact grown faster since the ceilings were instituted than in the years immediately prior. But because cyclical components such as unemployment benefits and outlays on labour market programmes are included, savings due to buoyant activity and low inflation have in turn been spent on other public programmes. Moreover, the original reserve provided within the ceiling for unexpected expenditure increases has been used up: projected spending for 2002 is almost at the same level as the ceiling. In the short run expenditure on benefit entitlements are beyond the authorities' control, and, without any cushion in the form of a safety margin, it might be necessary to offset unexpected cyclical increases with cutbacks in discretionary spending. A prudent approach to budget planning would call for identifying now which programmes would be cut back in 2002, if needed to remain within the ceilings. Postponing remedial action risks putting the government in the difficult situation of either violating the limits or making hasty and painful cuts to bring expenditure in line, thereby failing to prune in the most effective manner.

Public expenditure ceilings could be made more effective

The expenditure ceilings have been a useful instrument for strengthening budget control. But they could be made more effective by incorporating a separate planning margin, to allow for policy initiatives, and an uncertainty margin, to absorb fluctuations in economic activity and inflation, as was

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recently recommended by a government-appointed expert group. It might be tempting to add the new margins on top of the projected spending for 2002, but a better approach would be to restore the margins within the current ceilings. While ambitious, this would make lower taxes a more justifiable proposition. In any case, the effectiveness of such changes would still depend heavily on the political commitment to restrain expenditure growth – something that has not yet been tested in a period of economic weakness.

Momentum for structural reforms should be maintained

Sweden's deep economic crisis at the beginning of the 1990s provided a strong motivation for undertaking major macroeconomic and structural reforms. The adjustment process was painful and difficult, but the benefits are now clearly manifest in improved economic performance. The present conjuncture should provide an ideal opportunity to carry out further structural measures, but the momentum for taking such steps seems to have slackened now that the difficult times have passed. With buoyant demand and accommodating macroeconomic policies, the emphasis should be on raising the supply potential of the economy through providing stronger incentives for people to work, save and invest. In addition, greater competition and flexibility in product markets would improve the allocation of existing resources and encourage dynamic efficiency through innovation. With ample expertise in information technology, Sweden should also be well positioned for "new economy" effects to emerge.

Unemployment benefits could provide better job search incentives…

Lowering structural unemployment is an important factor in easing upward pressure on wages. In light of this, the government's proposal to increase the generosity of unemployment benefits by raising the ceiling for the first 100 days seems inappropriate, since it may prompt extended job search duration and spur pressures for a more general increase in benefits. The new activity guarantee is designed to help the long-term unemployed acquire better work skills as well as to provide a stronger incentive to find a job. It is too early to assess the impact of this measure, but it could be improved by imposing a time limit on the guarantee itself, instead of leaving it open-ended. Part-time unemployment is a particular feature of the Swedish labour

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market, with more than 10 per cent of all part-time workers collecting supplementary unemployment benefits. The rules governing these payments are generous, and anyone receiving them faces weak incentives to work longer hours or shift to a full-time job. The recent extension of the activity guarantee scheme to cover the part-time unemployed may help to sift out those who are not genuinely willing to work longer hours, but it would be better to reduce the generosity and duration of the supplementary benefits at the same time.

… while programmes that keep people out of the workforce ought to be scaled back…

Although unemployment has fallen sharply and employment growth has been impressive, labour force participation has only recently picked up appreciably and remains well below the levels before the early-1990s crisis. Encouraging participation should therefore be a key priority. But at present, some 5 to 6 per cent of the working-age population is tied up in labour market programmes and adult education, keeping them out of the workforce. A significant downscaling of these measures will be required for the government to fulfil its objective of increasing the unsubsidised employment rate from its current 77 per cent level to 80 per cent by 2004.

… and ALMPs better targeted and more efficiently run

Active labour market policies (ALMPs) have long been a key feature of Sweden's policy settings. While ALMPs were probably important in containing problems of permanent exclusion from the labour market when unemployment was high, there is little evidence that these programmes have had any significant effect on unemployment itself. A three-pronged approach to improving the programmes would help:

– provide stronger encouragement to participants to search for work themselves, by tightening the criteria for participation in programmes and applying strong sanctions for non-compliance;

– exercise tighter control over active labour market programmes to ensure that funds are allocated to those providers that deliver the best results; and

– target more closely those specific groups of unemployed who really do face employability problems, for example because of poor language or literacy skills, and tailor programmes to those specific needs.

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Labour supply would benefit from boosting the efficiency of the education system…

A further contribution to raising the employment rate could be achieved by encouraging students to complete their schooling at a faster pace so that they can enter the labour market at an earlier age. While the education system is performing well in many respects, the good results are often achieved at high cost. On average, students take too long to complete their studies. A greater wage dispersion would help provide a clearer signal about the real value of skills and a stronger incentive for individuals to invest more in their own human capital.

… while a carefully planned broad package of tax cuts could provide better incentives to work and save

Sweden's very high tax burden presents a number of challenges affecting both the labour market and resource allocation more generally. Given the way high marginal tax rates discourage work efforts, lowering taxes on labour should be an important medium-term objective, especially since average marginal effective tax rates have increased over the past decade. The income tax cuts in the last two budgets should stimulate labour supply and market-based activity more generally, although larger effects could have been achieved if the increase in the threshold for the state income tax had been greater. Lowering marginal effective capital tax rates and abolishing the wealth tax would also encourage savings, as well as diminishing the appeal of earning income offshore in order to avoid taxes. This could be achieved without a significant loss of revenue because it would also reduce households' allowances for mortgage interest payments, although the induced wealth effect through the negative impact on house prices would dampen household consumption along with a direct effect on disposable incomes. However, lowering the capital tax rate in isolation would exacerbate the incentives for income shifting in the dual-income taxation regime. It would therefore be most appropriately introduced as part of a broader package including income tax cuts. The recent reduction in property tax, in contrast, does not seem to be consistent with the standard taxation principle of taxing immovable bases. In any case, tax reforms will need to take into account a number of factors likely to erode tax bases over the coming years, including the spread of electronic commerce, and the room to manoeuvre will also depend on developments in public expenditure.

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Competition is improving in utilities, but some areas of weakness remain

Promoting competition has been another priority area for structural reforms, and the government has continued its efforts to liberalise product markets, particularly in the utilities sector. The recent abolition of the hourly metering requirement on households has made it easier for them to switch electricity suppliers. In addition, the expiry of long-term contracts for the supply of electricity to households should provide greater opportunity for active competition at the wholesale level. However, the power of the local monopolies is still evident in electricity distribution, even if the new obligation to publish network tariffs helps customers judge whether prices are reasonable. In telecommunications, competition has increased significantly since 1993, but although Telia's overall market share has fallen, it still holds a dominant position in most segments. Pre-selection and number portability have been unable to deliver on their promise to reduce concentration because of Telia's ownership of the local loop. "Unbundling" the local loop is a high priority, and draft legislation has been prepared, but it faces constitutional hurdles. Several legislative changes have been made for mobile telephone networks, including national roaming rights for new entrants and number portability, but it remains to be seen how much difference these changes will make to the intensity of competition. UMTS licenses have been awarded to four operators after a "beauty contest", the government having opted to forgo the large revenues associated with auctions held elsewhere in the belief that the network would become operational at an earlier date. However, subsequent legal challenges threaten to lengthen the process.

Obstacles to greater competition in retail distribution should be removed

Greater efforts are needed to strengthen competition in retail distribution, since the overall price level remains some 15-20 per cent higher than the OECD average, even if the two are slowly converging. For food and groceries, existing firms charge high prices, but that has not elicited potential competitors. The particular ownership structures of the three dominant chains, together with the block exemption from competition rules, provides a high degree of protection, while building and planning restrictions exercised by local authorities may work against new entrants building supermarkets and especially hypermarkets. Until

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these barriers are reduced and the threat of entry is real, Swedish consumers will continue to face higher food and grocery prices than elsewhere. Adhering to the abolition of the block exemption scheduled for June 2001 will help to free up the market, and industry arguments to extend it again should be firmly resisted.

Housing rents should become market-determined

The housing sector presents a number of concerns. The system of *de facto* rent controls has led to a poor allocation of rental housing and weak incentives for investment in new rental stock. This has contributed to some acute shortages in metropolitan areas, especially around Stockholm, and housing investment relative to GDP is easily the lowest in the OECD. Municipal housing rents, which are cost-based and low because the stock is largely old and free of debt, effectively set a ceiling for rents in the private sector. In addition, the gap between official rents and what market rents would be in some localities provides a strong incentive to bend, or even break, the rules on sub-letting apartments in order to obtain windfall gains. Allowing the owners of private and municipal properties to charge market rents would go a long way towards promoting a more rational allocation of the existing stock as well as providing a reasonable return to new developments. In addition, some authorities are reluctant to provide space for new housing development, especially because it reduces space for public amenities while doing little to expand their local tax base, given the inter-governmental equalisation arrangements and the absence of a municipal property tax. A few municipalities have started to sell their existing properties to their tenants, at cost price. While this may help improve the allocation of the housing stock in the long run, it gives sizeable windfall gains to some purchasers, given the current high prices for owner-occupied housing.

Long-term sustainability will depend on the soundness of environmental policies

In a longer-term perspective, potential rates of growth will be determined not only by the quality of macroeconomic policy setting and the institutions and structural frameworks already mentioned but also by sustainability considerations in the environmental domain. The recently adopted 15 environmental objectives provide a comprehensive structure for shaping policy. This has highlighted

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some of the more difficult trade -offs that it faces in trying to achieve all these goals, most notably the challenge of simultaneously lowering greenhouse gas emissions and reducing reliance on nuclear power without any new hydro-electric development. At the same time, the country has a depth of experience with economic instruments to improve environmental outcomes, although systematic and rigorous cost-benefit analysis needs to be applied in choosing policies. Such an approach would help environmental policy makers to identify which options would offer the greatest net benefit to society as a whole.

Sweden has been successful in reducing sulphur and NO_x emissions

The Swedish landscape is particularly vulnerable to acid rain, and an extensive liming programme is required to mitigate the damage. Sweden's harmful sulphur and NO_x emissions have been significantly reduced over the years, and damage now comes mainly from offshore sources. The Gothenburg protocol represents a significant step forward in international co-operation to reduce emissions in all signatory countries, and Sweden continues to press for further efforts in other countries. It has also taken steps to reduce the emissions of sulphur and nitrogen oxide from Baltic shipping, which is responsible for a significant share of depositions falling on Sweden. The Swedish system of differentiated fairway dues is an innovative approach providing economic incentives to shipping companies to reduce emissions. It would be even more successful if adopted by other Baltic countries. Some other domestic sources of emissions (for example, off-road machinery) may be more difficult to influence through economic instruments, and stricter regulations and controls may be the most cost-effective strategy, if further reductions are desired.

Higher CO_2 taxes or tradable emissions permits will be needed to meet Kyoto targets…

Climate change is clearly a key focus of Swedish environmental policy, although its own emissions are among the lowest in the OECD on a per capita basis, thanks mainly to the limited use of fossil fuels in electricity generation. Under the Kyoto Agreement and the burden-sharing arrangements within the European Union, Sweden is pledged to limiting its greenhouse gas emissions in the period 2008-12 to no more than 4 per cent above 1990 levels. But projections by the National Institute of Economic Research suggest that, on

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present policies, emissions could be 15 per cent higher. A carbon tax is already in place, but it is clear that to reduce domestic emissions further, either the tax must be raised substantially or a domestic cap-and-trade system of emissions permits needs to be instituted instead; the relative merits of these two options are now being examined by the Swedish authorities. However, with already low per capita emissions, the marginal cost of reducing emissions in Sweden is likely to be higher than in many other countries, and the inclusion of flexible mechanisms within the global arrangements would make it possible for Sweden to achieve its commitments in the most efficient way. A swift and clear resolution at international level would make it easier for Sweden to move forward with its own strategy to combat global warming and address the question of whether it should aim for further reductions beyond present commitments and, if so, how these should be achieved.

… and the emphasis should shift from energy taxes to CO_2 taxes…

In the meantime, the government has announced a further rise in carbon taxes and a realignment of the energy taxes in the 2001 budget, as part of the green tax swap. Energy taxes are to be based on energy content and in the longer term will be extended to biofuels. Their restructuring is intended to provide a clearer and general incentive for energy conservation as well as making them a more effective fiscal instrument for raising revenue. Thus, energy taxes play a limited and indirect role in the climate strategy, where the focus is on reducing harmful emissions. In this respect, making energy sources that emit greenhouse gases relatively more expensive makes more sense, as reflected in the shift towards higher CO_2 taxes. However, to the extent that raising carbon taxes lifts prices for all electricity producers, this would generate an economic rent for hydroelectricity and nuclear generators that would be appropriate to tax away – at least if the present policy of not allowing expansion in these sectors is maintained.

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… with the same incentives to reduce emissions applying across the whole economy

Another feature of the Swedish tax system is the favourable treatment of industry compared to other users, with respect to both energy and carbon tax rates, and also the special rules applying to energy-intensive industry. The approach is intended to ensure that producers in Sweden face energy prices that are competitive in an international context and for carbon taxes, the policy is based on addressing the consequences of Sweden having gone further in applying these taxes than many other countries. If Swedish industry faced the full force of these environmental measures, Swedish emissions would be reduced in the least costly way, but some loss of competitiveness would occur and some production might move offshore. This could result in higher global emissions, although the extent of such leakage is essentially an empirical matter. However, such concerns are no longer relevant if all countries are making similar efforts to address global warming, as would be the case once the Kyoto protocol is implemented, even if not all countries are bound by it. The main challenge for Sweden is to find the best strategy in the interim period until the protocol is applied. Under the current approach, some enterprises receive what amounts to a significant tax expenditure, while incentives to reduce emissions and improve energy efficiency more generally are distorted. This produces an inefficient allocation of resources across the economy as a whole and places a greater share of the adjustment costs of meeting Sweden's environmental targets onto remaining energy consumers. It would be better to equalise marginal abatement costs to help ensure that domestic emissions were reduced in the least costly way, while providing compensation for the loss of profitability that businesses would incur from the removal of the present exemptions and reduced rates applying to some sectors. Two alternative ways of dealing with this would be either to apply a uniform tax rate and provide an offsetting general tax credit, or to shift to a cap-and-trade system within which existing emitters would receive tradable permits under a grandfathering arrangement.

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A better set of economic incentives for reducing nitrogen leaching needs to be found

Sweden's waterways have been badly affected by excess nutrient run-off from agriculture, which has in turn been exacerbated by the substantial amount of agricultural support provided. More market-based price signals for agricultural production at the European level would clearly have an impact on the environment, and provide a sounder base for developing environmental policies to address the externalities that would remain. In the meantime, Sweden has used a combination of fertiliser taxes, regulations concerning farming techniques and subsidies to encourage organic farming. However, the relationship between nitrogen application, farming methods and leaching is complex, and none of the present instruments seems to be particularly effective or cost-efficient. Two alternatives merit further study. One option would be to move to a system of taxes or permits applying to nitrogen losses or residuals from each farm. The other would be to develop a charging system for agriculture whereby a standard fee was collected from all farms and then redistributed back to those that performed best according to a key set of environmental indicators. Both schemes might involve significant compliance costs, but this would be at least partly offset by easing the current tight regulations governing farming methods. A rigorous evaluation of the costs and benefits of the present arrangements would provide a useful starting point for evaluating alternative approaches.

Net social benefits need to underpin policies on waste management

Waste management is an area where cost-benefit analysis has not been used to a large extent. Policies are based on environmental impact studies, while the actual cost of waste management is not always reflected in the cost to consumers. The extended producer responsibility approach assumes that recycling is always worthwhile, despite some studies that question the value of recycling certain materials. The approach has achieved its recycling targets, but without taking into account overall environmental and economic costs. An evaluation of the full costs and benefits of the existing schemes is now underway. Municipal waste policy follows the "waste hierarchy" approach: landfill of combustible materials will be banned from 2002 and organic waste from 2005. But both the environmental and economic rationale for this ranking of waste disposal methods have

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been questioned. It would therefore seem prudent to rigorously assess the net social benefits to be gained before implementing the announced bans. Indeed, regional variations and local preferences for different disposal methods need to be at the forefront in the assessment of both costs and benefits, since much of the environmental damage is local.

Summing up

Sweden is now profiting from the macroeconomic and structural reforms undertaken in the wake of the severe recession of the early 1990s. Output has been growing at around 4 per cent per year, while inflation has been lower than in almost all other OECD countries. With healthy external and fiscal balances, the main risk to the immediate outlook is inflation drifting upwards and some overheating, since the economy is still growing faster than can be sustained in the medium term. The fiscal stimulus provided once again in this year's budget is, therefore, unhelpful, and higher interest rates are appropriate to start to relieve the pressure. Public spending strains likely in 2002 merit immediate attention, while the expenditure ceilings need more effective safety margins. At the same time, the momentum for further structural reform has weakened, even though current buoyant activity would make it easier for the economy to adjust. But the reform agenda remains unfinished, and emphasis should be placed on improving the incentives to work, save and invest in order to boost the supply of labour and capital. Competition remains weak in a number of product markets, thereby inhibiting the efficient allocation of resources across the economy as a whole. Environmental issues present a number of challenges, and although in some areas Sweden has led the way in the use of economic instruments, in others more rigorous analysis of the net social costs and benefits of alternative policies should be undertaken. Altogether, Sweden is well placed to maintain admirable economic outcomes and to achieve an even higher standard of living as long as it continues pursuing sound macroeconomic policies and satisfactorily addresses its remaining structural weaknesses.

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I. Recent developments and prospects for the near term

Overview

The Swedish economy expanded at an annual rate of 4 per cent in 1999 and 2000, well above its estimated potential growth rate of some 2½ per cent.[1] The rapid increase in total demand has been broadly based, with significant contributions from both domestic demand and exports. Strong growth in real disposable incomes, historically low interest rates, abundant credit availability and favourable developments in households' balance sheets due to significant increases in house and equity prices have combined to boost private consumption. At the same time, exports have benefited from accelerating market growth. Business investment has risen considerably as well, underpinned by the strength of other demand components and low interest rates. Housing investment, on the other hand, remains at historically low levels despite some recent recovery. Against the background of robust and balanced growth, the fiscal balance has continued to improve, while the current account has deteriorated only modestly; both were in comfortable surplus in 2000 (Figure 1). At some 2 per cent, economy-wide labour productivity gains over the most recent three years have been ½ percentage point above the average of the last two decades. Likewise, the employment growth has been significantly above trend. This paved the way for a sharp decline in unemployment to around 4 per cent of the workforce by the end of 2000. The overall picture of a benign macroeconomic situation is completed by the fact that inflation has remained remarkably subdued, with consumer prices having increased by a mere ¾ per cent in 1999 and 1¼ per cent in 2000. At the same time inflation expectations have been solidly anchored to the inflation target of 2 per cent. However, nominal wage increases in Sweden have remained slightly higher than in its trading partners, even if they have been low by historical standards.

In sum, the Swedish economy is probably stronger and better balanced than at any other point during the past three decades. With savings-investment balances and international competitiveness both in reasonably good shape, there are no imminent risks to activity and a repetition of earlier sharp recessions is very

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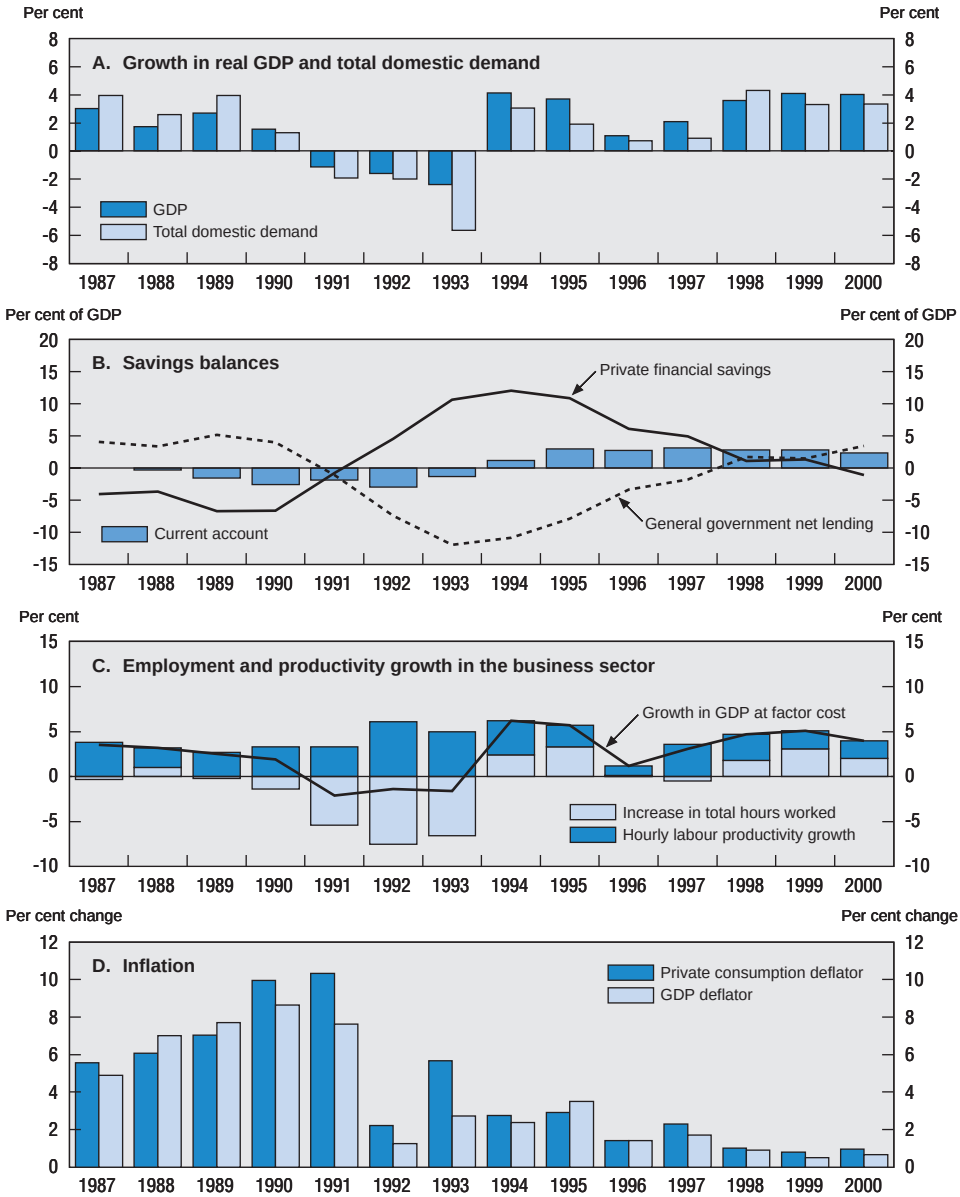
Figure 1. **Macroeconomic performance**



Note: 2000 data are projections from *Economic Outlook 68*. Data from 1997 to 1999 are revised national accounts figures published after *Economic Outlook 68*.
Source: Statistics Sweden; OECD.

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unlikely. Nonetheless some risks can be envisaged, mainly for the medium term, and a key objective for policy makers should be to safeguard what has been achieved till now. Considerable uncertainty surrounds estimates of the current level of potential GDP and hence the output gap. There is widespread consensus amongst Swedish analysts that there remains a certain amount of excess capacity and that the output gap will close only later in 2001 or even in 2002, even if only a few institutions produce and publish their own estimates. On the other hand, OECD estimates suggest that the output gap may have closed in 2000.

A key question related to the assessment of the output gap concerns the sustainability of low inflation at the current set of interest rates. While inflation has been impressively well contained so far and inflation expectations are in line with the inflation target, this should be weighed against indicators of increasing labour shortages. Suppression of rent increases due to what amounts to *de facto* rent controls has also contributed to low inflation. In addition, profit margins have been squeezed in recent years, which contrasts with developments in most other countries. Thus, the likelihood is that inflationary pressures will gradually build up if the economy continues to expand at rates above potential. The low level of housing investment also constitutes a particular risk. Investment is well below warranted levels, and a continuation of current low construction rates could induce further strain on the housing market. This, in turn, might push house prices in the fast-growing regions further beyond their long-term equilibrium, while in the rental market it would manifest itself in higher rents, although rent increases might temporarily be suppressed by the rent controls mentioned above. The opposite alternative – a significant pick-up in building activity – could aggravate labour shortages in the important construction sector and might put pressure on the current balance if domestic savings were not to increase in parallel.

Economic growth is likely to remain above potential in 2001, although at an expected 3¼ per cent it should abate from the very high rates in 1999 and 2000.[2] Recent information has indicated a moderation of private consumption and consumer and business confidence towards the end of 2000. Supportive fiscal policies and low interest rates (see Chapter II) are the key factors underpinning continued vigorous growth in domestic demand. An expected turnaround in the contribution to GDP from net exports – from positive to negative – is the main force acting to moderate growth in 2001. This is in part related to capacity constraints, which are expected to become gradually more binding. Under the assumptions of neutral fiscal policies and an increase in policy-controlled interest rates, GDP growth is expected to moderate further, to below 2½ per cent, in 2002. Inflation is projected to move above 2 per cent by 2002, although by a modest margin. The implications for macroeconomic and structural policies of this relatively buoyant outlook are discussed in Chapters II and III, respectively.

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Domestic demand and foreign trade

Buoyant private consumption

Real private consumption expanded 4 per cent per year in 1999 and 2000,[3] albeit with some signs of a slowdown in growth rates in late 2000. Cyclical components such as cars, other durables and foreign trips showed the largest increases (Figure 2). The recent buoyancy in private consumption is in sharp contrast with the average annual increase of 1.7 per cent in the period from 1994 to 1998. Consumer confidence has strengthened in line with actual consumption; it reached a record-high level in mid-2000 and, despite some moderation, it remained robust throughout the rest of the year. The underlying macroeconomic environment has indeed been conducive to increased consumer spending. While there was an outright fall in real disposable incomes during the fiscal consolidation period from 1994 to 1998, households enjoyed an increase in real income of some 3½ per cent in both 1999 and 2000, spurred by higher employment, historically large gains in *ex post* real wages and expansionary fiscal policies. As growth in consumption outstripped that in disposable incomes, the downward trend in the household savings rate has continued; at 2¼ per cent of disposable incomes in 2000 it was down by some nine percentage points from its peak in 1994.

Increases in households' wealth, stemming from higher house and equity prices, are likely to have added to growth in consumption in recent years. Although regional disparities have been considerable, average real house prices by the end of 2000 were up some 35 per cent on 1997 levels. To a large extent this is a natural recovery from the very low prices in the middle of the 1990s, but low interest rates, abundant credit, growth in real disposable incomes and stronger consumer confidence have reinforced and accelerated the boom in house prices. The fluctuations in the Swedish stock market have followed those of the US Nasdaq index quite closely in recent years, reflecting the dominance of ICT companies in Sweden. Despite a 25 per cent decline in equity prices since March 2000, they remain significantly higher than before 1999; the broad index (*generalindexet*) was still 45 per cent higher in December 2000 than two years earlier. In sum, gross household wealth was 20 to 30 per cent higher at end-2000 than at end-1998. The responsiveness of private consumption to fluctuations in household wealth is uncertain and may have changed in line with the increased share of equities in households' portfolios and the volatility in equity prices. The impact of changes in house prices is likely to be somewhat stronger than that of equity prices. Assuming a moderate short-term elasticity with respect to wealth, of 0.02,[4] the increase in asset prices would have contributed approximately 1 percentage point to the cumulative growth in consumption during 1999 and 2000. The future impact on consumption from recent changes in household wealth is difficult to assess. On the one hand, the long-run elasticity is likely to be higher than the

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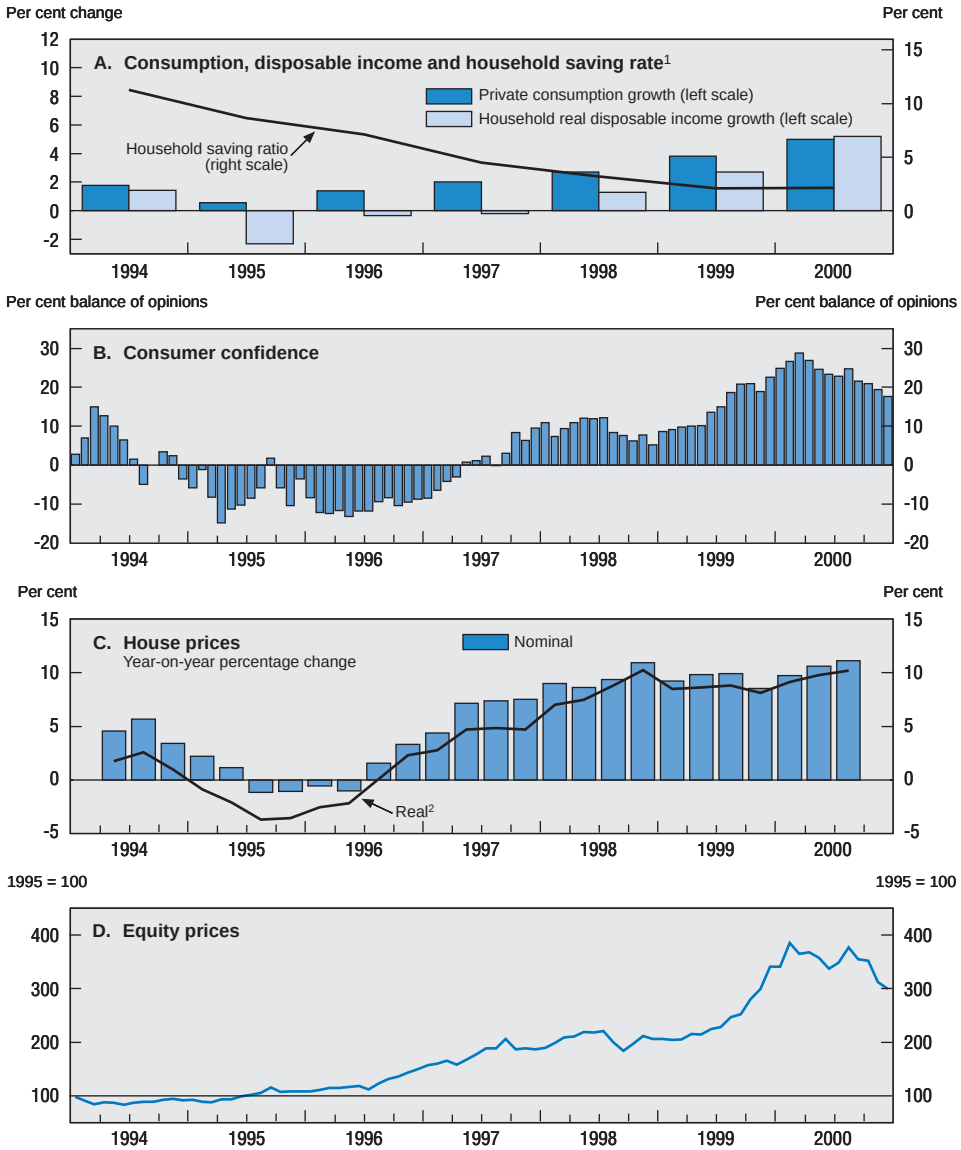
Figure 2. **The forces shaping private consumption**



1. 2000 data are projections from *Economic Outlook 68*. Data from 1997 to 1999 are revised national accounts figures published after *Economic Outlook 68.*
2. Deflated by the private consumption deflator.

Source: Statistics Sweden; OECD, *Main Economic Indicators, National Accounts.*

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first-year effect, which would suggest that the increase up to the peak in March 2000 might continue to stimulate consumption. On the other hand, the short-term effect of the recent decline would work in the opposite direction.

Private investment and financial savings

Following brisk growth of 9 per cent in 1998 and 1999, business fixed investment expanded at a more modest rate of 6 per cent in 2000. Share issues and bank lending have traditionally been the main funding sources, while corporate bonds have played only a minor role. As banks' lending rates have been very low and equity prices have surged, access to cheap capital has been ample; the resulting decline in capital costs has helped to cushion firms' profitability that might otherwise have suffered from the recent fall in profit shares. In addition, the market for venture capital is relatively well developed. In combination with increasing capacity utilisation and accelerating export market growth, these factors might have facilitated a stronger investment boom than has been the case. Although the investment ratio (business fixed investment as a per cent of the business sector's GDP at factor costs) has indeed risen during the current upswing the increase has been less than in earlier cycles. At 20 per cent in 2000 it remains near the average of the last two decades. The general trend in real private investment masks significant dissimilarities across industries; while increases have been relatively modest in manufacturing, they have been rather stronger in rapidly expanding sectors such as financial and business services.

Demand for housing has increased rapidly during the upswing as manifested in the considerable increase in house prices. However, the supply response mechanism in the form of residential construction has suffered from a set of policies characterised by comprehensive public regulation and weak investment incentives. Even though real residential investment expanded by more than 20 per cent in 1999, this was from a very low starting point, and it levelled off in 2000, resulting in a modest 5 per cent increase for the year as a whole (Figure 3). Housing starts and completions remain near historical lows; the 17 000 housing starts undertaken in 2000 should be seen in the perspective of a yearly "objective requirement" of 25 000 to 30 000 dwellings estimated by the Swedish authorities. At 1.6 per cent of GDP in 2000, residential investment in Sweden is by far the lowest in any OECD country, and 4 percentage points below the OECD average. The inadequacy of current investments is steadily putting additional strain on the housing market, with sub-letting and black-market transactions spreading in the rental sector and reinforcing price increases in the owner-occupied segment. The increasing housing shortages are concentrated in the fast-growing regions around the major cities, Stockholm in particular, while housing is still in excess supply in many rural areas. Thus, regional imbalances are mounting at the same time as the shortfall of housing starts in metropolitan areas risks impeding growth; spreading labour shortages outside the construction sector may already to some extent result directly from a

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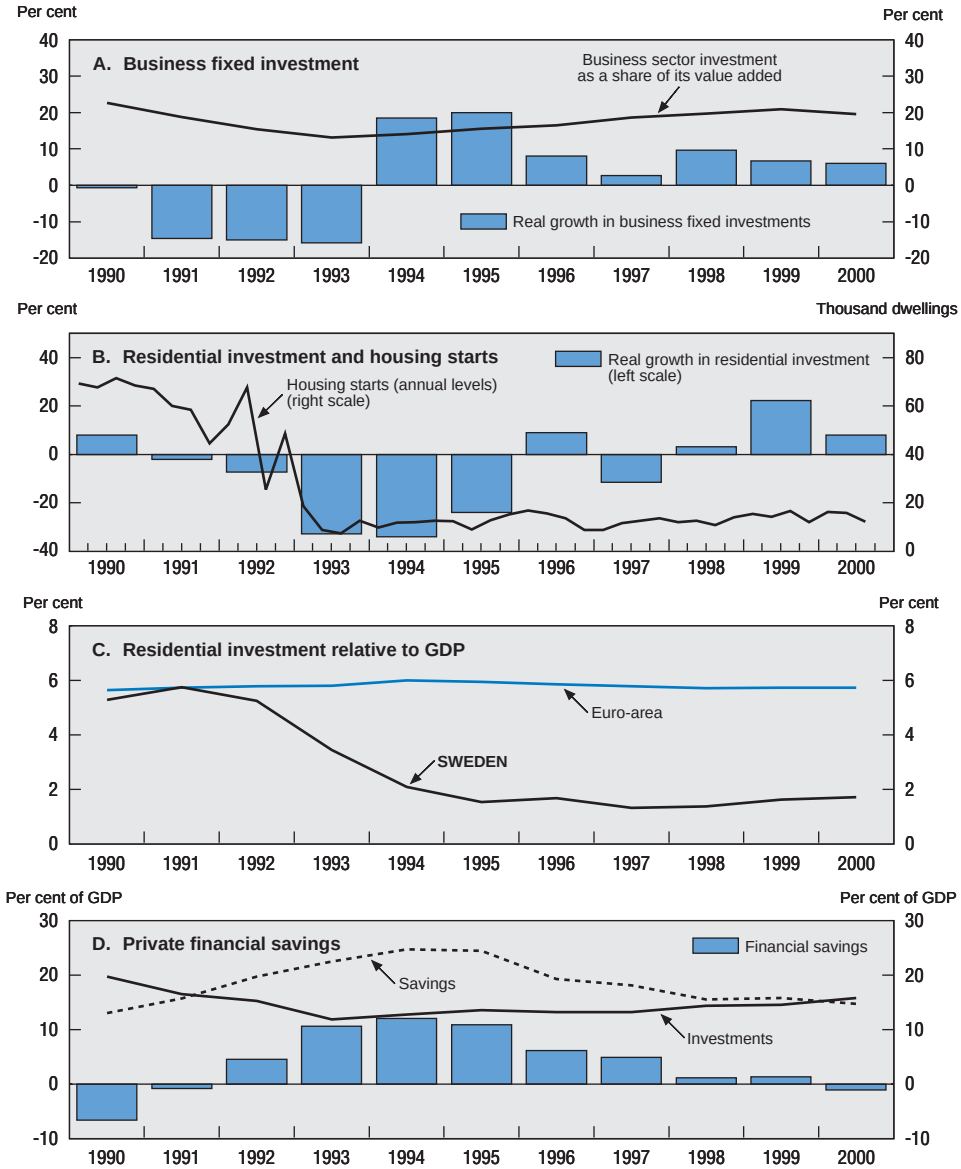
Figure 3. **Business and housing investment and private financial savings**



Note: 2000 data are projections from *Economic Outlook 68.* Data from 1997 to 1999 are revised national accounts figures published after *Economic Outlook 68.*
Source: Statistics Sweden; OECD, *Main Economic Indicators* and *National Accounts.*

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lack of accommodation restraining rural-urban flows and immigration. A number of factors have combined to explain the current unsustainable situation (see Chapter III for further details). *First*, the significant cutbacks in construction subsidies implemented during the fiscal consolidation period have virtually eliminated municipal investment in social housing, which accounted for more than 50 per cent of total housing starts as late as the first half of the 1990s. *Second*, private investment in rental housing is unprofitable due to *de facto* rent controls.[5] Finally, investment in owner-occupied housing, which has been the dominant segment in recent years, may to some extent be restrained by the fact that the absence of a municipal property tax and the design of the inter-municipal equalisation scheme have combined to weaken local authorities' incentives to supply suitable building sites.[6]

The net result of these household and business developments has been a rather sharp decline in private financial savings, *i.e.* total private savings minus total private investment, from 10 per cent of GDP in 1994 to –1 per cent of GDP in 2000. This was the first deficit on the private sector's savings-investment balance since 1991. Relative to GDP, total private savings are near the very low levels registered during the late-1980s upswing. Investment, on the other hand, is still much lower. Looking ahead, the risks may be mostly on the side of a further decline in private financial savings. Any marked increase in households' savings rate would probably require a somewhat higher real interest rate and/or a decline in the wealth-to-disposable-income ratio, neither of which seem imminent. Furthermore, the current low level of housing investment is unlikely to be sustainable, and a return to more normal levels might over the medium term contribute to raising the overall investment-to-GDP ratio.

Public-sector consumption and investments

After a number of years of zero or negative growth, real public consumption recovered in 1998 and 1999, increasing some 1¾ per cent per year (Figure 4). While some employment indicators have indicated a further increase in public-sector employment through 2000, preliminary national accounts data point to a more modest growth in consumption.[7] A new requirement that municipalities and counties balance their budgets as from 2000, adopted by the Parliament in 1998, is generally seen to have restrained local governments' spending somewhat. Even so, the increase in real public consumption and employment has exclusively taken place among the municipalities and counties. The total increase in the period 1998 to 2000 amounted to more than 10 per cent, although this was in large part due to the fact that costs for subsidies to pharmaceuticals were transferred from the central government to the counties. Central-government consumption, on the other hand, has been more subdued, and for the three most recent years as a whole it has even declined. In part, this is attributable to the system of expenditure ceilings that has been in place since 1997 (see Chapter II). The downscaling of active labour market programmes, as well as the compositional shift of such

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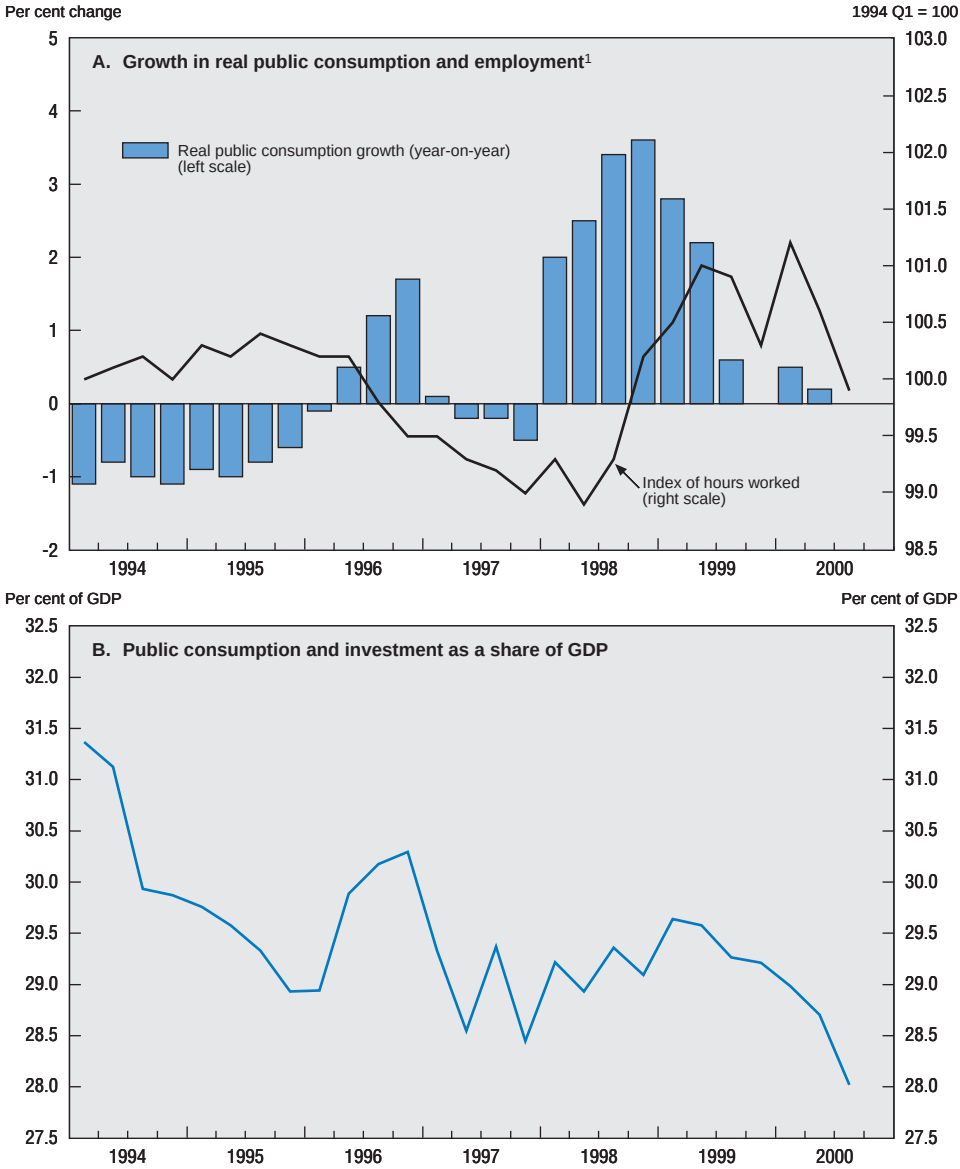
Figure 4. **Real public-sector consumption, investment and employment**



1. The data for the first three quarters of 2000 have been adjusted for the reclassification of the Swedish church in order to facilitate comparison with the preceding years.

Source: Statistics Sweden; OECD, *National Accounts.*

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programmes from job training to education, are additional contributing factors. Notwithstanding the pick-up in real growth, nominal public consumption has been unchanged relative to GDP, reflecting the vigorous output growth. Public investments have shown a broadly similar pattern, declining up to 1997, increasing fairly rapidly in 1998 and 1999 before levelling off in 2000.

Foreign trade and current account developments

With relative unit labour costs in a common currency having been fairly stable since 1997, real exports have evolved broadly in line with market growth. Thus, as the latter picked up significantly in 2000, the growth rate of real exports accelerated from 6 per cent in 1999 to almost 10 per cent. Sweden has a strong foothold in many ICT markets, and in general its exports are relatively knowledge intensive. Exports of telecommunications equipment have expanded particularly rapidly in recent years; their share in total exports has increased from 5 per cent in the early 1990s to around 15 per cent in 2000. While export performance has been largely unchanged in volume terms in the last few years (Figure 5), it has weakened more than 10 per cent when measured in current prices partly because of the declining relative prices of ICT equipment and weak export prices more generally (see below). Comparing present market shares with those of a decade ago, Sweden is rather unusual in having suffered losses in current-price terms despite gains in constant prices.

Real imports rose 4 per cent in 1999, which was relatively modest when considering the buoyant growth in domestic demand; this growth rate was below the trend increase in import penetration, *i.e.* the ratio of real imports to real GDP. The moderation was particularly pronounced for goods imports and may have been related to the slowdown in stock building, which reduced GDP growth by ½ percentage point. In contrast, real imports increased rapidly during 2000, almost 9 per cent for the year as a whole.

The terms of trade have been trending down in recent decades and continued to do so in recent years; they fell 4 per cent from 1995 to 1999, and, partly related to the surge in oil prices, dropped another 5 per cent in 2000 alone. However, the continued deterioration of the terms of trade is not only the result of high import price increases (before 2000, they grew only modestly). Very low increases in export prices and output prices in general have also played a role; export prices fell by an average of 1 percentage point per year from 1995 to 1999 but rose by 2 to 3 per cent in 2000.

The surplus on the current account was relatively stable from 1997 to 1999 but declined from 3.5 per cent of GDP in 1999 to a projected 2.4 per cent of GDP in 2000 (Table 1). In particular, the non-factor services deficit has increased, in large part resulting from net travel expenditures picking up. In addition, the merchandise trade surplus has edged down relative to GDP. The erosion of the current account surplus is not surprising, considering the strength of domestic

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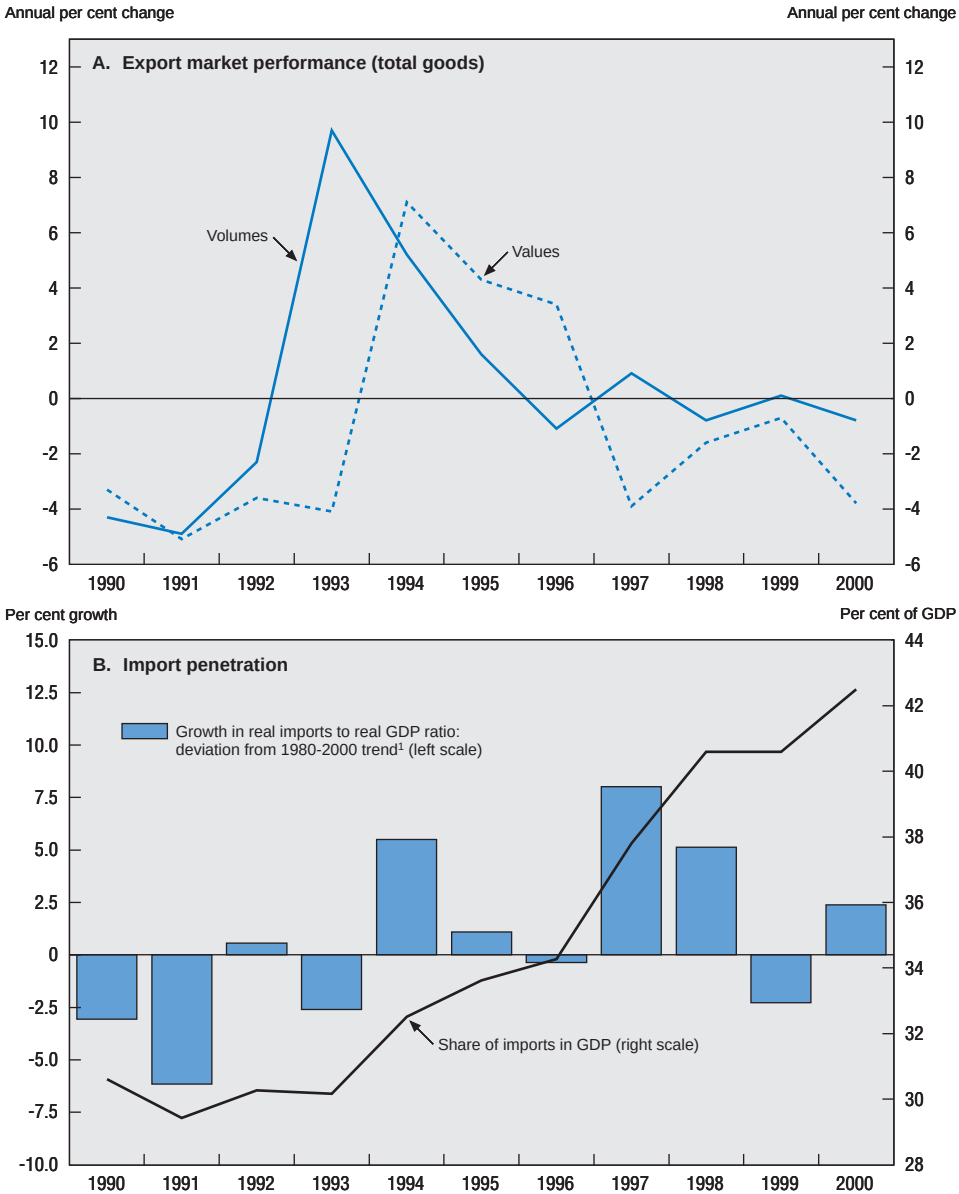
Figure 5. **Export market performance and import penetration**



1. The trend is the average annual increase in the ratio of real imports to real GDP from 1980 to 2000.
Source: OECD.

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Table 1. **The current account, 1997-2000**
Billion SKr

	1997	1998	1999	2000[1]
Goods	145.1	137.2	137.2	142.7
Non-factor services	–14.0	–18.4	–17.5	–34.7
Factor income[2]	–44.7	–35.0	–19.6	–30.9
Current transfers	–25.0	–27.7	–29.7	–28.2
Current account	61.3	56.1	70.4	48.9
(in per cent of GDP)	3.4	2.9	3.5	2.4

1. OECD estimate (*Economic Outlook* 68).
2. Includes compensation of employees.
Source: Sveriges Riksbank.

demand in recent years. The weakening of the goods and non-factor services balance has been partly offset by a shrinking deficit on factor income, related, most importantly, to lower debt-servicing costs, which themselves are the combined result of declining foreign debt and lower interest rates. In terms of savings balances the modest decline in the current account is the net result of a marked improvement of the fiscal balance more than offset by an even sharper decline in private financial savings.

Price and wage developments

Inflation has remained remarkably low

Recent years have produced a string of pleasant surprises on inflation, which is attributable in part to a more rigorous framework for competition policies and deregulation of utilities sectors. The headline CPI rose by a mere 0.5 per cent in 1999 and – despite higher bank and mortgage interest rates,[8] mounting labour market pressure and the tripling of oil prices – it only accelerated moderately in 2000 to 1¼ per cent. The 2000 increases in the deflators for private consumption and GDP appear to have been even lower (Table 2). Since 1995 the Riksbank has

Table 2. **Various measures of inflation, 1997-2000**

	1997	1998	1999	2000
Consumer price index	0.9	0.4	0.3	1.3
Underlying inflation (UND1X)	1.8	0.6	1.9	1.4
Domestic underlying inflation (UNDINHX)	1.7	1.8	1.8	1.1
Harmonised index of consumer prices (HICP)	1.8	1.0	0.6	1.3
Private consumption deflator	2.3	1.0	0.8	1.0[1]
GDP deflator	1.7	0.9	0.5	0.7[1]

1. OECD estimate.
Source: Statistics Sweden.

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pursued the objective of keeping annual CPI inflation at 2 per cent, although with a tolerance band of1 per cent. However, the increase in the CPI has been consistently below 2 per cent since 1996 and even below the lower limit of the tolerance band for a substantial part of this period (Figure 6). In large part this is attributable to steadily declining interest rates on mortgage debt and reductions of indirect taxes. Measures of underlying inflation, which exclude such components, have been less volatile and, except for short periods, within the tolerance band. Measured by UND1X (see Box 1), underlying inflation rose from 1 per cent in early 1999 to around 1¾ per cent by the end of that year to a large extent as the result of increasing oil prices. It subsequently moderated to around 1½ per cent during most of 2000. Domestic underlying inflation, UNDINHX, declined during 1999 and the first half of 2000, from above 2 per cent to 1 per cent or less. This resulted in part from the fact that the direct effect of the increase in oil prices is not included in this measure and no second-round effects have been visible. Moreover, housing rents, for which the annual increase has been on a clear downward trend, have a greater weight in UNDINHX than in other measures. Various measures of inflation expectations have fluctuated between 1 and 2 per cent since 1998. The increase in UNDINHX picked up again in the second half of 2000 to 1½ per cent by the end of the year. Financial-markets expectations implicitly reflected in the yields for nominal and real bonds have been near the 2 per cent target rate, while households' and manufacturers' surveyed expectations have varied more, with a modest increase over the year.

Sweden has not only followed the general trend towards lower inflation over the past – but even seems to have outperformed most other countries in this respect. From above-average inflation until the mid-1990s, HICP inflation – the most suitable measure for comparison with other EU members – has mostly been lower since 1996. In 1999 and 2000 the difference was some –¾ percentage point (Table 3).

A number of temporary, and partly technical, factors help to explain this consistently lower increase in the harmonised index of consumer prices in Sweden than in the EU as a whole. These include:

- In contrast with most other EU countries, the profit share (*i.e.* the share of gross profits in value added) in Sweden has declined since 1996 in manufacturing, construction and private service industries because unit labour costs have risen faster than output prices (Table 4). While profit margins were unusually high in the middle of the 1990s, in 2000 they were near the average level since 1970 and lower than in the rest of EU (Figure 7). Although there may be room for profit margins to narrow a little more in the short-term, any additional decline in the profit share is likely to be more modest than in the last five years; further increases in unit labour costs will eventually show up in prices. Thus, unless the increase in unit labour costs moderates in the coming years, cost-induced inflation is likely to pick up.

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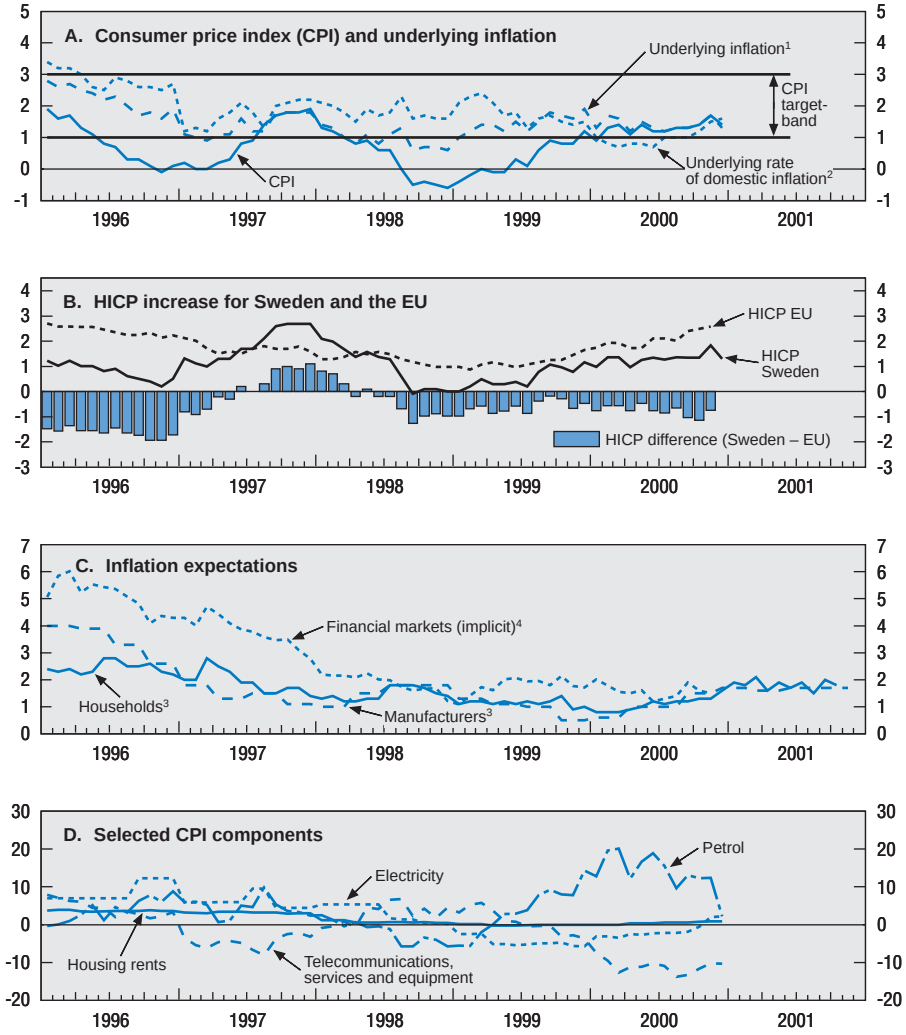
Figure 6. **Inflation and inflation expectations**
Year-on-year percentage change



1. CPI excluding interest payments and direct effects of changes in indirect taxes and subsidies.
2. CPI excluding interest payments, goods that are mainly imported and direct effects of changes in indirect taxes.
3. The curves for expectations have been shifted twelve months into the future so that they coincide with the period to which the expectations refer.
4. Based on the difference between implied 4-14-year real and nominal bonds.

Source: Statistics Sweden; Sveriges Riksbank; OECD.

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Box 1. Inflation measures frequently used in Sweden

The consumer price index (CPI) is the most broadly used inflation measure and the basis for monetary policy decisions. Thus, the objective of price stability has been defined by the Bank as a 2 per cent increase in CPI with a tolerance interval of 1 per cent. Compared with most other countries, Sweden has traditionally been more consistent in following a cost-of-living index approach to calculating the consumer price index. Thus, rather than measuring prices on actually traded goods and services, the CPI aims at measuring user costs. One reason for this is that the CPI is also used as the basis for indexing transfers and other expenditures as well as threshold values in the tax system, and that the authorities have wanted these adjustments to match the increases in living costs as closely as possible. For most goods and services this difference is trivial, but in a few cases it is not. In particular, the user costs of owner-occupied housing are difficult to measure and involve a number of uncertain imputations such as the rate of depreciation and amortisation profiles of the stock of owner-occupied housing. Another consequence is that changes in interest rates, property taxes, and the tax assessment of houses influence the calculations. In a recent report from the CPI Commission (SOU, 1999*a*) a number of changes to the CPI calculations were proposed, notably to the housing component, but these have subsequently been widely criticised. Except the inclusion as from 2002 of households' expenditures for publicly provided day care, more fundamental changes are thus not expected in the near term.

Besides the CPI and HICP the measures mentioned below are also calculated by Statistics Sweden and published on a monthly basis.

UND1X is usually taken as a measure of *underlying inflation*. In contrast with the CPI, changes in indirect taxes and subsidies as well as interest payments on owner-occupied housing are ignored in its calculation, but the relative weights of their components are identical to those in the CPI. By virtue of implicitly assigning zero-rate increases to these components UND1X tends to be less volatile than CPI in the short term. UND1X has been the basis for monetary policy decisions for several years, but the CPI is still the central target variable in the long term. UND1X is calculated by Statistics Sweden and reported together with the monthly CPI data. Separate information on the contribution to CPI from interest payments and indirect taxes respectively is not published by Statistics Sweden.

UNDINHX is a measure of *domestic underlying inflation* also calculated at the request of the Bank. As well as excluding changes in indirect taxes, subsidies and interest payments, UNDINHX also excludes goods which are predominantly imported.

A monthly *net price index* (NPI) is calculated by excluding indirect taxes and subsidies from the CPI. In contrast with UND1X the weights for all goods and services are adjusted in accordance. The NPI excludes a wider range of indirect taxes and subsidies than UND1X. For example, subsidies to interest payments for housing and changes in payroll taxes have been excluded from the NPI in recent years but not from UND1X.

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Table 3. **HICP inflation in Sweden and EU, 1996-2000**

	1996	1997	1998	1999	First half of 2000
	Annual increase, per cent				
HICP, Sweden	0.8	1.9	1.0	0.5	1.2
HICP, EU-15	2.4	1.7	1.3	1.3	1.9
	Percentage points				
Difference	**–1.6**	**0.2**	**–0.3**	**–0.8**	**–0.7**
of which: (price-weight split)					
Different weight for sub-groups[1]	0.2	0.3	0.0	–0.2	–0.2
Price changes within sub-groups[1]	–1.8	–0.1	–0.3	–0.6	–0.5
of which: (by components)[2]					
Food	–1.5	0.0	0.0	0.2	0.0
Alcohol and tobacco	..	0.4	0.2	–0.4	–0.1
Housing[3]	0.6	0.4	0.0	–0.4	–0.3
Transport[3]	..	–0.1	–0.1	–0.1	0.0
Postal services, telecommunication	0.0	0.0	0.1	0.1	–0.2
Recreation and culture	0.0	–0.2	–0.1	0.0	0.1
Restaurant and hotels	..	–0.2	–0.2	–0.1	–0.2
Others[4]	–0.7	–0.1	–0.2	0.0	0.0

1. The decomposition is undertaken with the use of the set of weights applying to EU as a whole. Using the Swedish set of weights would give a slightly different result.
2. The table reports each of the component's contribution to the difference in the increase in the overall HICP index in Sweden and EU, respectively. This is the combined result of different increases within each group and differences in their weights in the HICP indices for Sweden and the EU.
3. There is not a separate sub-group for energy in the COICOP classification. Energy is included in the two sub-groups "housing" and "transport".
4. Includes for 1996 (implicitly) values for sub-groups for which information is not available.

Source: Eurostat and OECD.

Table 4. **Costs, prices and profitability in the business sector**

Average annual increase in per cent, 1995-99

	Total	Manufacturing	Construction	Non-financial services
Unit labour costs	1.9	–0.1	4.3	2.0
Hourly compensation	4.9	4.5	5.4	5.3
Labour productivity	3.0	4.4	1.1	3.3
Output price	0.2	–1.1	1.9	1.1
Profit share in value added[1]	–1.6	–2.0	–1.7	–0.7

1. Percentage points change.

Source: National Institute of Economic Research.

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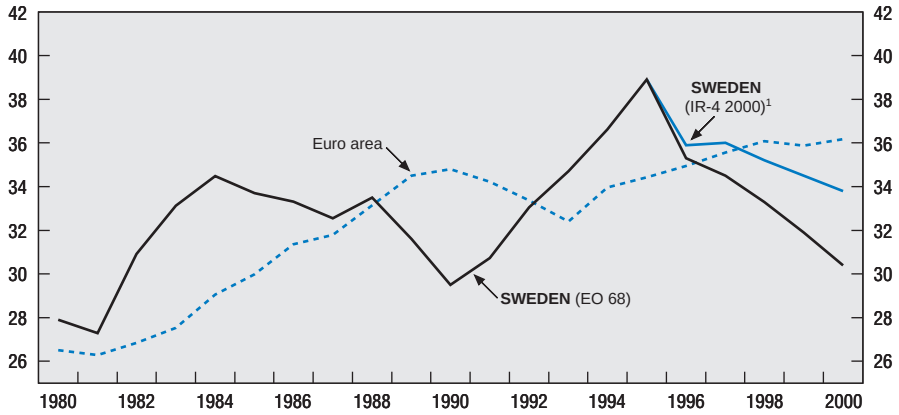
Figure 7. **Profit share in the business sector in Sweden and the euro area**

Per cent of value added



1. From the Bank's fourth quarter 2000 *Inflation Report*; includes revised national accounts data for 1997 to 1999.
Source: OECD.

– As mentioned above the increase in the housing component of the consumer price index has fallen remarkably since 1996 (see Figure 6 above). Even disregarding the contribution from lower interest rates, the annual increase in the housing component has fallen from some 4 per cent in 1996 to around zero in 2000 and has even been slightly negative for a period. In 1999 and the first half of 2000 the housing component accounted for almost half the difference in HICP increases between Sweden and EU (Table 3 above). Considering the still more tangible shortages of accommodation in recent years, especially in the bigger cities, this is probably more than anything a reflection of *de facto* rent controls and regulations in the housing market.[9, 10]

– Exchange rate developments have played a role as well. In particular, the 15 per cent appreciation of SKr *vis-à-vis* the euro during 1999 and the first half of 2000 (see Chapter II) has contributed to lower inflation in Sweden relative to the euro countries during most of this period. The reversal of this exchange-rate trend in the second half of 2000 may work to narrow the differential between inflation in Sweden and the EU as a whole in the near term.

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– The increase in energy prices has affected Sweden less than most other countries. While the energy component in the HICP rose almost 20 per cent from early 1999 to mid-2000 in the EU, it increased by only 10 per cent in Sweden. For one thing, higher-than average taxes on energy (see Chapter IV) reduce the pass-through into consumer prices. For another, by virtue of hydropower and nuclear energy Sweden is largely self-sufficient in electricity, and therefore less exposed to fluctuations in world market prices for oil and natural gas.

Wage developments

The reduction of unemployment and the evidence of mounting labour-market pressure (see below) has not been manifest in nominal wage increases, even if the latter may have passed the turning point. Private-sector hourly earnings increased 3.3 and 4.1 per cent in 1999 and 2000, respectively (Table 5). The pickup in 2000 was in part technical, reflecting the timing of pay increases already decided in the collective agreements from 1998. Wage drift at the enterprise level remained modest and stable in 2000. In the public sector, hourly earnings increases reached a low of 3.1 per cent in 1998 and have accelerated since then to 4.0 per cent in 2000. Taking the economy as a whole, the increase in wages was 4.1 per cent in 2000, higher than in 1999 but comparable to the increase in 1998. Total labour costs have risen faster than hourly earnings in each of the years

Table 5. **Hourly earnings and labour costs, 1997-2000**

	All employed			
	1997	1998	1999	2000[1]
	Annual increase, per cent			
Hourly earnings				
Private sector	4.4	4.1	3.3	4.1
Manufacturing	4.5	4.2	2.5	3.8
Public sector	4.7	3.1	3.7	4.0
Total	4.5	3.8	3.4	4.1
Implicit hourly wages[2]	4.6	4.3	3.3	–
Hourly labour costs				
Private sector	4.8	4.6	3.4	–
Manufacturing	4.8	4.0	2.8	–
Public sector	5.1	3.6	3.8	–
Total	4.9	4.2	3.6	–

1. OECD estimates.
2. From the national accounts.
Source: National Institute of Economic Research and OECD.

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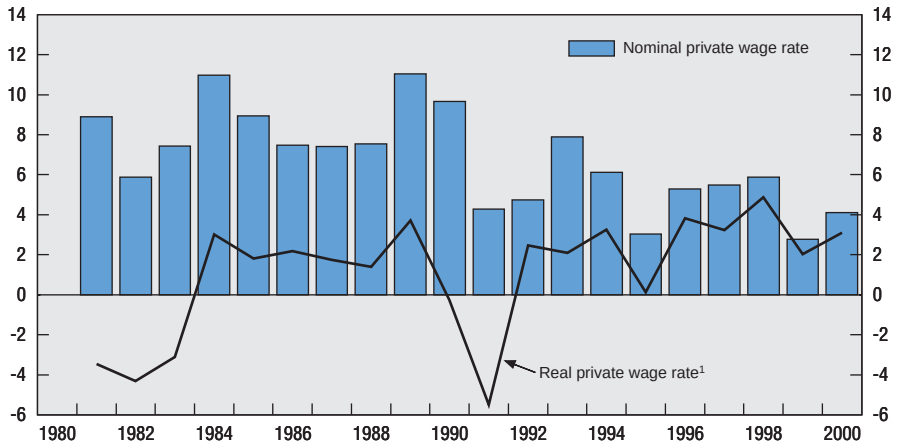
since 1997. On average the annual addition from indirect labour costs has been around ½ percentage point for this period. This pattern is in contrast with those in many other European countries, (for example, France and Italy) which have reduced payroll taxes recently.

It is ironic that the average increase in nominal wages in the three most recent years has been the lowest in several decades, while real wage increases at the same time have been high by historical standards, in large part due to unforeseen price stability (Figure 8). Fundamentally this is a benefit of the painful but now fully accomplished transition from a high-inflation economy to one where price stability is a widely accepted and credible core objective for economic policy (see Chapter II). Nevertheless, with current levels of labour shortages and the upward trend in labour demand as yet unabated, caution still needs to be observed. Moreover, the relevant benchmarks against which wage developments should be assessed are wage increases abroad and the consistency with the inflation target:

– Low wage and price inflation has been consolidated across most of the OECD area during the 1990s, and the absolute margin for excesses in individual countries may have narrowed as a consequence. In this perspective, it should be observed that even the historically modest

Figure 8. **Nominal and real wage increases**

Per cent change



1. Deflated by the private consumption deflator.
Source: Statistics Sweden; OECD, *National Accounts* and estimates for 2000.

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nominal wage increases in Sweden in recent years have been slightly above those in competitor countries. Thus, while private-sector wages rose at an average annual rate of 3¾ per cent from 1998 to 2000 in Sweden, the comparable increases were 2½ and 1¾ per cent for the euro- and EU areas, respectively.

– Consumer price inflation has been consistently below the 2 per cent inflation target in recent years, indicating that recent wage performance has been well aligned with the monetary policy objectives. However, as discussed above, output prices in the business sector have not risen in line with increases in unit labour costs. While this has been helpful in terms of keeping consumer price inflation low, it has been at the cost of shrinking profit margins in most industries (see Figure 7 above). Even if the profitability of businesses is not of concern at present, profit margins are not likely to decline much further. Rather, nominal wage increases, corrected for productivity gains, should henceforth be expected to carry over into output prices and, in turn, consumer prices.

Notwithstanding these reservations, the modest nominal wage increases in recent years are encouraging, and the risk of a sudden shift in wage inflation is limited. This is due in large part to the inflation target, which serves as a firm anchor for inflation expectations and, therefore, restrains the claims put forward in wage negotiations. The wage bargaining system has developed along two lines in recent years; one is stronger central co-ordination following the 1997 "Industrial Agreement" and the other is a greater role for bargaining at the enterprise level. As a consequence the system has become more dual. The move towards stronger central co-ordination is probably crucial for the impact of the inflation target on wage formation. Wage drift at the enterprise level is less likely to be influenced by the inflation target and remains an important risk factor in the coming years. The most likely risk would seem to be a scenario with total wages increasing moderately, but consistently, above those of competitor countries over a number of years.

Output and productivity at the aggregate and industry levels

Potential output in Sweden has grown at an estimated annual rate of 2½ per cent in recent years compared with a lacklustre 1½ per cent in the first half of the 1990s. While trend productivity growth has been rather constant over the last decade, the non-accelerating inflation rate of unemployment (NAIRU) and the trend participation rate have shown distinctly different developments between the first and the second half of the 1990s. The trend participation rate fell markedly between 1990 and 1995, in line with the collapse of the actual participation rate, but has stabilised since then. Likewise, the NAIRU is estimated to have declined moderately since the mid-1990s, reversing part of the increase in the first

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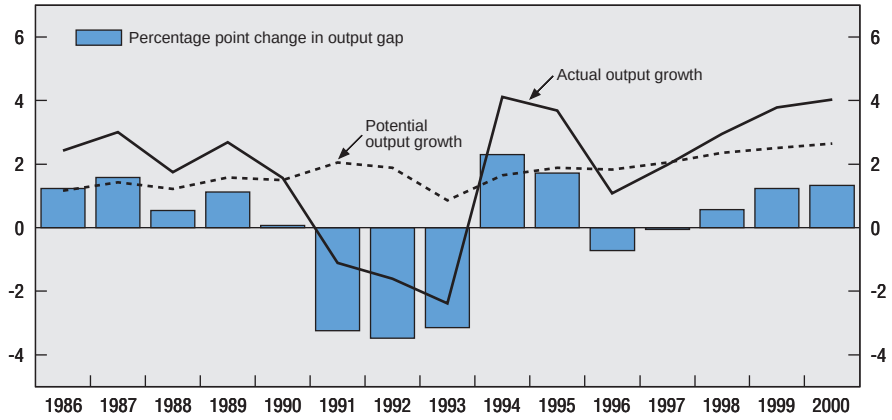
Figure 9. **Actual and potential output**
Per cent



Note: The data used in this figure are identical to those in *Economic Outlook 68* and do not take into consideration the revised national accounts data for 1997 to 1999 which were published in November 2000.
Source: Statistics Sweden; OECD, *National Accounts* and estimates.

half of the decade (Richardson *et al.*, 2000). Despite this recent improvement in the underlying sustainable growth rate, actual output has expanded at rates well above potential for some years, in particular in 1999 and 2000 (Figure 9). Consequently, the output gap has increased considerably; the accumulated change in the gap from 1994 to 2000 amounted to more than 6 percentage points. OECD estimates suggest that the output gap may have closed and turned positive in 2000. However, such estimates are subject to considerable uncertainty, not least in the case of Sweden where the amplitude of business cycle fluctuations over the last decade has been larger than in most other countries. Thus, it is easily within the margin of uncertainty that the output gap was still slightly negative in 2000. This is indeed the consensus view in Sweden where most forecasters project the output gap to close only during the course of 2001.

Output growth has been broadly based, with most sectors of the economy booming. For the business sector as a whole, the growth rate has been in the range of 4½ to 5 per cent in the years 1998 to 2000 (Table 6). An increase in productivity growth might potentially explain the recent combination of buoyant growth and low inflation. Productivity growth has certainly been high in some industries, most notably within the ICT sector and even at the aggregate level signs of a pickup relative to the historical average are beginning to appear.[11]

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Table 6. **Output and hourly labour productivity 1981-2000**

	Average 1981-1989	Average 1990-1996	1997	1998	1999	2000[1]
	Annual increase, per cent					
Business sector						
Output	2.5	1.4	3.3	4.6	4.9	4.8
Productivity	1.8	2.3	3.9	2.9	1.8	2.4
Manufacturing						
Output	2.3	2.4	5.3	6.8	4.1	7.0
Productivity	2.7	4.1	6.4	5.3	2.9	4.5
Construction						
Output	2.0	–2.5	–4.4	1.0	5.9	4.4
Productivity	1.4	1.8	–2.7	0.2	1.0	0.9
Private services						
Output	2.9	1.9	3.8	4.0	5.1	4.1
Productivity	1.1	1.6	3.7	1.8	1.1	1.6
Whole economy						
Output	2.3	0.7	2.4	3.6	4.1	4.0
Productivity	1.2	1.6	3.2	2.3	1.3	2.0

1. NIER estimates.
Source: National Institute of Economic Research (NIER).

Average annual labour productivity growth in the business sector was somewhat higher in the 1990s than in the 1980s – 2.4 per cent in the 1990s against 1.8 per cent in the 1980s. However, this should be viewed in the perspective that the 1980s average in itself was quite low and that the 1990s outcome was lifted by the scrapping of the least efficient companies and activities during the reconstruction period (OECD, 1999*a*). Considering just the most recent three years, the average annual increase in business-sector labour productivity was ½ percentage point higher than the average increase since 1980. This masks considerable differences across sectors; while productivity growth has been below the historical average in construction and private services, it has been above average in manufacturing. Although it is probably too early to conclude firmly that a durable acceleration in aggregate productivity growth has occurred, such a scenario – related to the spreading of ICT across the economy and continued gains from enhanced competition – might certainly be possible in the near future.

Improved labour market performance

Employment and unemployment

Against the backdrop of buoyant demand and rapid growth in output, employment has risen solidly in the last three years, and labour shortages have

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spread (see Box 2). Some 250 000 new jobs were created, corresponding to an average annual increase of more than 2 per cent and the fastest expansion in employment in three decades. Most of the increase has been in the private sector and particularly in private services. Regional differences have been evident here too; the increase has been much more pronounced in the metropolitan areas. However, in 2000 employment growth accelerated in the less densely populated regions as well. Labour supply continued to shrink until late 1997 but has recovered somewhat since then, partly reflecting the policy of downscaling active labour market programmes and the return of discouraged workers. By the end of 2000, the labour force was up 2½ per cent per cent on the level two years earlier but still well below the levels in the early 1990s. The medium- and long-term trends in employment and labour force participation are discussed more thoroughly in Chapter III.

Box 2. Survey indicators of labour shortages across industries

Evidence that the labour market has tightened has become steadily clearer, although pressures apparently abated somewhat in the fourth quarter of 2000. The number of unfilled vacancies has risen quite rapidly during 1999 and most of 2000, indicating that labour shortages are spreading (Figure 10). The same picture emerges from survey data, even if there are significant differences across industries and some reversal lately.

- In manufacturing, the share of companies reporting shortages of labour increased until the third quarter of 2000 and reached its highest level in 10 years. While in level terms shortages are still much below earlier peak levels for manual workers, they have approached them for salaried employees with technical skills.
- In other private services, the share of companies facing labour shortages has climbed in most sectors in recent years, and some acceleration in that trend was visible in 2000. It is not possible to compare the significance of bottlenecks in private services with the upswing in the late 1980s, as the relevant surveys started only during the 1990s.

When assessing the importance of labour shortages at present and their possible policy implications, the focus should not primarily be on the levels themselves. Rather, the direction and the speed of recent changes might be of concern, in particular if actual GDP continues to expand significantly more rapidly than its potential. If, as expected by households, unemployment continues to trend down, capacity constraints might become more apparent in the near term. Past experience shows that labour shortages can spread very rapidly; thus, in the 1980s' upswing they doubled from a level comparable to the current reading in only three quarters.

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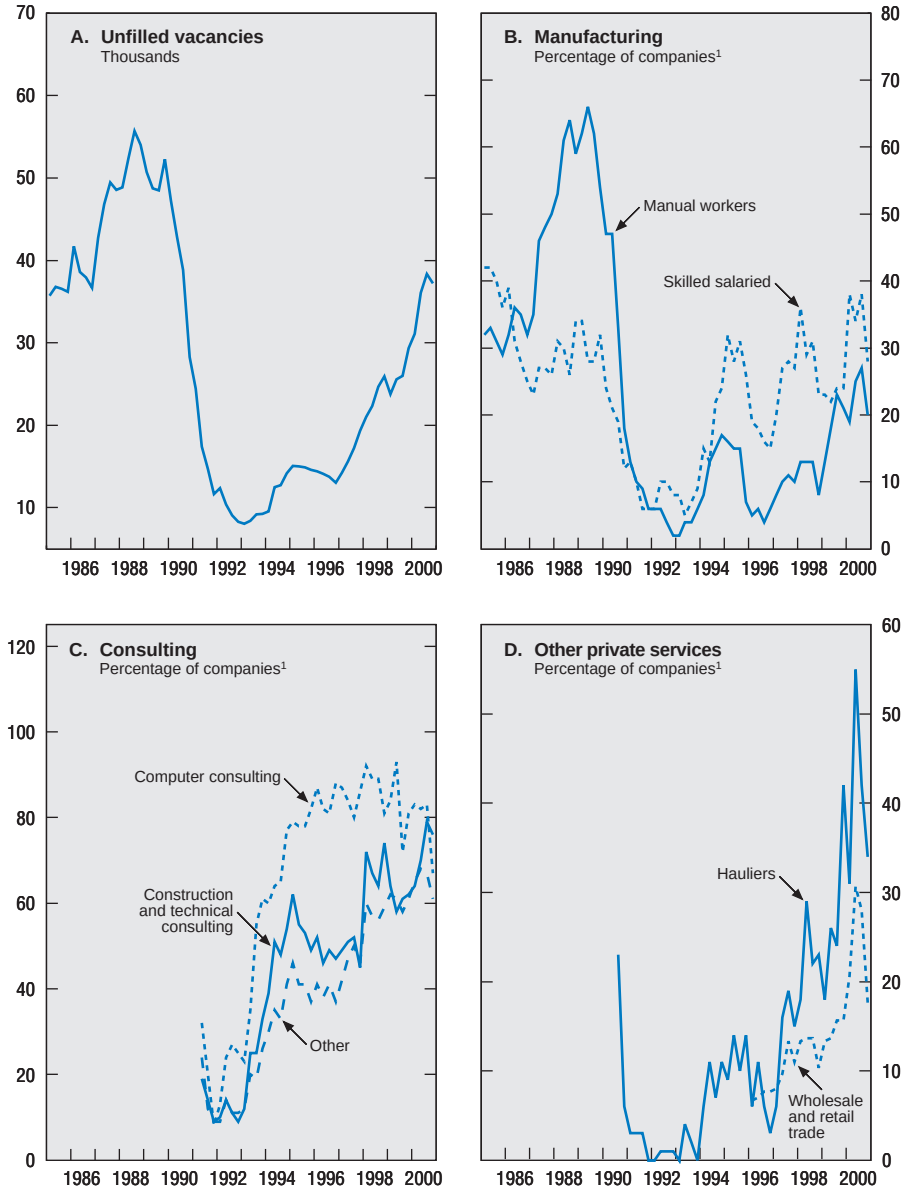
Figure 10. **Indicators of labour shortages**



1. Percentage of companies reporting labour shortages.
Source: Ministry of Finance; OECD, *Main Economic Indicators.*

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Since peaking in the period 1993 to 1997, unemployment has been on a downward trend; according to the official definition,[12] it reached 4 per cent by the end of 2000, 4 percentage points lower than the level four years earlier (Figure 11). The decline in unemployment in 2000 was reinforced by the fact that enrolment in labour-market programmes, which had been declining in previous years, started climbing again over the course of 2000, although enrolment for the year as a whole was lower than in 1999. Registered unemployment, *i.e.* unemployed registered at the public employment service (PES), fell in parallel with the official unemployment rate during 1997 and 1998 but less rapidly during most of 1999 and 2000. Looking ahead, several indicators suggest that unemployment may continue on its downward path in the near term. In particular, newly reported vacancies increased throughout most of 2000, and recent survey data indicate that household expectations of a further reduction in unemployment have been more pronounced and widespread than at any other time during the current upswing.

Has the growth-inflation trade off improved in recent years?

The recent years' combination of high growth and low inflation has come as a surprise for most analysts in both Sweden and elsewhere, and there is no doubt that earlier assessments – including those of the OECD – of the economy's capability of reconciling these two key objectives have proved too pessimistic. The inflation-targeting regime and the more independent status of the Bank have obviously been important in anchoring inflation expectations at the current, historically low level, and may have interacted positively with recent changes in the wage bargaining system (see Chapter II). While policy-initiated labour-market reforms seem of second-order importance, structural changes on product and capital markets are likely to have played a role (see Chapter III). Deregulation of utilities sectors, notably electricity and telecommunications, have put downward pressure on prices, albeit probably not more so than in other countries. Finally, competition is likely to have strengthened gradually from such factors as the new competition framework established in 1993, Swedish EU membership from 1995 and the general trend towards more global production and consumption patterns. Although difficult to quantify, these trends may have combined to give perceptible contributions to reduced inflation. However, an important question is to what extent, and under which conditions, the recent favourable developments can be extrapolated into the future.

Supplementary explanations of the recent favourable growth-inflation performance include the possibility that idle resources were more abundant at the outset of the recovery than has been envisaged hitherto, *i.e.* the output gap was underestimated. In addition, this chapter has identified at least two significant factors that have helped to reduce inflation in recent years, both of which are likely to be temporary. One is that that gross profit margins (markups) rose until

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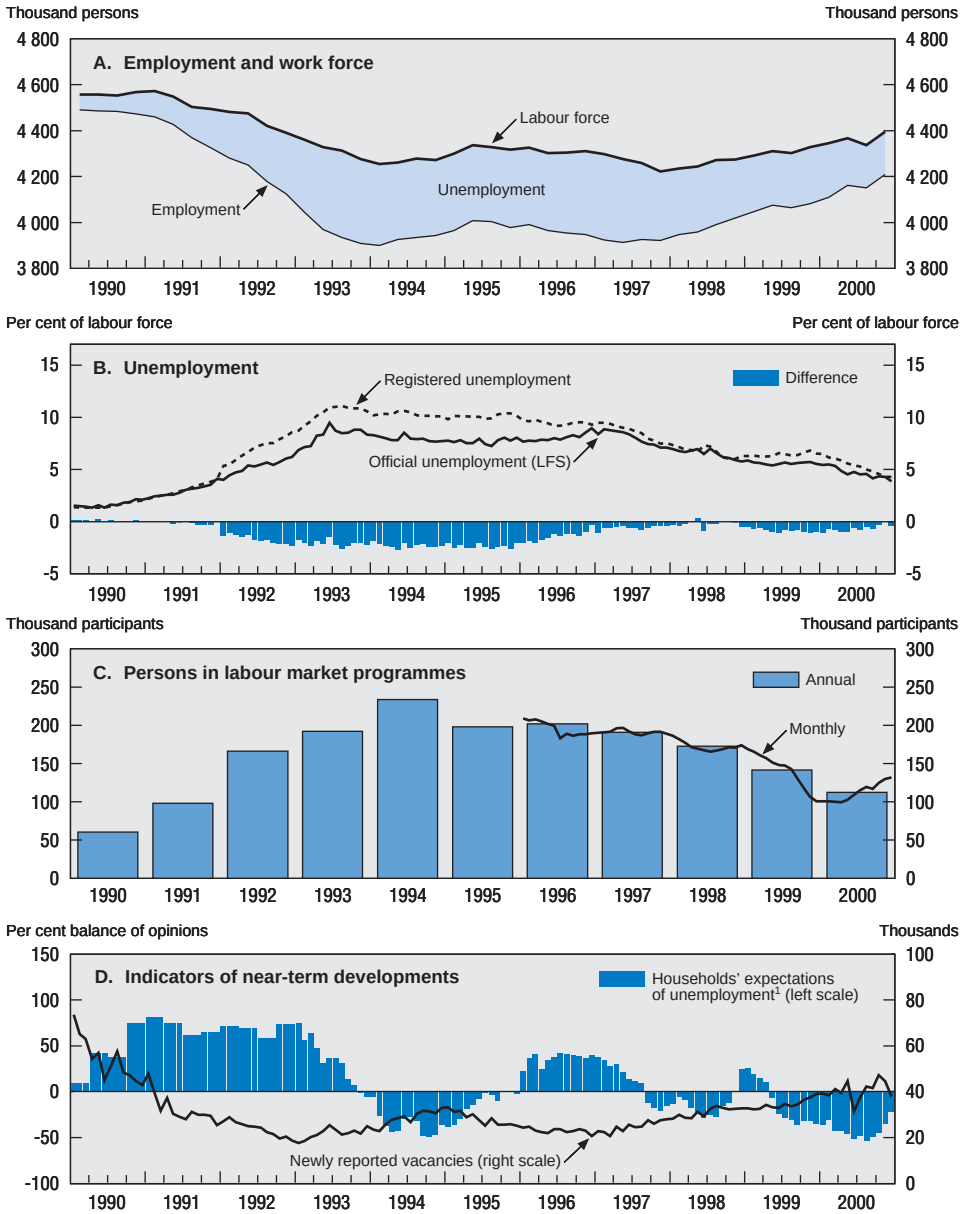
Figure 11. **Labour market developments**



1. Negative values indicate that unemployment is expected to decline.
Source: Arbetsmarknadsstyrelsen, Statistics Sweden and OECD.

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the mid-1990s but have declined since then. The latter is likely, in part, to have been offset by declining capital costs and may therefore not have weakened business profitability. Nonetheless, the likelihood is that any further decline in the business-sector profit share will be more modest and that, henceforth, increases in unit labour costs will carry over into prices more than they have done recently. The second notable factor is the decline in housing rent increases from more than 4 per cent in 1996 to zero in 1999 and 2000, which seems related to regulation akin to rent controls. These explanations might suggest that temporary factors have to some extent masked the fact that the economy has moved into a situation of excess labour demand, which would seem consistent with the spreading of labour shortages and high real wage increases. On the other hand, productivity growth in the business sector has been above the historical average in recent years. If this is a durable change, it should – all other things being equal – contribute to a higher potential growth rate henceforth.

The outlook to 2002[13]

With the fiscal balance and the current account both in comfortable surplus and inflation below the average level in EU, there are no immediate risks of serious imbalances emerging. The assumed persistence of relatively high oil prices and the decline in equity prices since March 2000 both point in the direction of a welcome moderation in demand growth in the near term. Recent external developments and slower growth in retail sales point in the same direction. However, most other underlying factors point to greater persistence in above-potential growth even in 2001 and 2002. In view of total tax cuts of more than 1 per cent of GDP in 2001 the fiscal policy stance is quite supportive. The resulting stimulus is expected to be reinforced by higher spending by municipalities; the requirement to balance the budget as from 2000 has restrained local governments' outlays in recent years, but that will cease now that balance has been achieved. The budget for 2002 is assumed to be neutral to economic activity, even if further tax cuts have been signalled and are widely expected. Monetary conditions are also supportive to demand. Nominal interest rates have remained very low throughout 2000 and will continue adding impetus to domestic demand for some time to come. Moreover, the weakening of the exchange rate in the second half of 2000 – which in part resulted from the easier monetary policy stance than in the euro area – should boost export demand in the near term, although slower export market growth is expected to work in the opposite direction. The current expansionary macroeconomic policy stance is reflected in strong consumer and business confidence (even if some moderation has taken place since mid-2000) and widespread expectations of a continued decline in unemployment.

As current macroeconomic policies are not exerting any restraint on domestic demand, capacity constraints, already visible in the spreading of labour shortages, are likely to be exacerbated in the near term. This, in turn, is projected

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to result in losses of market shares by Swedish producers at home and abroad. GDP growth is expected to moderate from 4 per cent in 2000 to 3¼ per cent in 2001 and around 2½ per cent in 2002 (Table 7). Thus, it will remain above potential in the near term and approach sustainable levels only towards the end of the projection period. Total domestic demand, on the other hand, is projected to keep expanding at growth rates above 3 per cent throughout the period. Inflation

Table 7. **Outlook for 2001 and 2002**

Percentage change from previous year (unless otherwise noted)

	1999[1]	2000[1]	2001	2002
Private consumption	3.8	5.0	4.8	3.0
Government consumption	1.7	–1.4	1.3	1.8
Gross fixed investment	8.1	4.9	6.4	5.9
Housing	22.3	8.0	15.0	10.0
Business	6.7	6.0	6.0	6.0
Government	6.1	–3.0	3.4	2.4
Final domestic demand	3.9	3.2	4.2	3.3
Change in stockbuilding[2]	–0.5	0.1	–0.1	0.0
Total domestic demand	**3.3**	**3.3**	**4.1**	**3.2**
Export of goods and services	5.9	9.7	6.8	5.1
Imports of goods and services	4.3	8.5	9.0	7.0
Net foreign balance[2]	0.8	1.3	–0.3	–0.4
Gross domestic product (constant, market prices)	**4.1**	**4.0**	**3.2**	**2.4**
Memorandum items:				
Private consumption deflator	0.8	1.0	1.8	2.3
GDP price deflator	0.5	0.7	1.9	2.7
Oil prices, $US per barrel[3]	17.3	28.3	29.5	27.5
Total employment	2.2	2.0	1.3	1.0
Labour force	1.2	1.1	0.7	0.6
Unemployment rate, per cent of labour force	5.6	4.7	4.1	3.7
Effective exchange rate	–0.5	0.2	–3.2	0.0
Interest rates				
Short-term, per cent	3.1	3.8	5.0	5.3
Long-term, per cent	5.0	5.5	5.7	5.8
Government balance, per cent of GDP	1.5	3.4	3.3	3.8
Current account, per cent of GDP	2.8	2.4	1.6	1.1
Household savings rate	2.1	2.1	2.0	2.2

1. The estimates for 2000 are unchanged since *Economic Outlook* 68; specifically, they do not take into consideration the national accounts data released on 13 December (first estimates for the third quarter and revisions to the first half). However, the 1999 figures are those revised and released on 20 November.
2. As a per cent of GDP in the previous year.
3. Crude oil prices are OECD average import unit prices, including cost, insurance and freight, but excluding import duties.

Source: OECD, *Economic Outlook* 68.

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is expected to pick up modestly from the unusually low rates of 1 per cent or less in recent years; the private consumption deflator is expected to rise 1¾ per cent in 2001 and 2¼ per cent in 2002. The main factors contributing to the higher inflation are upward pressure on import prices due to the currency weakening since mid-2000 and a steadily tighter labour and product market. With unemployment edging down towards 3½ per cent by the end of 2002, wage inflation is expected to gradually pick up and exceed 5 per cent in 2002. In addition, the recent years' fall in the business sector's profit share is expected to come to an end, implying that increases in labour costs will feed into price increases more than of late. Against the background of some acceleration in inflation, policy-controlled interest rates are expected to be raised. This should dampen output growth in 2002, but the tightening may come too late to prevent inflation from exceeding the 2 per cent target slightly that year. The budget surplus is likely to remain at nearly 3½ per cent of GDP in 2001 and increase to 3¾ per cent in 2002. A further deterioration of the private sector's savings-investment balance is projected; while the savings rate is expected to be a largely constant, business and housing investment is expected to expand rapidly. The result is a decline in the current account surplus from almost 2½ per cent of GDP in 2000 to just above 1 per cent in 2002.

Weakening of equity markets in Sweden and abroad as well as recent signs of global slowing are factors that might, if continuing, lead to lower than projected growth. On the other hand, a number of risks to the projection are on the side of higher growth in demand and output. For example, oil prices may fall faster and stronger than assumed, which would in event also reduce inflation in the short term. Moreover, the losses of market shares in 2001 may be less pronounced than projected if the recent weakening of the effective exchange rate persists. Also, with house prices still on the rise the household savings rate may continue declining, particularly if the Riksbank raises interest rates only modestly. The real-economy effects of smaller interest-rate increases would, however, primarily show up in 2002, and the stimulus to demand and output would be reinforced if fiscal easing were also to continue. With the government's commitment to tax cuts, the risk is that, like in 2001, these will not be offset by lower spending. To what extent such a more accommodating policy stance would feed through into actual growth is uncertain. On the presumption that output was already above potential in 2000, it might primarily increase inflationary pressures above what is projected, leading in particular to added wage drift at the enterprise level, or cause even higher growth in imports and hence a more rapid dwindling in the current account. On the other hand, the OECD's assessment of the output gap at the outset may be too pessimistic, or additional inputs might become available at a faster pace than projected.

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II. Macroeconomic policies

Comprehensive reforms of the frameworks for both monetary and fiscal policies took place during the 1990s. This reflected a growing recognition that the poor economic performance in the 1970s and 1980s, manifest, for example, in high inflation, several currency crises and rising public and foreign debt was in large part attributable to shortcomings in the earlier macroeconomic policy settings. The policy paradigm has been changed repeatedly during the last 25 years; the evolution of stabilisation policies in Sweden was characterised in a recent report as having been "looking ahead through the rear-view mirror", indicating that past failures have been a major impetus for reforms (Jonung, 1999). The current framework for macroeconomic policies has been in place for a number of years by now; monetary policy has been targeting a 2 per cent inflation rate since 1993, and fiscal policy has been governed by expenditure ceilings and a target for the cyclically-adjusted budget surplus since 1997. This chapter reviews recent performance with respect to meeting monetary and fiscal policy objectives.

To summarise, inflation has remained well below the target during 1999 and 2000, and the monetary authorities have projected it to stay below 2 per cent up to at least the end of 2002. Against this backdrop monetary policy has been accommodating. The budget surplus has increased from 1¾ per cent of GDP in 1998 to more than 3 per cent in 2000, because savings on social transfers resulting from improved labour market performance and lower interest payments have offset discretionary stimulus. Notwithstanding the record of low inflation hitherto, the output gap may have turned positive in 2000, and the expansionary macroeconomic policy stance could therefore risk pushing demand further beyond the supply capacity of the economy. This would cause inflationary pressures to build up, and a tighter overall macroeconomic policy stance would, under these conditions, be clearly warranted. In addition, structural policy initiatives to increase the economy's production capacity would also be most helpful.

Monetary policy

The framework for monetary policy

The framework for monetary policy was strengthened in 1999 through an amendment to the Central Bank Act, which made the *Riksbank* (henceforth the

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Bank) independent from the political executive and brought the Swedish legislation into closer compliance with the Maastricht Treaty. The management structure of the Bank was changed: the former Governing Board was replaced by a new body, the Executive Board which is now responsible for operational matters in the monetary and exchange-rate domains.[14] Furthermore, price stability was formally stated as the objective for monetary policy. Although the new legislative framework was primarily an adjustment of formal rules to already established practice and has not directly affected the conduct of monetary policy, it should help to further underpin the already firm and well-established credibility of the inflation-targeting regime.

Since the previous S*urvey*, the Bank has retained the objective of keeping inflation one to two years ahead at 2 per cent with a tolerance band of 1 per cent. The inflation target has gained full support from the employee organisations, and, more generally, the confidence in the Bank's commitment and ability to maintaining price stability has strengthened at the political level, on the financial markets and amongst consumers and businesses.[15] Thus, the inflation-targeting regime and the more independent status of the Bank are key factors to recent years' favourable inflation performance. However, the definition of inflation has been subject to debate at times. For the past five years, headline CPI inflation has remained well below the mid-point of the target interval. Based on these outcomes monetary policy has occasionally been criticised as being too tight.[16] However, interest-rate fluctuations as well as frequent changes in indirect taxes have added to short-term volatility in headline CPI inflation, and the Bank has typically chosen to ignore such temporary effects.[17] This was eventually formalised in mid-1999 when the Bank announced that for the time being it would base its monetary-policy decisions on underlying inflation as measured by UND1X; however, for the long-term the CPI remains the key variable.

The Bank has been successful in promoting transparency in several ways, not least by publishing its inflation forecasts. In addition, the minutes from the meetings of the Governing Board, including the voting of individual Board members, are available to the public approximately two weeks after the meetings have taken place. However, the above decision to focus on UND1X involves a trade-off. Even though the undershooting of the CPI relative to the target has often been attributable to temporary factors – typically with insignificant effects on the real economy and on inflation expectations – the shift of the short-term target variable adds an *ad hoc* element to monetary policy inasmuch as it is temporary. In any case, the distinction between the CPI and UND1X should be considered important only if there is a noticeable difference between the Bank's forecast one to two years ahead for the two measures (Apel *et al.*, 1999). While this was the case in mid-1999, the official forecasts for the two inflation measures have been quite close more recently.[18] Although of second-order importance, transparency might also suffer from the fact that UND1X is published only as an aggregate measure;

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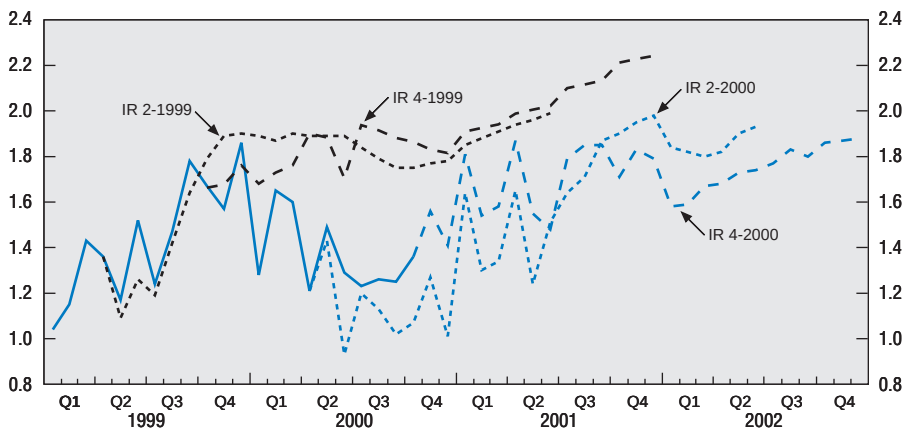
information on subgroups or even a breakdown of the difference between this measure and the CPI on interest rates and indirect taxes is not published by Statistics Sweden. However, this information is unveiled with some delay by the Bank in its quarterly Inflation Reports.

Implementation of monetary policy during 1999 *and* 2000

Monetary policy decisions are taken on the basis of the Bank's inflation forecasts. These have a two-year time horizon and are compiled under the technical assumption of unchanged policy-controlled interest rates. A closer look at the inflation forecasts presented during 1999 and 2000 reveals that while the modest increase in underlying inflation in the second half of 1999, to rates of around 1.7 per cent, was in line with the Bank's expectations, the subsequent moderation in 2000 came as a surprise (Figure 12). The Bank's recent inflation projections have all been very close to the 2 per cent inflation target at the end of the projection period; in the fourth-quarter 2000 Inflation Report, which covers the whole of 2002, inflation (UND1X) is forecast to stay marginally below the 2 per cent target, even though growth in demand is expected to remain robust and unemployment to fall to 3.7 per cent. The continued buoyancy of demand even in 2002 is partly explained by the assumption of tax cuts of more than ½ per cent of GDP even if no

Figure 12. **Selected inflation forecasts by the Riksbank, 1999-2000**
Year-on-year percentage change



Note: Projections of underlying inflation (UND1X) published in Riksbank's Inflation Reports (IR).
Source: Sveriges Riksbank.

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decision on tax cuts for 2002 has been taken as yet. It should also be observed that technical effects stemming from fiscal policy decisions already taken – a cap on childcare charges and a reduction of the property tax on rental housing – amount to a reduction of the increase in UND1X by ½ percentage point in 2002.

Since the last *Survey*, monetary policy easing has ended, and a certain tightening has taken place. The repo rate was cut twice by ¼ percentage point at the beginning of 1999 to 2.90 per cent (Figure 13), marking the end of the series of downward interest rate adjustments that had begun in mid-1998. The same rate was increased by 0.35 percentage point in November 1999 and by a further 0.50 percentage point in February 2000, reflecting concerns at that time that the strong momentum in the upswing might give rise to increased inflationary pressure. Subsequently, underlying inflation remained well below 2 per cent throughout 2000, and the Bank abstained from further repo rate changes until December 2000, when it was raised by ¼ percentage point. While motivated by domestic considerations, these shifts in monetary stance have been much smaller than interest rate developments elsewhere. The Swedish repo rate has been lower than that of the euro zone since June 2000, and Sweden was one of only three countries in the OECD (along with Japan and Switzerland) with policy interest rates below 4 per cent in the second half of 2000. Since August 2000 the repo rate spread *vis-à-vis* the euro has been –¾ to –1 percentage point. Reflecting these changes in policy-controlled interest rates, short-term money-market interest rates in Sweden moved below the level in the euro area in mid-2000. That differential was also in the range –½ to –1 percentage point during most of the second half of the year. This is likely to have impacted on capital flows and consequently on the value of the krona.

Under the current inflation-targeting regime the exchange rate is market-determined and impinges on interest-rate decisions only to the extent it influences the assessment of future price developments. The nominal effective exchange rate strengthened by over 7 per cent in the 18 months to June 2000 but then weakened by an almost similar amount until the December repo rate move was fully discounted. The currency depreciation in the second half of 2000 was in contrast with the Bank's expectations of an appreciation against the dollar and a stable rate against the euro. In its second-quarter 2000 Inflation Report the Bank expected the nominal effective exchange rate index (TCW) to strengthen gradually from 122 in May 2000 to 118 two years later;[19] however, the index weakened to 128 in December. Although the longer-term requirements, in terms of price stability and sound public finances, resulting from inflation targeting are basically the same as those applying under a fixed-exchange-rate regime, short-term fluctuations in the exchange rate may be significant. The Swedish krona has fluctuated *vis-à-vis* the euro in recent years (and the German mark before 1999), with the rate ranging from 8.1 to 9.5 SKr. In terms of the nominal effective exchange rate, the volatility of the Swedish currency has been less than that for currencies in other countries

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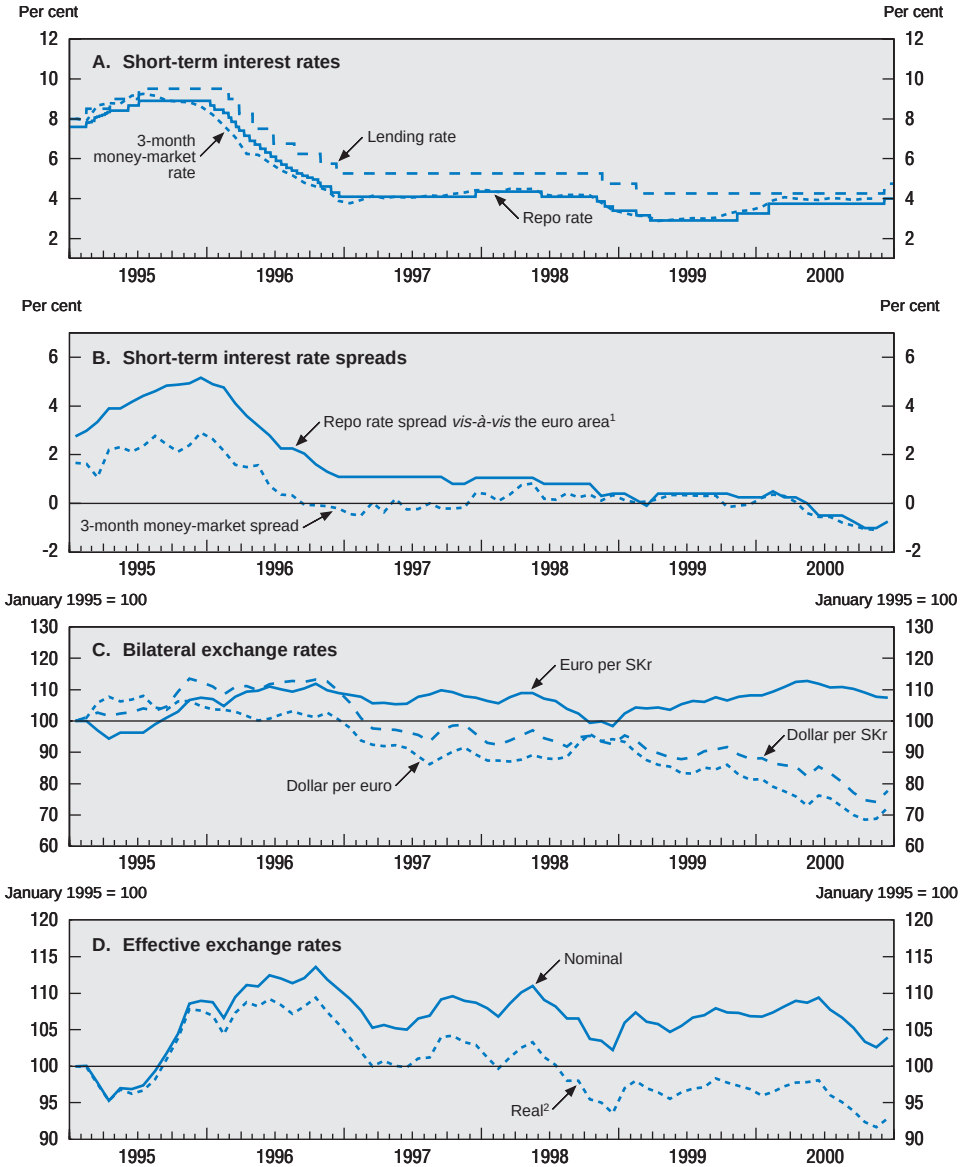
Figure 13. **Short-term interest rates and exchange rate developments**



1. For the euro area, before 1999, German government interest rates are used.
2. CPI deflated.

Source: Statistics Sweden; Sveriges Riksbank; OECD.

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operating an inflation-targeting regime. Having weakened somewhat during the second half of 1998, the krona appreciated almost 15 per cent against the euro during the course of 1999 and the first half of 2000, partly reflecting higher growth in Sweden than in the euro area and the fact that that the strength of the dollar and sterling against the euro during this period is likely to have rubbed off on the Swedish krona as well. However, the krona fell back somewhat in the second half of 2000 even against the euro, which itself continued to depreciate against the dollar until late in the year. The Bank's decision to leave the repo rate unchanged until December and the resulting negative interest-rate spread against other currencies is likely to have contributed to this development.

The exchange-rate fluctuations in 1999 and 2000 exhibit some correlation with capital-flow patterns. Thus, the krona's appreciation from early 1999 to mid-2000 coincided with the tremendous boom in the stock market and the simultaneous reversal of equity outflows as foreign investors increased their holdings of Swedish equities (Figure 14). The pattern of capital flows reversed in the second half of 2000, and the krona's depreciation in that period was manifest in a deterioration of the balance on portfolio investments in the form of debt securities and reflects the negative interest-rate spread *vis-à-vis* the euro area and, in particular, the United States. Moreover, the shift in net capital flows has been reinforced by the levelling-off of and subsequent fall in the stock market and portfolio readjustment resulting from more liberal provisions for households' placement of pension savings which has led to a higher share of foreign holdings.[20]

The stability of the krona *versus* the euro would be an issue if entry to the EMU were under active consideration. The government is supportive of EMU membership at a later point but has set a number of preconditions to be met before drawing up a timetable for the country to make a decision by referendum. These include, most importantly, firmer evidence that the wage-formation process has improved, *i.e.* a sustained period of wage increases comparable to those of the euro area. Meanwhile, the formal criterion of a stable exchange rate for two years up to an eventual entry has according to the European Commission's assessment not been complied with so far. The government has reiterated, most recently in the November 2000 Convergence Programme, that there are no current plans to enter the European exchange rate mechanism (ERM2). Therefore, changes in the short-term interest-rate differential are likely to continue to have a significant influence over the bilateral exchange rate against the euro.

Long-term interest rates and mortgage financing

Long-term interest rates have, by and large, followed the cycle of short-term rates during 1999 and 2000, even if their fluctuations have been greater. The yield on 10-year government bonds reached a low in January 1999 of close to 4 per cent but increased quite sharply – by almost 2 percentage points – during 1999 (Figure 15).

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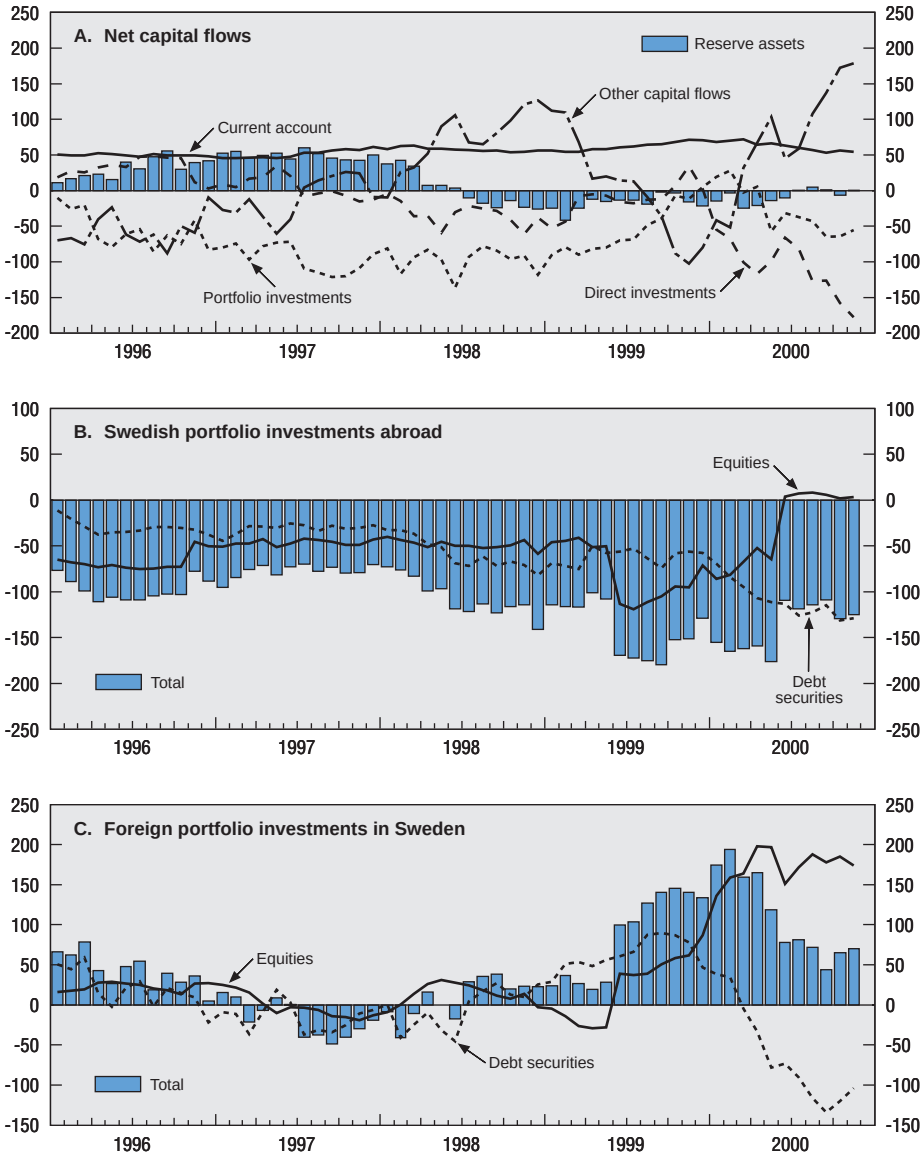
Figure 14. **Capital flows**
Billion SKr, 12-month moving sums



Note: Adjusted in April 1999 for the merger between Astra and Zeneca.
Source: Sveriges Riksbank.

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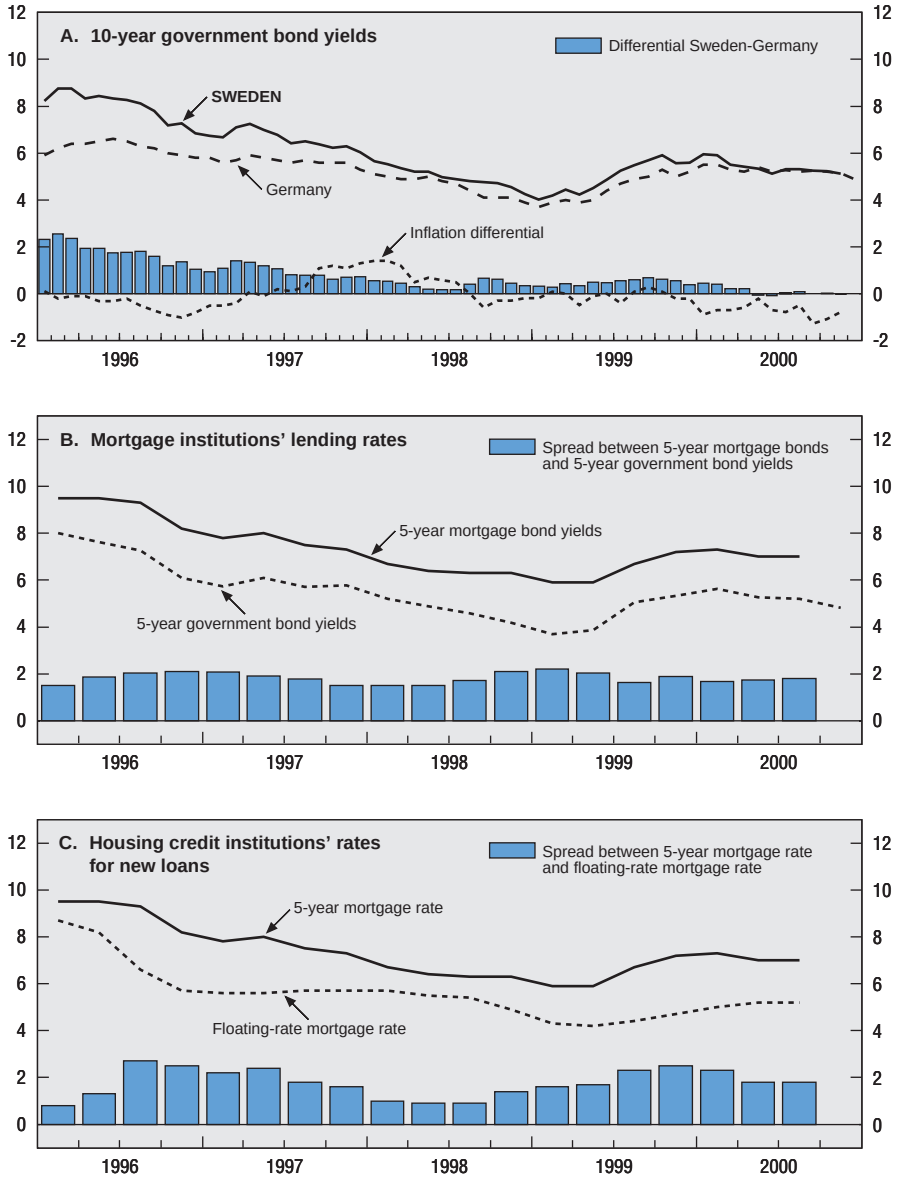
Figure 15. **Long-term interest rates**
Per cent



Source: Sveriges Riksbank, National Debt Office.

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This increase was very close to the rise in the long-term euro bond rates, and the spread *vis-à-vis* Germany remained fairly stable around ½ percentage point during this period. In most of 2000 the 10-year interest rate on government bonds hovered around 5½ per cent. The long-term interest rate differential against the euro narrowed during the first half of 2000 and was even slightly negative during May and June following the announcement of a government bond buy-back programme and the appreciation of the krona against the euro during that period. In the second half of 2000 the interest-rate differential remained minuscule by historical standards (around 0.1 percentage point on average), a clear demonstration of the credibility that Swedish monetary policy has earned on financial markets and of the significant payoff from the low inflation and fiscal consolidation achieved.

As the yield gap between short- and long-term bonds widened from mid-1998 to end-1999, it became more attractive for households to finance house purchases and refinance existing fixed-rate mortgages with floating-rate loans based on securities with a short maturity. The interest-rate difference between fixed-interest-rate mortgage loans – typically of five years' duration – and those with floating rates widened from 1 to 2¼ percentage points between the third quarter of 1998 and fourth quarter of 1999 and remained near 2 percentage points throughout 2000 (Figure 15, Panel C). The increased attractiveness of floating-rate loans relative to fixed-rate loans is clearly reflected in mortgage-finance patterns. One quarter of all newly issued mortgage loans had floating interest rates in the period 1996 to 1998, but by 2000 the fraction had risen to nearly three quarters (Table 8). At the end of 1996 only 10 per cent of mortgage institutions' outstanding loans to households were based on floating rates, but this proportion had increased to over a third by October 2000. While this shift is a good illustration of

Table 8. **Mortgage institutions' lending to households, 1996-2000**

	1996	1997	1998	1999	2000[1]
All new loans (billion SKr)[2]	107.2	113.5	113.2	126.7	98.7
of which: (per cent)					
Floating interest rate	18.4	23.0	31.5	53.4	72.3
Fixed interest rate, 1-5 years	56.8	54.8	21.5	21.5	16.8
Fixed interest rate, 5+ years	24.7	22.1	46.8	25.1	9.6
All outstanding loans[3] (billion SKr)	489.7	507.6	537.2	581.6	604.6
of which: (per cent)					
Floating interest rate	9.8	13.0	16.2	25.4	36.0
Fixed interest rate, 1-5 years	35.7	44.0	34.7	28.0	21.8
Fixed interest rate, 5+ years	54.5	43.1	49.1	46.6	42.2

1. January-October.
2. Yearly average.
3. End of period.
Source: Sveriges Riksbank.

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households' low inflation expectations and widespread confidence in sustained low policy-determined interest rates, it also exposes them to greater risk in case of a marked and general increase in interest rates. The shift in the financing structure, which results to some extent from the monetary policy stance in place since 1998, has the potential of influencing the transmission mechanism for monetary policy, as changes in policy-controlled interest rates might feed through to households' consumption behaviour faster than they did previously. However, households retain the possibility of shifting back to fixed-interest loans on short notice. Moreover with inflation expectations anchored around the target, a possible offsetting factor would be a general tendency of prolonged contracts in society resulting in increased price rigidity.

Financial stability and credit developments

As the recovery has matured, the financial sector's lending to households and enterprises has gradually picked up. Thus, following very subdued growth rates during the years of consolidation, lending to businesses and, in particular, households grew much more vigorously in 1999 and 2000, thereby causing money demand to pick up as well (Figure 16). Even though the pickup in lending rates indicates that risk-taking in the banking sector may have increased in recent years, there are no signs as yet of financial stress. While the ratio of households' gross debt to disposable incomes has increased, it is still well below the levels reached at the previous business cycle peak, and borrowing has been based on more realistic expectations of real interest rates under the current upswing than during that of the late 1980s. Furthermore, households' interest payments are at a historically low level relative to disposable incomes. Even though they remain higher than in the second half of the 1980s, bankruptcies have been steadily declining since 1993, indicating that the average quality of business loans is not an imminent concern. Overall, recent credit losses in the banking sector have been negligible. However, bearing past experience in mind, such losses may increase very rapidly in case of a sharp deterioration of the general economic situation.

The banking sector's profits before credit losses were relatively stable, at some SKr 30 billion per year, between 1994 and 1997 but fell in 1998. They then rose sharply in 1999 and the first half of 2000, essentially related to the stock-market boom, and the trend may, therefore, have been partly reversed again in the second half of 2000 with the fall in equity prices. The fact that the banking sector has not managed to increase gross profits in line with increased lending is in part attributable to sharpened competition within the sector, the latter being manifest in a steadily declining differential between deposit and lending rates; starting from an unusually high level this differential has fallen some 1½ percentage points for both households and businesses since the mid-1990s. Increased investment in ICT equipment has, in the short term, curbed profits as well but may have started to pay off recently in the form of reduced costs.

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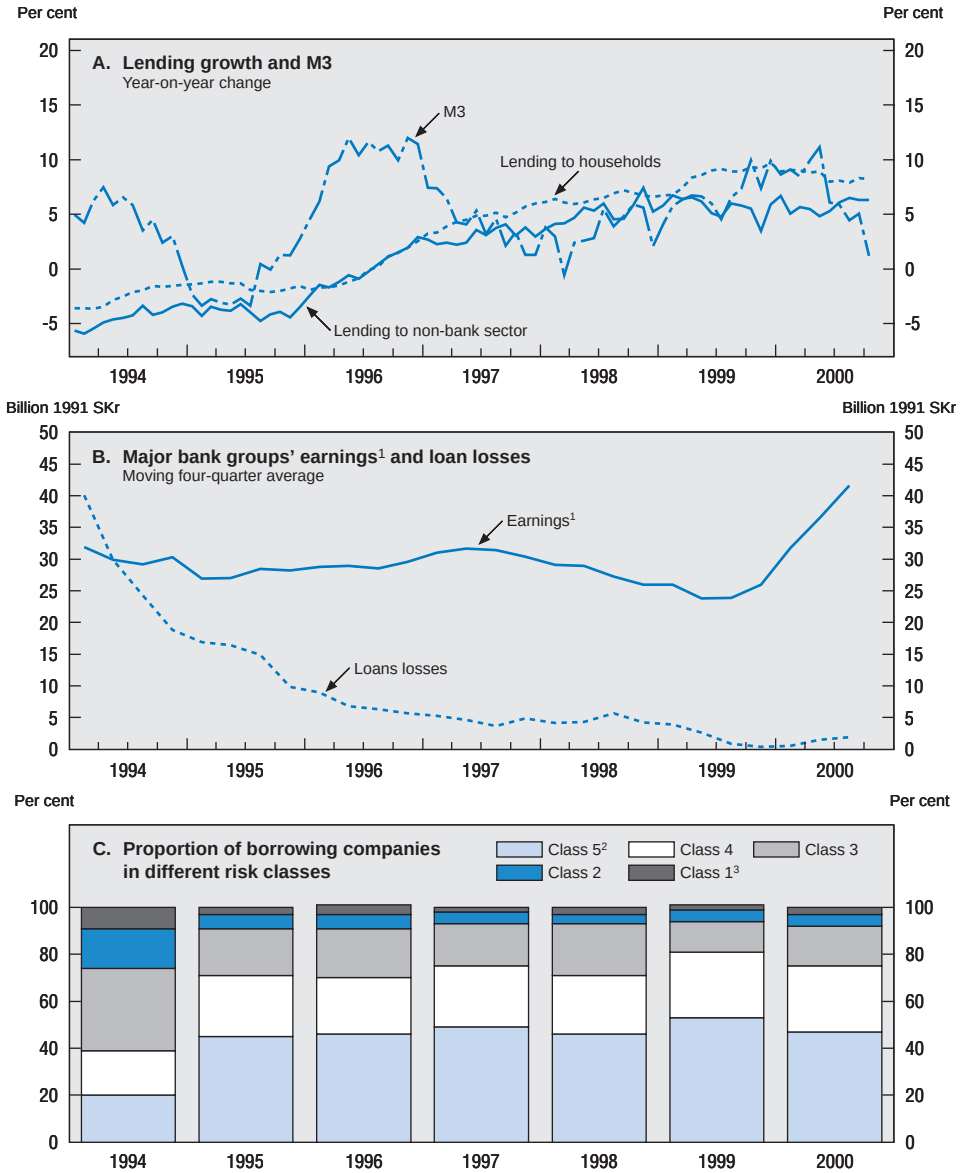
Figure 16. **Lending, money growth and financial stability**



1. Before loan losses.
2. Lowest risk.
3. Highest risk.

Source: Sveriges Riksbank; OECD, *Main Economic Indicators.*

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Table 9. **Households' financial balance sheets, 1990-2000**

	1990	1993	1996	1997	1998	1999	2000[2]
	Per cent of disposable income, period						
Assets							
Coins, notes and bank deposits	61.4	53.5	54.3	52.4	52.5	51.4	50.1
Equities	51.9	46.6	61.4	80.8	86.6	134.7	139.9
Insurance saving[1]	21.5	24.6	31.0	70.1	77.1	82.5	80.7
Other[1]	29.8	38.3	49.7	19.9	16.6	14.6	12.5
Total	**164.7**	**163.1**	**196.3**	**223.2**	**232.9**	**283.3**	**283.1**
Liabilities							
Public sector	10.9	9.8	10.5	11.4	12.0	12.5	12.2
Financial sector	111.6	84.6	80.1	84.4	86.1	89.9	86.8
of which:							
Banks	44.0	24.0	19.8	21.9	22.4	23.2	22.8
Mortgage institutions	60.7	58.9	59.2	55.4	56.4	58.4	56.3
Other	6.8	1.7	1.0	7.1	7.3	8.2	7.8
Total	**123.4**	**95.3**	**91.5**	**95.7**	**98.1**	**102.4**	**99.0**
Net financial assets							
Total	41.3	67.7	104.8	127.5	134.7	180.9	184.1
Excluding equities	–10.6	21.1	43.5	46.7	48.1	46.2	44.3

1. As from 1997, a considerable amount of assets was reclassified from "others" to "insurance saving".
2. First quarter.
Source: Statistics Sweden.

Despite higher borrowing, households' balance sheets still look comfortable. In fact, their net financial assets have risen during the recovery, notably in 1999 as a result of surging equity prices. Relative to disposable incomes, households' net financial wealth has strengthened continuously during the recent decade, from 41.3 per cent in 1990 to 180.9 per cent in 1999 (Table 9). This is not just an equity effect, as other net financial assets rose from –10 per cent of disposable incomes in 1990 to 45-50 per cent in the second half of the 1990s. As a measure of the short-term financial vulnerability of the household sector, gross liabilities may be more relevant, as its assets are to some extent illiquid on short notice. Even the increase in gross financial liabilities – from 91.5 per cent of disposable income in 1996 to 102.4 per cent in 1999 – is fairly modest, and the debt income ratio remains well below that at the peak of the previous business cycle, when it rose to more than 130 per cent.

Fiscal policy

The government's budget balance has moved from a peak deficit of 11 per cent of GDP in 1993 to a surplus of more than 3 per cent of GDP in 2000. The years from 1994 to 1998 were a period of major fiscal consolidation, and the magnitude

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of the restraining measures undertaken to bring down the huge budget deficit was extraordinary. With the public finances back in surplus, Sweden is now enjoying the benefits of the policy pursued during that period. Besides having achieved the restoration of sound public finances, another important result from the consolidation period is the change in the operational framework for fiscal policy that was put in place. Since 1997 fiscal management has been driven by two criteria. *First*, policy has aimed to establish a cyclically-adjusted budget surplus of 2 per cent of GDP in order to ensure a steady reduction of public debt before the ageing of the population starts to put pressure on the public finances. Moreover, a cyclically-adjusted surplus of this magnitude is seen to provide sufficient cushion for the deficit limit fixed under the Stability and Growth Pact to allow the automatic stabilisers to work during a recession.[21] *Second*, a system of ceilings for central-government expenditures was introduced as part of a broader reform of the budget process intended to enhance overall expenditure control.

These two complementary constraints work together to avoid any further upward drift in tax rates, which are already the highest in the OECD (OECD, 2000*a*). By establishing a top-down approach to expenditure policy and setting a medium-term horizon for current policy decisions, the new framework is certainly a better basis for ensuring good fiscal conduct than its predecessor. However, it may still be too early to assess its merits thoroughly, as it has not yet been in place over a full economic cycle. In particular, the political commitment to comply with the expenditure ceilings has yet to be tested during a period of less favourable economic developments than the buoyant conditions of the last three years.

The achievements of the consolidation process briefly revisited

Following a budget deficit of 11 per cent of GDP in 1993, Sweden achieved a surplus of 2 per cent of GDP only five years later (Figure 17). As discussed in previous *Surveys*, this remarkable improvement was mainly a result of the consolidation programme from 1994 to 1998 – the cyclically-adjusted balance improved by 11 percentage points of potential GDP during this period and was in 1998 comparable to the actual surplus. Since then, the latter has increased further to more than 3 per cent of GDP in 2000. While the cyclically-adjusted surplus of 2 per cent of GDP in 2000 was largely the same as two years earlier, the cyclically-adjusted net primary balance weakened from a surplus of 6¼ per cent to 4¾ per cent in this period. Gross public debt, having jumped from 43 per cent of GDP in 1990 to 78 per cent in 1994, stabilised around the middle of the 1990s and started to come down again more significantly as from 1999. Reinforced by one-off revenues from the sale of shares in Telia (the former state-owned telecommunications operator), gross public debt fell below the key level of 60 per cent of GDP in 2000. On present projections, net debt will be eliminated altogether within a few years.

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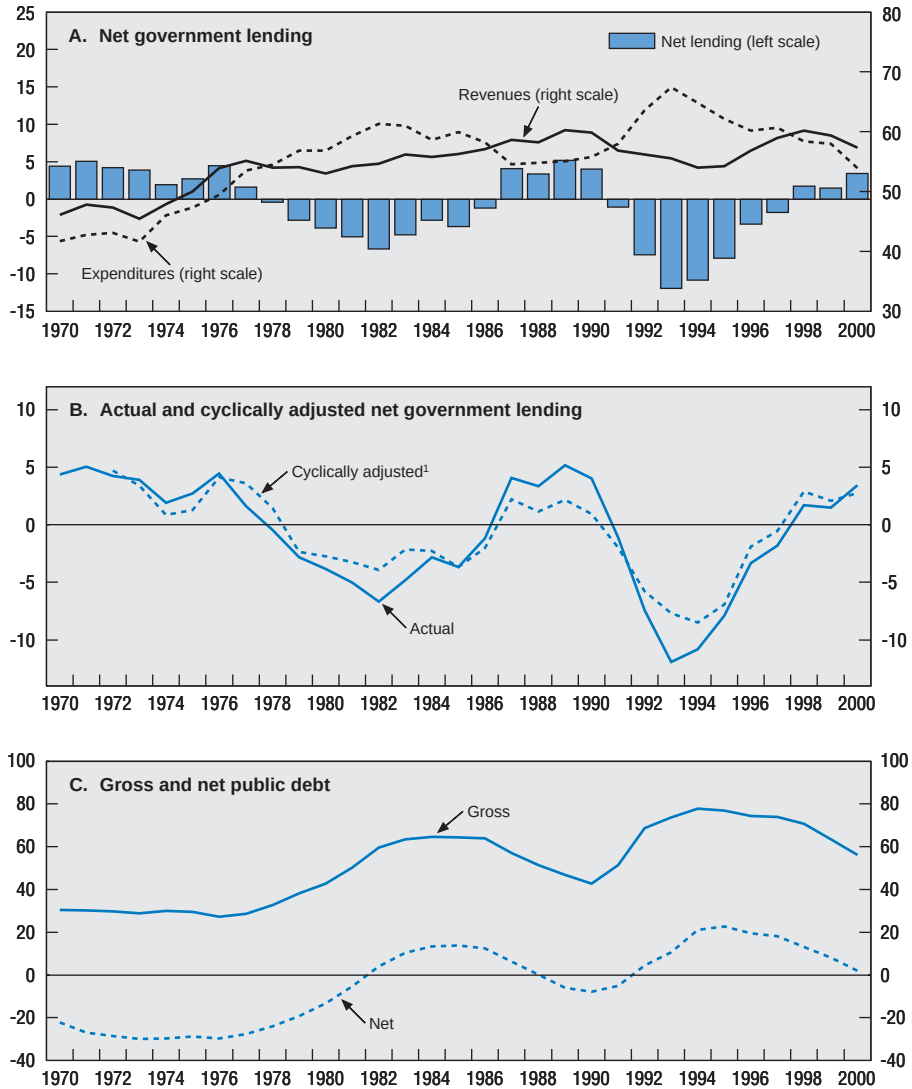
Figure 17. **The fiscal balance and the public debt**
Per cent of GDP



1. As per cent of potential GDP.

Note: Data are from *Economic Outlook 68* unless otherwise mentionned. The series in Panel A and actual net government lending in Panel B are, for the years 1997 to 1999, based on revised national accounts figures released after the cut-off date for information to *Economic Outlook 68.*

Source: Statistics Sweden; OECD.

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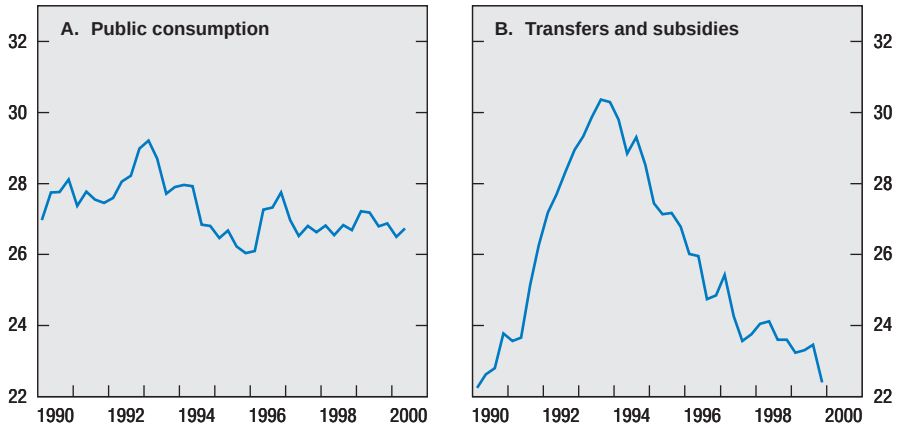
Figure 18. **Ratio of public consumption, transfers and subsidies to GDP**

Per cent of GDP



Source: Statistics Sweden; OECD.

Expenditure cuts and tax increases contributed in roughly equal amounts to the fiscal consolidation programme. Having declined quite considerably in the first half of the 1990s, the tax-to-GDP ratio rose by more than 5 percentage points between 1995 and 1998 and approached the 1990 level of more than 55 per cent of GDP. Total expenditures, on the other hand, have declined steadily from the peak level in 1993/94. Although general-government consumption was curbed in the early phase of the consolidation process, it has stabilised at between 26 and 27 per cent of GDP (Figure 18), still the highest in the OECD area. The sharp reduction in transfers and subsidies from 30 per cent of GDP in 1994 to 22 per cent in 2000 made the greater part of the contribution on the expenditure side.

Implementing fiscal policy in 2000 *and* 2001

Latitude for reform?

The current fiscal situation closely resembles the budget position that prevailed ten years ago. The budget surplus has not quite reached the 5 per cent of GDP achieved in 1989, but it would have approached it in 2000 and 2001 if fiscal policy had been neutral to activity. In addition, the surplus is being achieved at

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tax- and expenditure-to-GDP ratios close to those existing at the time. Also, the gross debt ratio is set to decline to levels not much higher than 10 years ago. Although the present surplus looks impressive and comfortable against the backdrop of a fairly recent double-digit deficit, it is no higher than those achieved at the business-cycle peaks of the early 1970s and late 1980s. In both these cases, the subsequent downturns took the budget balance to a deficit far greater than the 3 per cent of GDP allowed under the Stability and Growth Pact (see Figure 17 above). Even if the character and magnitude of the economic shocks that caused those earlier recessions may not have much relevance in the present situation, these past events should be kept in mind. The automatic stabilisers are stronger in Sweden than in almost all other OECD countries,[22] largely because of the very high tax and public spending ratios, and the public finances are therefore more sensitive to slowdowns or recessions. While this fact in itself means that Sweden needs a larger cushion than other EU countries to stay within the 3 per cent deficit limit, it also warrants a prudent and conservative assessment of both the output gap and the relevant elasticities used to calculate the structural budget balance. Unfortunately, these can be estimated only with considerable uncertainty.

With the government's stated objective of a 2 per cent budget surplus over the business cycle, the assessment of cyclically-adjusted net lending by the Ministry of Finance has come to play a crucial role as a basis for fiscal policy decisions. A structural surplus above 2 per cent is often presented as *"latitude for reform"* by the government.[23] In this context, transparency about fiscal policy decisions would be enhanced if official estimation results for the output gap and the structural budget balance were available to the public. Likewise, current calculations and sensitivity analysis reflecting estimation uncertainties should be published, and deviations in the assessment from the underlying estimation model should be documented. Without such information, the scope for fiscal-policy easing (or the required tightening) assessed by the government presents itself as an assumption more than an analytically based assessment.

The OECD has recently updated estimates of rates of structural unemployment (NAIRUs) which serve as an important input in output-gap calculations (Richardson *et al.*, 2000). In the case of Sweden, the NAIRU for the first quarter of 1999 was estimated at 5.8 per cent, close to the level of actual unemployment at that time. However, the results for Sweden may be more uncertain than for many other member countries,[24] implying that for policy use any output-gap estimate should be interpreted with caution. Although recognising the risk of overheating associated with output growing above potential, the Swedish authorities have taken a somewhat more optimistic view on the size of the structural budget surplus. On their assessment the economy has not yet reached full capacity, so that, following the logic of targeting a 2 per cent structural surplus, there would still be scope for some discretionary stimulus without imparting upward pressure to inflation.

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Fiscal policies supportive of economic activity

The budgets for 2000 and 2001 have both added to already high demand growth. Thus, initiatives coming into force in 2000 and 2001 amounted to a fiscal easing, measured by their direct budget impacts, of SKr 16 and 27 billion respectively, *i.e.* a total of more than SKr 40 billion, an average of more than 1 per cent of GDP (Table 10). Most of the easing has been on the revenue side. A multi-year tax-cut programme was announced in the budget for 2000, although with the reservation that its full implementation would depend on future assessments of the output gap. The main elements of the tax-cut programme are (see also Chapter III):

– A *de facto* phase-out of employees' social-security contributions as these are offset at the individual level through tax discounts linked to the contributions paid. Employee-paid social-security contributions were introduced in 1994 as a part of the fiscal consolidation programme; in 2000 they raised revenues equivalent to 2 per cent of GDP. The budget for 2000 held out the prospect that the contributions would be fully compensated over a period of four years, and with the adoption of the budget for 2001 this is half-way to being met. A formal decision on the eventual continuation along this path in 2002 has not yet been

Table 10. **Discretionary fiscal policy measures in the budgets for 2000 and 2001**[1]

Billion SKr

Gross budget impact	Measures implemented in		
	2000	2001	2002
Revenues			
Income taxes	–12.0	–18.9	–1.8
Business taxes	–2.8	–1.8	2.4
Energy taxes	1.4	3.0	–
Property taxes	–	–0.6	–
Others	–2.1	–2.3	3.0
Expenditures	–0.9	–6.4	–7.5
Child benefits	–	–2.5	–
Parental leave	–	–	–1.0
Educational grants	–	–2.1	–3.5
Cap or childcare charges	–	–0.2	–4.2
Others	–0.9	–1.6	1.0
Total	**–16.4**	**–27.0**	**–3.9**
(per cent of GDP)[2]	–0.8	–1.3	–0.2

1. Includes initiatives presented in the budgets for 2000 and 2001. Negative values indicate a deterioration of the budget balance, *i.e.* higher expenditures or lower taxes. Discretionary measures taken at local levels of government are not included.
2. Per cent of GDP as estimated by the OECD.

Source: Ministry of Finance and OECD.

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taken. However, in its recent Inflation Reports the Bank has assumed further tax cuts of some SKr 15 billion (0.7 per cent of GDP) in 2002, which illustrates the widespread expectations of further tax concessions.

– An increase in the threshold in the state income-tax scheme, which is the most progressive element of the Swedish income tax system; it comes on top of the proportional municipal taxes and employers' social-security contributions. The tax-cut programme aims to reduce the proportion of taxpayers who pay any state tax to 15 per cent, thereby restoring the intentions of the 1991 tax reform.

Furthermore, the state property tax on owner-occupied houses is being reduced in 2001 from 1.5 per cent to 1.2 per cent of assessed value. However, this is merely compensation for bringing the assessments back to levels reflecting their market values after they had been frozen since 1997. This change should be approximately neutral to the budget but will redistribute tax payments towards the Stockholm region and other larger cities which have experienced the greatest increases in house prices in recent years. Finally, as a first step in a so-called *green tax shift* (see Chapter IV) environmental taxes – mainly on diesel fuel and carbon dioxide – are being increased in 2001 by SKr 3.4 billion, which is offset by an equal increase in the basic personal allowance in the income tax system (the threshold income above which municipal income tax is levied) and a reduction of employers' social-security contributions. In sum, total tax concessions amounted to SKr 16 billion in 2000, and additional cuts of SKr 21 billion become effective in 2001. The extent to which these tax cuts will induce further stimulus to private spending in the near future is uncertain. Inasmuch as they were, for the larger part, announced in the spring of 1999 in the context of a multi-year programme, it is possible that they have already to some extent been factored in to expenditure decisions in 2000. This is indeed the consensus view in Sweden (Box 3). However, it would still seem most likely that the bulk of the demand effect will take place as the tax cuts are actually implemented.

Discretionary government spending increases were negligible in 2000 but amount to SKr 6½ billion in 2001 (0.3 per cent of GDP), mainly as a result of higher child benefits and more generous educational grants. Apart from the overall expenditure ceiling (see below), fiscal policies have not yet been laid out in detail for 2002. However, some elements have already been decided upon; these include an extension of the duration of parental leave by one month (reserved for the father) and further enhancing the generosity of educational grants. Moreover, as from 2002, a cap will be put on parents' payments for municipal childcare. Such charges are currently set by individual municipalities and are typically dependent on households' incomes. The cap will reduce the average payment by some 40 to 50 per cent and is particularly favourable for families with two or more children.[25] The direct costs are estimated to be SKr 3½ billion in 2002 on the assumption of unchanged demand for childcare. Any increase in demand is to be financed by the

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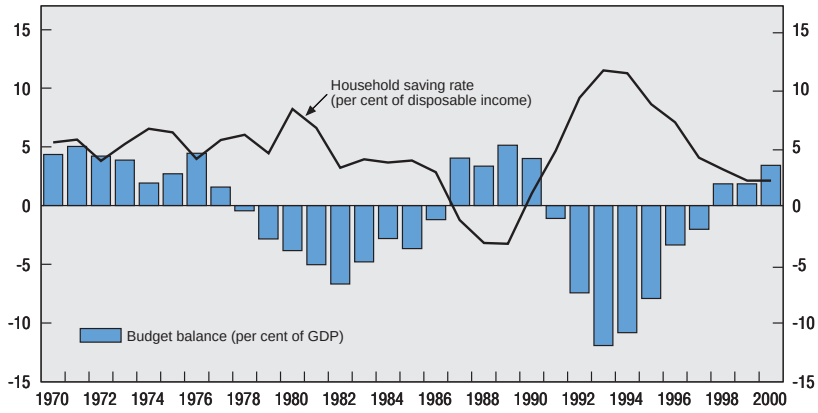
Box 3. Tax cuts and private consumption

The impact of the current tax cuts on private consumption, or more precisely the *timing* of this impact, has been discussed in Sweden recently. A study by Becker and Paalzow (1997) on Swedish data from 1970 to 1995 would seem to give some support to the prevalence of Ricardian consumption behaviour, although the results are very sensitive to the restrictions imposed. In line with the Ricardian equivalence hypothesis some have argued recently that the improvement of public finances *per se* has strengthened private consumption, because households have been expecting future tax cuts (Handelsbanken, 2000; Häggström and Kinnwall, 2000). If so, tax cuts may not add much further stimulus and inflationary pressure to the economy.

The argument relies primarily on the close historical correlation between households' saving rate and the general government's budget balance (Figure 19). The period of fiscal consolidation (1994 to 1998), for example, was accompanied by a marked decline in the saving ratio. There may be other explanations to the strong correlation between households' saving rate and the budget balance; one is the strength of automatic stabilisers in Sweden. This would imply that the causality runs mainly from changes in the saving ratio to changes in the in the budget balance rather than *vice versa*.

Figure 19. **The budget balance and household saving rate**

Per cent



Source: Statistics Sweden; OECD.

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Box 3. **Tax cuts and private consumption** (*cont.*)

The now-established fiscal surplus indeed makes future policy easing more likely. Moreover, inasmuch as the current tax cuts have been announced in the context of a multi-year programme, they may have influenced private consumption to a larger extent than normally before actually being implemented. More generally, however, the character of a fiscal policy easing following a comfortable budget surplus would be difficult for individual households to envisage; bearing past experience in mind it might equally well take the form of increased public spending, not least in the case of Sweden. In any case, unless countered by spending cuts, it might be hazardous to implement tax cuts on the assumption that the income elasticity of private consumption is significantly below unity.

municipalities and might, therefore, put upward pressure on their expenditure and tax levels. The direct budget impact of the above decisions amounts to SKr 7½ billion (nearly 0.4 per cent of GDP) in 2002; still, total spending must be held within the central government's expenditure ceiling (see below).

The cyclically-adjusted primary balance, which is the OECD's basic fiscal-stance measure, reached a surplus of 6¼ per cent of GDP in 1998 but fell back to 4¾ per cent in 2000 and is expected decline by a further 1 percentage point in 2001. This reflects discretionary fiscal easing, which has outbalanced positive effects from lower structural unemployment. Net interest payments have continued to decline relative to GDP, and, in cyclically-adjusted terms, the overall surplus declined less markedly in 1999 and 2000 than did the primary balance. In 2001 the cyclically-adjusted surplus is expected to decline to some 2 per cent of GDP, *i.e.* close to the government's target. The fiscal stance measure published by the National Institute of Economic Research (NIER) paints a broadly similar picture. The NIER's measure of fiscal stance compares budgets and outcomes for various subgroups of revenues and expenditures with a benchmark for neutral policies and adds up deviations for each category to give an overall indicator of the impact on the fiscal balance from discretionary initiatives (Braconier and Holden, 1999). According to this indicator, discretionary initiatives improved the budget balance in 1997 and 1998 by some 1.5 percentage points of GDP per year as both revenues and expenditures contributed to restraining demand (Table 11). In contrast, such contributions worsened the fiscal balance by 1.0 percentage point of GDP in 1999, as higher discretionary spending exceeded revenue gains from higher tax rates. Likewise policy decisions – primarily tax cuts – are estimated by the NIER to entail a cumulative reduction in the fiscal balance of almost 2¼ percentage points of GDP during 2000 and 2001.

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Table 11. **Indicators of fiscal policy**

	1997	1998	1999	2000	2001
	Per cent of GDP				
Fiscal balance					
Actual[1]	–1.8	1.7	1.5	3.4	3.3
Cyclically-adjusted[1]	–0.5	2.9	2.1	2.7	2.1
Cyclically-adjusted net primary balance[1]	2.9	6.2	5.1	4.8	3.9
Change in fiscal balance					
Actual[1]	1.6	3.5	–0.2	1.9	–0.1
Cyclically-adjusted[1]	1.4	3.4	–0.9	0.7	–0.5
Cyclically-adjusted net primary balance[1]	1.4	3.3	–1.1	–0.3	–0.9
Budget impact of discretionary policy decisions[2]	1.5	0.9	–1.0	–1.1	–1.1
of which:					
Revenues	0.2	0.7	0.8	–0.4	–1.0
Expenditures[3]	1.3	0.2	–1.8	–0.7	–0.2

1. OECD estimate, *Economic Outlook* 68. The data for the actual fiscal balance in 1997 to 1999 are national accounts data published after *Economic Outlook* 68.
2. Estimate by the NIER based on the method described in Braconier and Holden (1999). Adjusted in 1998 and 1999 for the separation of properties held by the AP pension fund into an independent company.
3. Measures the effect on the fiscal balance. A negative number is equivalent to increased expenditures and, thus, a negative impact on the balance.

Source: OECD, National Institute of Economic Research.

Expenditure ceilings

The financial crisis in the early 1990s provided the impetus for radical changes to the budget process. Several reports from the early 1990s pointed out that the Swedish budget process at that time was weak by international standards. For example, it was emphasised that the Ministry of Finance had insufficient influence, that the budget process was characterised by a bottom-up rather than a top-down approach and that co-ordination, control and follow-up procedures were, in general, inadequate (SOU, 1993). A new framework was established to address these shortcomings, with the Budget Act as its foundation; it came into force on 1 January 1997. The Act clarified the government's and the parliament's competencies and responsibilities in the budget process. Lengthening of the electoral cycle from three to four years was another important step conducive to setting a longer budgetary time horizon.

The cornerstone of the so-called *new budget process* is the expenditure ceiling for central government expenditures. The overall ceiling is set in nominal terms and is determined three years in advance in the Spring Budget. The detailed appropriations for each of 27 expenditure areas are set 2½ years later in the Autumn Budget in the year prior to the relevant fiscal year. The basic principle is that budgeted increases in some programmes must be offset by saving on

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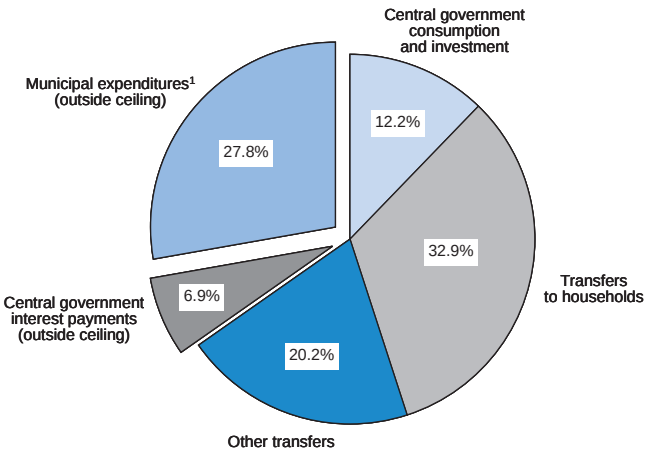
others, through discretionary measures if necessary. In 2000 the ceiling applied to around two-thirds of total public expenditures – of which roughly 50 per cent were transfers to households, 30 per cent were other transfers and 20 per cent were consumption and investment (Figure 20). The ceiling includes old-age pension outlays outside the central government but excludes interest payments and expenditures at local-government levels.[26] As counties and municipalities make up some 70 per cent of total public consumption and investment outlays, a large proportion of expenditures subject to discretionary political decision is left outside the ceiling.[27] In contrast, cyclically-dependent budget components such as unemployment benefits, expenditures on active labour market programmes and social security are included. This type of outlay is typically an entitlement for anyone meeting the criteria set down in legislation. This demarcation of ceiling-restricted expenditures, *i.e.* having most transfers within the envelope and most consumption and investment expenditures outside it, may have the rather perverse consequence that if expenditures are lower than expected because of a strong economy,[28] these savings may fuel additional discretionary spending during an upswing. Conversely, unexpected cyclical expenditure increases during a period of low growth may oblige programme cut-backs elsewhere to comply with the ceiling. The risk of cyclical surprises is quite important in the current framework, as macroeconomic forecasting with a three-year time horizon is obviously subject to considerable uncertainty (Box 4).

Figure 20. **Demarcation of ceiling-restricted expenditure, 2000**

Per cent of total public expenditure



1. Net of block grants from the central government. Predominantly consumption and investment expenditures.
Source: Ministry of Finance.

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Box 4. Nominal expenditure ceilings and macroeconomic forecasting

In the process of determining the cap on expenditures it is crucial to observe and deal with the uncertainty involved, in particular as the ceiling is set in nominal terms and includes cyclically-dependent expenditure components. Macroeconomic forecasts three years ahead are uncertain, and budgeting safety margins must be included to ensure some short-term flexibility. However, determining the appropriate size of such margins is difficult. On the one hand, they should be large enough to absorb normal forecast errors, allowing the automatic stabilisers to work. On the other hand, they should not be so broad that they almost never restrain spending and they should not allow "compensation" for unexpected inflation.

The sensitivity of actual outlays to unforeseen developments in real aggregates and inflation is high in Sweden, reflecting both widespread use of indexing rules and the fact that unemployment-related expenditures constitute a large proportion of GDP. From 1988 to 1998 the mean absolute forecast error three years ahead on the government's forecast of real GDP was 1.8 per cent. The similar figure for the level of consumer prices was 2.5 per cent (Table 12). Against this background a recent study requested by the expert group evaluating the new budget process concluded that to fully reflect the uncertainties involved, a safety margin of 6 per cent in year t + 3 would be required (Braconier, 2000).* For years t + 1 and t + 2 the necessary margins were assessed to be some 2 and 5 per cent, respectively.

Table 12. **Absolute mean forecast errors in the spring budgets, 1988-98**

Per cent

	Year 0	Year 1	Year 2	Year 3
GDP	0.7	1.5	1.6	1.8
Consumer price index	0.5	1.7	3.2	2.5
Hourly wages	0.7	1.2	1.4	1.4
Employment	0.6	1.8	3.2[1]	1.8[1]

1. Three observations only.
Source: SOU (2000).

* To allow automatic stabilisers to work, a safety margin of this size would be necessary if real GDP three years ahead falls short of *ex ante* expectations by 3 percentage points and prices simultaneously rise 5 per cent above expectations. Underlying this assessment are also estimated elasticities of total ceilings-restricted expenditures with respect to GDP and the CPI of –0.9 and 0.3 respectively. The study assumes unchanged discretionary policy, which in the case of active labour market policy is taken to mean constant spending. This may tend to understate the cyclical sensitivity of the expenditure side of the budget.

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The new budget process was first applied in 1996, when overall expenditure ceilings for the years 1997 to 1999 were set. A recent evaluation by a government-appointed expert group concluded that, by and large, the new process has worked well and has been pivotal in strengthening budget discipline (SOU, 2000*a*). This assessment basically results from the fact that the ceilings set out in 1996 were in fact subsequently adhered to. However, the report also pointed to some problems. Most important the *ex ante* safety margins have not served their purpose as a cushion against unforeseen developments, but have rather been treated by policy makers as an additional reserve for discretionary spending.

While total outlays were kept well below the expenditure ceiling in 1997, actual spending in 1998 and 1999 was well above the original forecast and only just within the ceiling. As the higher level of spending carried over into subsequent years, by 2000 there was virtually no safety margin left despite the fact that outlays corresponding to some ¼ per cent of GDP were technically transferred to the revenue side of the budget.[29] The *ex ante* safety margins are also very small for 2001 and 2002, while the margin of SKr 20 billion (2.4 per cent) in 2003 is more comfortable – even though that still does not fully reflect the "objective" uncertainty as to the expenditure estimate for that year (see Box 4 above). The ceiling for 2003 allows a 3.7 per cent nominal increase in the ceiling-restricted expenditures that year, which is slightly higher than that set for 2001 and 2002 (Table 13).

A closer look at what caused actual outlays in 1999 to diverge from what had been expected in 1997 reveals the shortcomings of the arrangements. Macroeconomic forecast errors, with actual economic growth outstripping expectations

Table 13. **Expenditure ceilings: experiences in 1997-99 and expectations for 2000-03**

Billion SKr

	1997	1998	1999	2000	2001	2002	2003
Expenditure ceiling[1] (1)	723	720	753	765	789	814	844
Expected outlays[1] (2)	708	702	731	764	788	805	827
Actual (3)	699	718	752				
Ex ante safety margin (1 – 2)	15	18	22	1	2	9	17
Actual minus ceiling (3 – 1)	–24	–2	–1				
Actual minus expected (3 – 2)	–9	+16	+21				
of which:							
Macroeconomic assumptions	1	–6	–15				
Discretionary spending increases	7	26	41				
Discretionary savings	–4	–1	–7				
Residual	–17	–3	2				

1. Data for 1997 to 1999 are from the spring budget 1996 while those for 2000 to 2003 are from the 2001 budget. The numbers for 1999 has been adjusted for a technical effect related to the reform of the old-age pension system.

Source: SOU (2000*a*); Ministry of Finance.

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and lower-than-predicted inflation, reduced spending relative to expectations. However, political decisions to increase spending far exceeded discretionary savings, resulting in additional net outlays of more than SKr 30 billion (1½ per cent of GDP). In 1999 favourable macroeconomic surprises yielded savings of almost 2 per cent of the ceiling-restricted expenditures, while additional net outlays resulting from discretionary decisions amounted to slightly over 4 per cent. This adds to the concern that the current boundaries of ceiling-restricted expenditures risk giving fiscal policy a pro-cyclical bias, which might be further reinforced by the fact that local governments' revenues (predominantly income taxes) are by nature quite sensitive to business-cycle fluctuations. Against this background of significant discretionary spending increases, it is noteworthy that the ceilings are generally considered very restrictive, and there is broad consensus in Sweden that they have been highly effective in restraining total spending. In contrast with recent experience, discretionary fiscal policy has – mainly due to decisions on the revenue side – been slightly counter-cyclical when assessed for the past two decades as a whole (Braconier and Holden,1999; Braconier, 2000).

Municipalities and counties are legally obliged, as from 2000, to balance their budgets, so that municipal spending does not constitute a risk to the fiscal balance as such; on the other hand the balance requirement is in itself not a guarantee against upward drift in local governments' tax and expenditure levels.[30] The expert group previously mentioned suggested only rather minor adjustments to the expenditure-ceilings framework. No changes were proposed to the definition of the types of spending included in the ceilings. Accordingly, local governments' expenditures remain outside the ceiling. To deal better with uncertainty the expert group suggested the separation of the current safety margin into a planning margin, a reserve for policy initiatives and an uncertainty margin to absorb forecast errors. The report suggested a 3 per cent uncertainty cushion be incorporated when the ceiling three years ahead is set. This is more modest than the past average forecast errors would suggest, but broader than what has been applied hitherto. The proposals from the expert group do not address the potential conflict between letting automatic stabilisers work and complying with the expenditure ceilings. Moreover, there are no built-in mechanisms to ensure that the uncertainty margin is not used in practice as a reserve for discretionary spending increases. Therefore, much comes down to a question of political commitment, which has not really been tested yet.

Assessment of the current policy mix

Both fiscal and monetary policies have been supportive of economic activity since 1999, thereby adding to growth rates that are already above potential levels. The apparent justification for this policy is sound public finances and low inflation. The actual fiscal surplus exceeds the government's medium-term

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objective by a comfortable margin, and consumer price inflation has remained below the Bank's 2 per cent target. However, a key question when assessing the current macroeconomic policy stance is whether the good inflation performance of the recent past can be considered durable. While the inflation outcome has, so far, clearly been better than expected by most observers, it remains unclear whether this is primarily due to temporary effects – including one-off gains from earlier structural reforms – or reflects a lasting increase in the growth rate of potential output. Recent aggregate labour productivity growth has been slightly higher than the average rate of the latest 20 to 25 years, while the fall in unemployment since 1996 is mainly attributable to cyclical factors, although structural unemployment has shown some decline as well, thereby helping to increase potential output. While this has facilitated modest wage increases in recent years, further progress in this area will be required if low wage increases are to be reconciled with a continuing decline in unemployment. OECD estimates suggest that the potential growth rate for Sweden is around 2¼ to 2½ per cent. This is close to or even slightly more optimistic than the government's assessment. But it is well below the growth rate experienced in recent years and envisaged for 2001.

The low inflation in recent years is to a large extent the result of sounder macroeconomic policy settings and a better framework for competition policies. However, it is also in part attributable to falling interest rates and lower indirect taxes, which technically reduce inflation in the short term. Rent controls and a decline in profit shares have played a role as well. The appreciation of the krona against the euro from early 1999 to mid-2000 helped to reduce inflation relative to other European countries. These effects are likely to be of a one-off nature. An assessment of underlying domestic inflationary pressures must also take wage developments into consideration. Sweden has improved on this account in recent years as well, but its performance is not as outstanding as it has been on price inflation. In fact, wage increases were well above those in competitor countries until a few years ago and have only fallen to a more sustainable level in 1999 and 2000. A modest outcome of the spring 2001 wage round could be important in maintaining both low inflation and low unemployment. Even so, a steadily tighter labour market increases the risk of wage drift at the enterprise level, and such risks could be exacerbated if shorter working hours are decided upon either through legislation or collective bargaining.

Real and, in particular, nominal short-term interest rates have remained low by historical standards. Moreover, they have been raised less than elsewhere and have moved from ½ percentage point above those in the euro area in late 1999 to nearly 1 percentage point below at end-2000. The monetary policy course followed in 2000 results from the Bank's assessment of continued favourable inflation prospects despite its projection of continued GDP growth above potential both in 2001 and 2002. In addition, the Bank has allowed the effects of the cap on childcare payments and of lower property tax on rented homes – which

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it estimates will reduce inflation by around ½ percentage point in 2002 – to show up in the inflation measure to be targeted. The latter seems especially difficult to justify, since the effects result from expansionary fiscal policy initiatives, which will most likely ultimately contribute to *higher* underlying inflation. The risk of increasing inflation is further spurred by the weakening of the krona in the second half of 2000, which was in part caused by the negative differential in short-term interest rates. Admittedly, the uncertainties about the future inflation path are significant, but the policy of maintaining easy monetary conditions throughout 2000 would seem to have involved unnecessary risks.

As for fiscal policy the achievement of a comfortable budget surplus of more than 3 per cent of GDP in 2000 has, ironically, left policy makers with a difficult strategic choice. On the one hand, sound public finances would seem to allow a fiscal easing. On the other hand, this might increase the risks of overheating. The further stimulus to demand in 2001 from discretionary fiscal policy could prove to be unhelpful inasmuch as it may reinforce the need for monetary tightening if labour-market pressure intensifies – even if this concern has so far not been shared by the monetary authorities. The risk of pushing aggregate demand beyond the level of potential output could be reduced if the concrete measures taken focused more on strengthening the supply side. While some complacency over the consolidation achieved is well deserved, the pro-cyclical fiscal policy track now embarked upon may be inappropriate, when considering risks and uncertainties in the present situation and bearing past experience in mind. While reducing tax rates should be a key element in the medium-term strategy for structural reform, tax cuts in the current phase of the cycle should be accompanied by lower spending.

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III. Structural policies for improving economic performance

Sweden's current economic success – as exemplified by high economic growth with low inflation – can be partly attributed to earlier reforms to structural policies. These started in the mid-1980s and continued through the deep recession of the early 1990s, although their economic benefits were obscured by the macroeconomic circumstances. It is only since the recovery got underway that the fruits have been enjoyed. However a major impetus for reform – the grave fiscal position of the early 1990s – has been eliminated, reducing the pressure to tackle remaining weaknesses. Further reform efforts are needed in a number of areas that are particularly important to the long-term challenges faced by the Swedish economy: an ageing population, the increasing importance of environmental objectives (see Chapter IV) and continuing internationalisation of the Swedish economy. This chapter first reviews long-term developments in the Swedish economy, before addressing outstanding policy issues in labour, capital and product markets.

Long-term decline and recent recovery in economic performance

Sweden has suffered a long-term decline in the average standard of living relative to other OECD countries, although the second half of the last decade showed a turn-around after the deep recession. GDP per capita, albeit an imperfect measure of economic well-being, grew in real terms on average only 1.3 per cent per year over the past three decades, compared with 1.8 per cent for the United States and 2.0 per cent for the European Union (Figure 21) and if the steady decline in Sweden's terms of trade were taken into account, the gap would be even larger. Real GNP per capita grew even more slowly – at around 0.9 per cent annually – because of rising interest payments on foreign debt.

Recent OECD work on understanding the causes of economic growth has highlighted the key mechanisms that may have been at work (OECD, 2000*b*). Economy-wide growth depends on the amount of capital and human resources allocated to production, together with changes in the efficiency with which these

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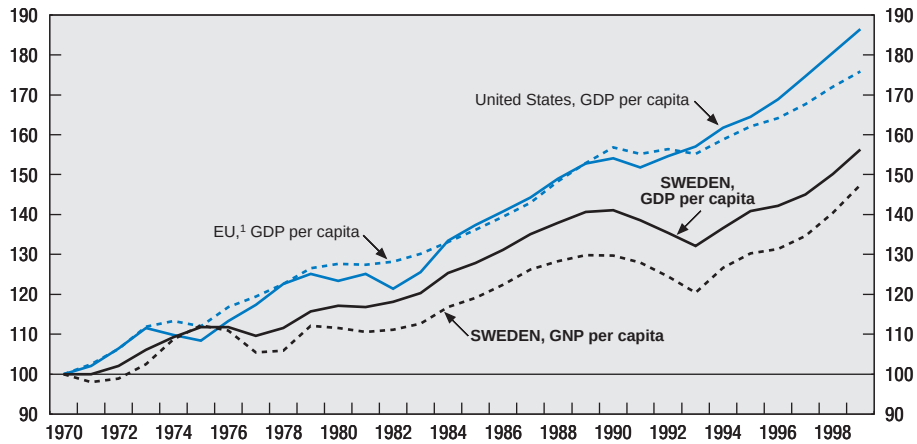
Figure 21. **Long-run economic growth performance**
1970 = 100



1. Calculated for current 15 members throughout the whole period.
Source: OECD.

inputs are used. In Sweden's case, OECD analysis comparing the 1980s with the 1990s indicates that:

– The shortfall in trend per capita GDP growth rates relative to the OECD average can be explained in large part by a lower investment share than average, the greater variability of inflation and a larger ratio of public consumption to GDP. These negative factors were offset to some extent by greater-than-average human capital and population growth.

– The slowdown in Sweden's real GDP growth per capita in the 1990s can be attributed to the falling ratio of investment to GDP, slower population growth and the variability of inflation, although it was attenuated by improvements in human capital and greater trade exposure.

An even clearer picture of the role of factors of production emerges when the business sector is examined: there the investment share rose steadily through the 1980s, overtaking a number of other OECD countries, before collapsing in the early 1990s (Figure 22). Since the trough in 1993, however, it has picked up significantly, and by 1999 had risen by 4½ percentage points, although this remains well below the average level of the 1980s. A broadly similar trend can be seen in private-sector employment.

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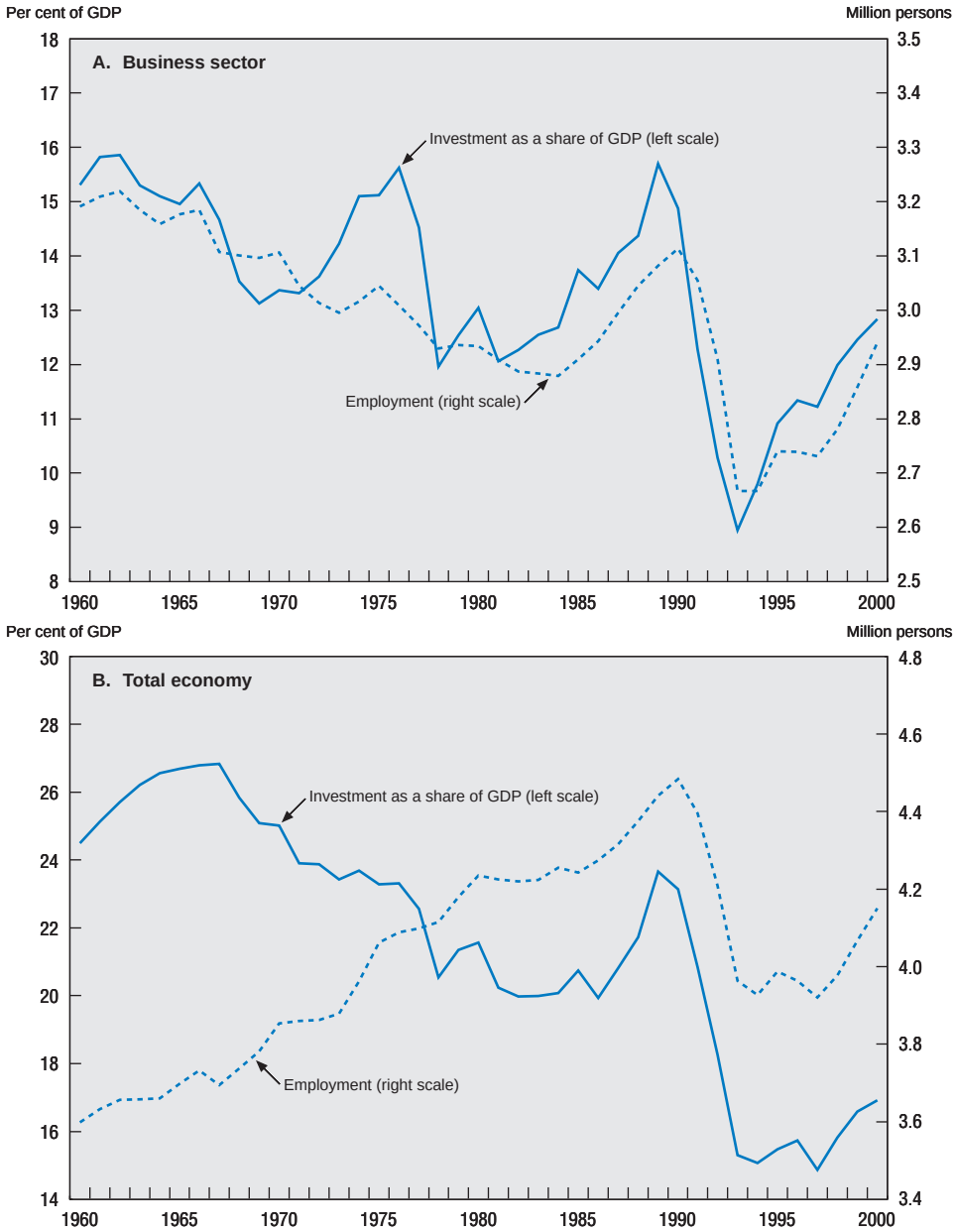
Figure 22. **The evolution of employment and investment**



Source: Statistics Sweden; OECD.

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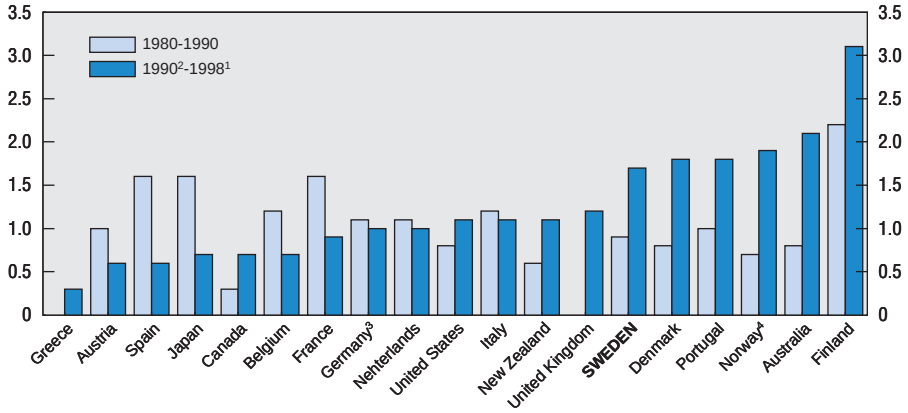
Productivity – as defined by output produced per hour worked and per unit of capital – is a crucial factor in determining economic performance. In Sweden's case, measured productivity for the economy as a whole is damped more than in most countries by the weight of public employment, for which productivity growth is assumed to be zero in the national accounts. Indeed, between 1960 and 1990, almost 1 million jobs were added to the public sector, while private-sector employment actually fell. Trend labour productivity growth in the business sector gradually improved in the 1980s, to just over one per cent per year, before jumping to around 2½ per cent in the 1990s. Sweden's annual multi-factor productivity (MFP) growth – *i.e.* the capacity to combine inputs more efficiently – has also improved by over half a percentage point between the 1980s and the 1990s (Figure 23), placing Sweden above the OECD average. However, this still lags behind a number of other OECD countries, including those of other Nordic countries, averaging only 1.6 per cent per year in the last decade.

An area where Sweden has achieved considerable success over the past 30 years is the upgrading of the average level of formal educational achievement of the working-age population: it has managed a larger rise than any other country.

Figure 23. **Multi-factor productivity growth**
Average annual growth rates



1. 1997 for Australia, Belgium, Canada, Italy, Japan, Norway, Spain and United States; 1996 for Austria, Finland, Greece, Ireland, Sweden and United Kingdom; 1995 for New Zealand; 1992 for Portugal.
2. 1991 for Germany.
3. Western Germany before 1991.
4. Mainland only.

Note: Method of estimation: time-varying factor shares. Based on trend series.
Source: OECD.

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A closer examination shows that most of the gains came in the 1970s, with little progress thereafter, in line with the shrinking returns to higher qualifications. Nevertheless, further improvements in human capital may have contributed around ¼ percentage point to annual MFP growth in the 1990s. A related positive factor should have been the rising proportion of national income dedicated to research and development (R&D), which was always relatively high and reached almost 4 per cent of GDP by the mid-1990s, a larger share than in any other OECD country. However, given the amounts invested and the generally accepted link in principle between R&D and output over the longer term, the impact on Sweden's performance seems to have been disappointingly small. Indeed, a simple correlation of changes in MFP and business-sector R&D intensity between the averages for the 1980s and the 1990s, indicates that Australia, Denmark and Finland achieved notably stronger improvements in MFP with smaller increases in R&D intensity, while the United States, Canada and New Zealand achieved virtually the same lift in MFP growth for very little additional efforts in R&D. One possible explanation for these results may be the concentration of research and development in a few large firms in Sweden. Another hypothesis is that Swedish firms have been less successful at bringing research to market (Lundberg, 1999). More recently, "new economy" effects from Sweden's ample expertise in information technology have been postulated. The evidence to date remains too scant to form a clear judgement. However, Sweden currently ranks third among OECD countries in terms of average installed personal computer base, fourth in cellular mobile subscribers and sixth in Internet hosts, all of which should make it well positioned for any such effects to emerge in the future.

A further issue that has been raised in interpreting the growth experience is the extent to which the exceptionally deep recession of the early 1990s produced a major shakeout of inefficiently used capital and labour. Although painful, this elimination of the poorest performing firms should leave only particularly efficient survivors, and some would argue that the lessons those enterprises gained from the process have made them more nimble and adaptable as a result – well placed to compete on international markets and to benefit from increased domestic demand once household incomes started to rise. This process has been suggested as a possible explanation for the surprisingly strong growth achieved in recent years without incurring significant capacity constraints or even noticeable speed-limit effects. At this stage, it remains an interesting but largely untested proposition and would require a careful examination of the data. However, while the rate of bankruptcies (relative to the existing stock of enterprises) more than doubled during the period 1991 to 1993, the rate of new establishment creation fell by only 1 percentage point. Since then, both start-ups and bankruptcies have been higher than they were in the buoyant years before the economy crashed (Table 14). To the extent that this represents a more dynamic business environment, it suggests a more rapid reallocation of resources between firms and sectors

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Table 14. **Start-ups and bankruptcies**[1]
Per cent, annual average

	1986-90	1991-93	1994-98
New establishments	6.5	5.7	7.7
Bankruptcies	1.7	4.3	2.1

1. As a percentage of the total stock of enterprises before, during and after the crisis in the early 1990s. Agriculture and forestry are not included.
Source: Statistics Sweden.

may be possible in response to economic signals and shifts in circumstances. Nevertheless, Sweden has a lower proportion of "nascent" entrepreneurs in the adult population than a number of other countries surveyed and only one-quarter as many as the United States (Table 15).[31]

The OECD's work on economic growth, while not being able to point to a direct link between structural policies and growth *per se*, does indicate the areas where Sweden might need to concentrate on improvements in order to raise its long-term economic growth: improving the quantity and quality of labour inputs and setting the right conditions for raising productive investment. Here, structural policies clearly play a role, and a range of important structural reforms have been carried out since the mid-1980s aimed in particular at improving the functioning of capital and product markets while labour markets were subject to less significant policy shifts. In a number of areas, including electricity and telecommunications markets, Sweden has been at the leading edge of reforms, while in others, it is in line with best practice within the OECD or has at least made changes that go in the right direction. However, there remains considerable scope for further improvement, and the present conjuncture would seem to provide a window of opportunity for undertaking reforms that might involve rather more painful adjustment costs at other, less favourable, points in the cycle.

Table 15. **Nascent entrepreneurs in different countries**

		Sample size	Per cent of population aged 18-64
Sweden	1998	30 427	2.2
Germany	1999	2 000[1]	2.9
Netherlands	1998	30 000[1]	3.5
Norway	1997	10 000[1]	3.8
United Kingdom	1998	2 000[1]	3.1
United States	1998	30 000[1]	8.4

1. Approximate sample size.
Source: Davidson and Henrekson (2000).

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Improving the labour market

Against the background of recent positive labour-market outcomes, this section provides an updated assessment of the strengths and weaknesses of the structural settings of the Swedish labour market and the policies pursued. Key future challenges are minimising obstacles to further reductions in structural unemployment, improving the cost-effectiveness of active labour-market policies (ALMPs), raising the employment rate and expanding labour supply.

Progress with reducing structural unemployment

In contrast with most other European countries, Sweden maintained an unemployment rate around 2 to 3 per cent of the workforce throughout the 1980s. However, with high and accelerating inflation it became steadily more evident that such low rates were not sustainable, and in the severe crisis in the early 1990s the unemployment rate surged to over 8 per cent. Thus, high unemployment is a relatively new phenomenon, and a return to "full employment" has been an important policy priority. In 1996, with unemployment at 8 per cent, the government set out its goal of reducing unemployment to 4 per cent in 2000. Although considered highly ambitious at that time, it was reached in the autumn of 2000. In the Budget for 2001 the government signalled its wish to achieve full employment, thereby implicitly aiming for an even lower unemployment rate.

A key question is whether unemployment at 4 per cent or less can be reconciled over the medium and longer term with low and stable inflation. Estimates of structural unemployment are likely to be more uncertain in the case of Sweden than for many other countries. This is due to the particular historical profile of the unemployment rate as well as the complex short-term interaction with the recorded participation rate because of the role played by transfer schemes and active labour-market policies (ALMPs). Nonetheless, OECD estimates point to a structural unemployment rate in 1999 of 5¾ per cent.[32] Although this indicates a fall in structural unemployment in recent years (Figure 24), it also implies that current unemployment is probably below its sustainable level.[33]

The recent years' decline in structural unemployment may, in part, be explained by sounder macroeconomic policy settings and reforms of product and capital markets that have strengthened competition (see below). As for labour-market policies proper, a 1998 follow-up study on member countries' actions with respect to implementing the recommendations from the OECD *Jobs Study* indicated that Sweden had made significantly less progress than average (OECD, 1998*a*):

- employment protection legislation (EPL) remained restrictive and had eased only marginally;
- the effective duration of the unemployment benefit period was longer than elsewhere due to the fact that eligibility for benefits could be restored through participation in job-training programmes; and

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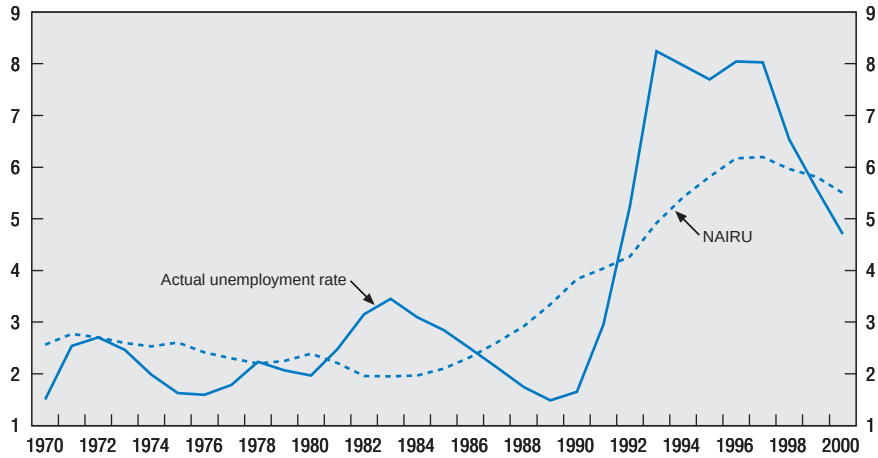
Figure 24. **Actual and structural unemployment**
Per cent of labour force



Source: OECD.

– while the formal eligibility and availability criteria were more stringent than in most other countries, their enforcement was weak.

The basic structural settings of Swedish labour-market policies (Figure 25) have not changed since 1998, but some adjustments have been undertaken. These include:

– A further modest easing of EPL; the general principle of first-in/last-out still applies, but from January 2001 companies can choose to exempt two employees from that rule.

– Eligibility and availability criteria have been changed. The previous criteria, which were formally very stringent but not enforced in practice, have been replaced with rules that are much less restrictive but may be enforced more vigorously. Thus, while the new rules may not ease eligibility *de facto*, the opportunity of a tightening has been missed.

– The government has proposed a change to the labour market code, implying that no person can be obliged to retire before the age of 67; collective agreements containing lower compulsory retirement ages will need rewriting.

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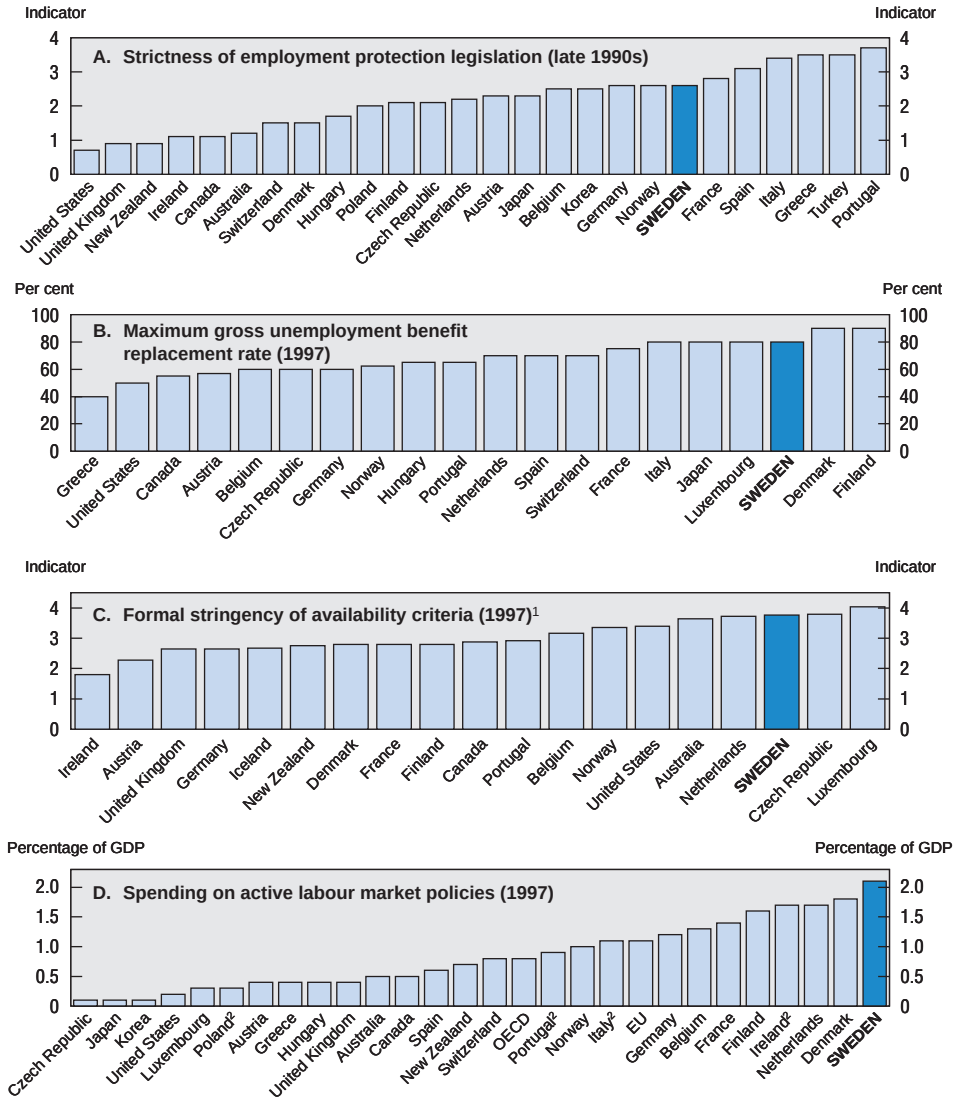
Figure 25. **Structural settings of Swedish labour-market policies**



1. The indicator is a weighted average of the eight following criteria: *i)* demand for job search activity; *ii)* availability while participating in ALMPs; *iii)* occupational mobility; *iv)* geographical mobility; *v)* valid reasons for rejecting job offers; *vi)* sanctions applied in case of voluntary job leaving; *vii)* sanctions applied when offer of job or ALMPs is rejected; *viii)* sanctions applied in case of repeated refusals. Higher value indicates stricter rules.
2. 1996.

Source: Danish Ministry of Finance (1998); OECD.

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- An activity guarantee has been introduced for the long-term unemployed (see Box 5). This applies to those who have been unemployed for 27 months and remain jobless. This group is required to participate in labour-market programmes (either job training or education). The activity guarantee resembles the compulsory activation policy that has been pursued in Denmark since 1996, except that in the Swedish system the activity period is open-ended.
- Industrial action directed towards one-person businesses with no employees has been prohibited.
- Recruitment firms, which were prohibited until recently, have been allowed to operate, which may enhance competition and efficiency in the area of job matching.

Box 5. Activity guarantee

The so-called *activity guarantee* came into force on 1 August 2000. It applies to persons registered at the public employment service (PES) who receive either unemployment or other social benefits and who have not had ordinary (unsubsidised) work for 27 months. This group comprised 50 000 persons (1¼ per cent of the workforce) at the end of 2000. They are obliged to participate in full-time activation to maintain their benefit entitlement. The PES may also decide, for example based on "profiling" exercises, that other unemployed with a high risk of becoming long-term unemployed must comply with the stricter activation demands under the activity guarantee. As from February 2001 it covers part-time unemployed receiving supplementary benefits as well.

The activity guarantee itself is a requirement that job seekers participate in an ALMP or some other education in order to build their human capital. Each activation period within the frame of the activity guarantee must be planned for a maximum duration of six months, at the end of which an evaluation must be done and, eventually, a new six-month period planned. There is no fixed time limit to the total duration of the activity guarantee period; it may in principle be open-ended.

A more generous two-year wage subsidy is offered to private or public employers hiring unemployed aged 57 or above during the activity guarantee period. This subsidy is 75 per cent of the wage, although subject to a ceiling of SKr 525 per day. The backdrop to this feature is that according to the assessment of the Swedish authorities, persons in this group have a very low probability of finding an ordinary job. Employees for whom the employer receives the 75 per cent are not covered by the normal strict EPL; for example, the last-in/first-out principle does not apply to this group.

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The unemployment benefit replacement rate has remained at 80 per cent of preceding wage income, subject to a ceiling (*i.e.* a maximum benefit level), which has been broadly unchanged in nominal terms since 1992. This has caused the *average* replacement rate to decline. However, in June 2000 the government proposed to raise the ceiling on the benefit level, with a first step to be taken in July 2001 and another in 2002. It has not yet been determined how much the ceiling will be raised, as this in part depends on how much room is left under the total expenditure ceiling (see Chapter II). However, a 10 per cent increase was suggested in a 1999 report from the Ministry of Industry, Labour and Communication, and in any case the political commitment to raising the ceiling is strong. This higher ceiling is to apply only during the first 100 days of unemployment and will be the first adjustment to the maximum unemployment benefit level for some years. Even so, if implemented, it may prove unhelpful inasmuch as it weakens job-search incentives by increasing the average replacement rate. The adverse effect on the sustainable rate of unemployment is probably limited if restricted to the first 100 days, but would be heightened if the higher ceiling were extended to the full benefit period at some point in the future.

Despite the favourable labour market performance in 2000, further reforms are necessary to facilitate a shift to lower structural unemployment; the effect of recent policy changes in this respect is uncertain and may prove to be rather small. As labour shortages have already risen and aggregate labour demand may well already exceed aggregate supply, it is especially important to adopt reforms that would ease these pressures. Two measures that should be considered in this context are significantly reducing the scope of supplementary unemployment benefits for part-time workers and reassessing the role and effectiveness of ALMPs.

Supplementary benefits for part-time workers sustain unemployment

The provisions regarding supplementary benefits allow individuals to work part-time and receive benefits for the remaining hours of a standard workweek, subject to some preconditions. The rules, which have changed frequently over the years, are generous and rather complex:

- Entitlement for such benefits in case of unemployment is acquired by working at least 17 hours per week during six months or more within a 12-month reference period.
- The replacement rate is the same as for full-time unemployed, but calculated on an hourly basis. Thus, the benefit level may be up to 80 per cent of the recipient's hourly wage, subject to a ceiling of SKr 73 per hour.
- The formal time limit applying to those who work part-time on permanent contracts is 300 full benefit-days; but since these recipients accumulate benefit-days at a slower rate, it means that supplementary

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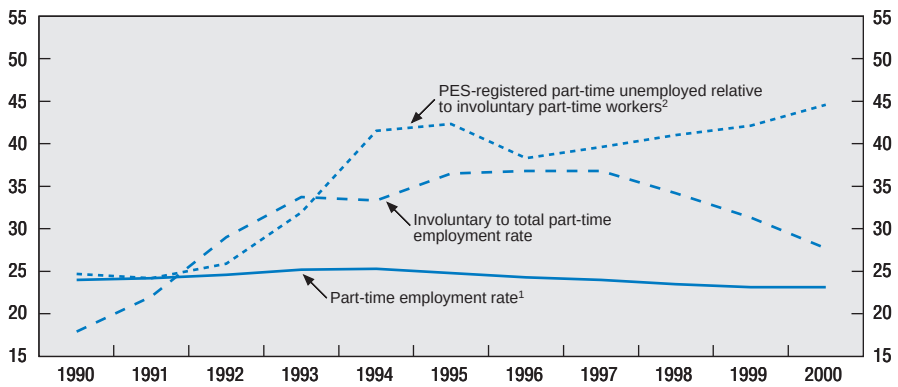
benefits can be received for several years, depending on the number of hours worked.[34] There is no time limit to supplementary benefits for part-time employed on temporary contracts.

Part-time work is common, with almost one quarter of all employees working less than 35 hours per week; women working in the public sector and in retail sales predominate. While the ratio of part-time to total employment has been stable at one quarter during the last 10 to 15 years, involuntary unemployed part-time workers as a share of all part-time employed increased from 15 per cent in 1989 to a peak of 35 per cent in 1997, but then declined to 25 per cent in 2000 (Figure 26). Some 250 000 people were involuntary part-time employed in 2000 (out of a labour force of 4.3 million). Half of these were registered at the Public Employment Service, of which 80 per cent – some 100 000 persons – received supplementary benefits.[35] Total outlays for such benefits amount to approximately SKr 4½ billion per year, some 15 per cent of total unemployment benefits.

It seems likely that the high level of part-time unemployment relative to total unemployment is in part attributable to a combination of policy settings. For one thing, stringent dismissal rules are likely –because of the greater flexibility involved – to have given employers a preference for imposing reduced working

Figure 26. **Employment and unemployment for part-time workers**

Per cent



1. Employees working less than 35 hours per week relative to total employment. Both series according to the Labour Force Survey.
2. Part-time workers registered as unemployed at the Public Employment Service, relative to all part-time workers according to the Labour Force Survey.

Source: Statistics Sweden, Public Employment Service.

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hours on staff instead of laying any off.[36] This could be reinforced if employers' marginal wage costs are lower for part-time than for full-time workers, for example due to less overtime supplements. Secondly, with the generous supplementary benefit scheme described above, the gain in disposable income from working longer hours is often very modest, at least in the short term. In addition, it is typically more difficult and inconvenient for one person to hold two part-time jobs than one full-time job. Therefore, while part-time unemployed may in general be ready to increase working hours in their current job, it may not be feasible for them to work additional hours elsewhere. Thus, the combination of strict EPL and generous provisions for supplementary benefits tends to reduce search incentives and search effectiveness and thus increase total unemployment. It is also likely that for some, the current hours worked match both employers' and employees' preferences, and, in that sense, these people are voluntarily unemployed. One recent measure that may test the extent to which part-time workers are actually willing to add more hours is the recent introduction of the activity guarantee, which will cover those who have received supplementary benefits for 27 months. However, this is still a very long duration, and it would have been more efficient to adopt the proposal from the DELTA Commission (SOU, 1999*b*) to set a limit of 52 weeks. The norm in most other countries that have a supplementary benefit scheme is a maximum duration of 6 months. Recent policy initiatives to curb the level of part-time unemployment also include various regulatory measures, including an obligation for the employer to certify to the PES that the employee cannot be offered full-time employment.

Active labour market policies

Swedish labour-market policies have traditionally emphasised the role of ALMPs, and this focus has even intensified further since the mid-1990s. Total ALMP expenditures amounts to more than 2 per cent of GDP, more than in any other OECD country. Including participants in the Adult Education Initiative (AEI),[37] 220 000 persons (full-year equivalents) took part in labour-market programmes in 2000, corresponding to 5 per cent of the workforce (Table 16). Although this was fewer than the peak of 7.4 per cent in 1998, it was still slightly higher than in 1995. Labour-market programmes have been scaled down broadly in line with the fall in unemployment, and as a result the activation intensity, *i.e.* the ratio of participants in labour market programmes to unemployment, has on average remained around 50 per cent. However, including participants in the Adult Education Initiative it has increased considerably.

The emphasis on ALMPs is deeply rooted in Swedish labour market policies. In the aftermath of the sharp increase in unemployment that followed from the recession in the early 1990s, ALMPs (and adult education) were boosted significantly. This is likely to have been helpful as a means of ensuring discouraged

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Table 16. **Participants in labour-market programmes and adult education**

Thousands

	1995	1996	1997	1998	1999	2000
Labour force	4 319	4 310	4 264	4 255	4 308	4 364
Registered unemployment	436	408	367	286	277	246
Participants in labour-market programmes	198	202	191	173	142	120
In the labour force (per cent)	32	21	17	22	24	28
Not in the labour force (per cent)	68	79	83	78	76	72
Adult Education Initiative[1]	0	0	113	144	143	136
Total	198	202	304	317	285	256
Per cent of labour force	4.6	4.7	7.5	7.5	6.6	5.9
Activation intensity[2]	45	50	52	60	51	49

1. Data for 1997 to 1999 are from Skolverket (2000) and represent the number of courses in municipally organised adult education completed on a full-year basis. Data for 1997 relate to the second half of the year. Data for actual activity in 2000 are not yet available. The number presented is derived from the appropriation in the government's budget plus an estimate of the activity level financed by local government.
2. Labour market programmes in per cent of unemployment.

Source: Public Employment Service; National Institute of Economic Research; Statistics Sweden; Skolverket and OECD.

workers' continued attachment to the labour market, and prevent problems of large-scale marginalisation.[38] If so, this might have facilitated an earlier return towards previous high participation rates than would otherwise have been possible. However, the merits of ALMPs in terms of reducing aggregate unemployment are more uncertain. Numerous micro-data studies have been undertaken to identify the effects of participation on subsequent labour-market outcomes. Even if measurement and identification problems may cloud the picture, cross-country evidence suggests that such effects are often small and/or insignificant (Martin, 2000). Similar results are found using Swedish data, although with some disparity across different types of measures (see, for example, Ackum-Agell, 1999; Ackum-Agell *et al.*, 2000). Moreover, a number of studies have found neutral or even adverse effects on aggregate wage pressures, suggesting that – on average – participants tend to be locked-in rather than helped out (Forslund *et al.*, 2000). In sum, these findings indicate that effects on aggregate unemployment have been insignificant in the past. While this does not exclude favourable effects on the earnings prospects of individual participants, it implies that any net benefits to society are smaller than to individuals.

Yet ALMPs' somewhat lacklustre record so far in terms of reducing unemployment does not imply that they have no useful role to play. In particular, they may be efficient when targeted at certain groups of unemployed with specific problems and needs. Recent analysis emphasises that different measures may not all have similar effects and that the circumstances under which ALMPs operate

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may be crucial (Martin, 2000). Thus, one possible explanation why it has been difficult to provide clearer evidence of a favourable impact on aggregate labour-market performance in Sweden may be that the framework conditions have not been conducive to success. ALMPs can work through two separate channels: by "taxing leisure time" they may enhance job-search incentives, and by upgrading individuals' skills they may improve the average competence of the workforce. At least until recently, ALMPs have done little to encourage participants to look for work, and the "tax on leisure" channel of requiring individuals to use their time in a specific way has been very weak. To be effective, activation should be continuous and compulsory after a certain period of unemployment, and sanctions in case of non-compliance should be firm.[39] To some extent, such rules should change behaviour by making it relatively more attractive to search for jobs before entering ALMPs in the first place, as well as encouraging transition from the programme(s) to work among current participants. While this has not been the situation so far, the combination of the *activity guarantee* and stricter availability criteria may be interpreted as a shift of emphasis in active labour market policies, giving more weight to stronger incentives to search for unsubsidised jobs.

Skills upgrading has traditionally been the primary aim and focus of ALMPs in Sweden. While this might be worthwhile in terms of increasing individuals' productivity and earnings prospects, such considerations have not been the rationale underlying this policy. Rather, the emphasis on upgrading skills and competencies reflects the implicit assumption that the most important reason for structural unemployment is widespread problems of mismatch between individuals' productivity levels and reservation wages. It is rather surprising that the validity of this crucial assumption has not been subject to more empirical research than has been the case. Although firm conclusions would be premature, this *lack-of-employability hypothesis* would seem at odds with the large turnover in unemployment and the empirical observation that changes in long-term unemployment clearly respond to changes in overall unemployment. Again, this need not exclude the possibility that specific groups of unemployed face employability problems, for example related to language skills for immigrants or functional illiteracy more generally, but it may suggest that these should be addressed through better targeted policies rather than the current generalised approach.

Employment and participation rates remain well below earlier peak levels

Despite the rapid increase in employment in the past three years, the employment and participation rates are still significantly lower than the levels seen ten years ago; 75 per cent of the working-age population (*i.e.* those aged 16 to 64 years) were employed in 2000 which was 10 percentage points below the 1990 level. Likewise, the participation rate of 78 per cent in 2000 was 7 percentage points less than the 1990 outcome; in fact, most of the accumulated contribution

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to the labour force during the 1970s and 1980s was lost in the 1990s (Box 6). The fall is spread across all age groups and has been roughly the same for men and women (Table 17). A noteworthy feature is that – unlike the trend in many other countries – the reduction in employment and participation rates for older workers has not been particularly pronounced in Sweden. This may in part be attributable to strict EPL, in particular the last-in/first-out principle, which has tended to favour older workers relative to other groups.

Thus, while the unemployment rate has reversed much of its rise, employment and participation rates remain far from the levels that existed before the crisis of the early 1990s. Those levels may not be realistic points of reference for current policies, at least not in the near term; they were extremely high both by international and domestic historical standards (Figure 28) and were obtained in a period with severe overheating. The government has nevertheless stated an objective of increasing the employment rate for persons aged 20 to 64 years in unsubsidised jobs to 80 per cent by 2004. This is some 3 percentage points above the level in 2000, which itself was an increase of 3 percentage points from the 1997 outcome. The medium-term objective is ambitious inasmuch as it requires continued high growth in employment for an extended period and would imply a return to levels near the highest amongst OECD countries. Yet, it is still somewhat lower than earlier peak levels, which suggests that it may be attainable if the necessary policy measures are undertaken.

However, as unemployment is already low, it will be crucial to ensure a steady increase in the participation rate if the 80 per cent target is to be met. While it is encouraging that labour supply is now increasing, the current participation rate remains rather modest; it is still lower than at the outset of the current upswing and no higher than in the late 1960s. At least partly, this results from an ambiguous set of policies that has probably been biased too much towards bringing unemployment down rapidly. Thus, while some recent policy initiatives have had a helpful impact on labour supply, others have not. The 1999 pension reform is a significant step in the right direction: individuals' incentives to remain in the labour market beyond the age of 61 is enhanced by the fact that future pensions are raised actuarially with the amount initially "foregone" by postponing retirement. However, the reform is being phased in gradually and is not likely to have significant effects on participation rates in the near term.[40] The tax cuts in 2000 and 2001 are likely to have positive effects as well, although probably more on average hours worked than on the participation rate *per se*. On the other hand, the one-month extension of the parental leave scheme (reserved for fathers) as from 2002 works in the opposite direction. The same is true for the decision to prolong the Adult Education Initiative to 2003 and the goal to keep the number of participants numbers at current levels until the end of 2002 (Ministry of Finance, 2000*b*). A continuation of the favourable economic conditions may eventually encourage an endogenous net inflow to the workforce; however, the magnitude of

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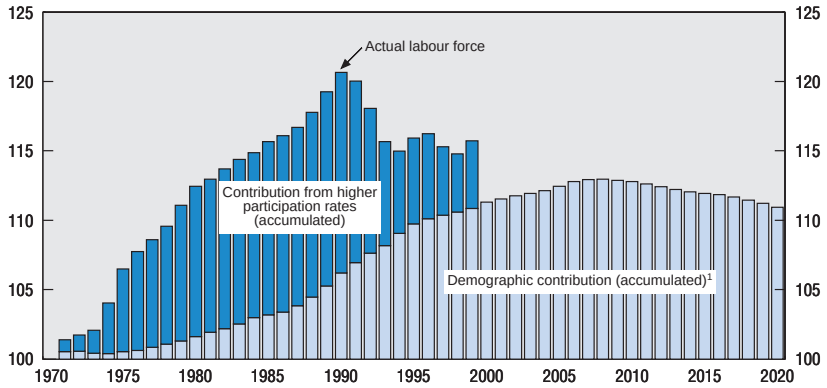
Box 6. A long-term perspective on labour-force developments

A closer look at labour-force developments in Sweden in the three decades since 1970 reveals a stunning picture; while policies were highly successful in raising labour supply during the 1970s and 1980s, they were equally unsuccessful during the 1990s. The endogenous contribution to increases in the labour force from demographics, *i.e.* the combined effect of an increasing working-age population and a favourable age composition, was considerable in all three decades. Demographic developments have added ½ per cent per year to the labour force since 1970, on average, implying an accumulated contribution of approximately 15 per cent up to 2000 (Figure 27). During the 1970s and 1980s, this was complemented by sizeable contributions from increases in age- and gender-specific participation rates; this boosted the labour force by 15 per cent from 1970 to 1990. Above all, the latter was the result of higher labour-force participation by women. However, most of the accumulated contribution from higher participation rates was lost during the 1990s. Thus, the 5 per cent decline in the labour force from the late 1980s to the late 1990s is fully attributable to lower participation rates and came about despite continued positive demographic contributions. Hence, comparing the current situation with that of 1970 reveals that the absolute increases in the labour force since then follows mainly from demographic developments, while higher participation rates have been much less important.

Figure 27. **Contributions to the labour force, 1970-2020**

1970 = 100



1. The accumulated contribution from demographics is taken as the difference between indices for the actual and a synthetic labour force, with the latter being calculated under the technical assumption of constant age and gender-specific participation rates. Thus changes in the synthetic labour force index reflect only changes in the size or the composition of the working age population. The participation rates from 1999 have been applied for the calculation.

Source: OECD, *Labour Force Survey.*

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Box 6. A long-term perspective on labour-force developments

The demographic trends of the past three decades can also serve to put the evolution of the public finances into perspective. Thus, it is noteworthy that the increase in the tax-to-GDP ratio from 35 per cent in 1970 to more than 50 per cent in 2000 and the simultaneous build-up of a gross public debt of 60 per cent of GDP have taken place in an environment of very favourable demographic developments, which, all other things being equal, should have entailed the opposite. This is, of course, the result of the extension of the welfare state with higher standards and coverage rates for publicly provided services and more generous transfer schemes.

Looking ahead, continued favourable demographic developments can also be envisaged for the immediate future, but the positive contributions will gradually fade and eventually reverse from around 2005. The disappearance of positive labour-supply contributions will gradually put more pressure on policies to find other ways to stimulate labour supply.

Table 17. **Changes in participation rates from 1990 to 2000**

	Age							
	16-19	20-24	25-34	35-44	45-54	55-59	60-64	**Total**
	Change in participation rates, percentage points							
Men	–21.8	–16.1	–5.8	–6.3	–4.7	–3.2	–8.0	**–7.1**
Women	–20.6	–21.3	–8.3	–6.4	–3.8	–0.1	–6.8	**–7.5**
Total	**–21.2**	**–18.6**	**–7.1**	**–6.3**	**–4.3**	**–1.4**	**–7.3**	**–7.3**

Source: *Annual Labour Force Survey*, 1999 *and* 1990, Statistics Sweden.

such a contribution is uncertain. In any case, endogenous contributions should be accompanied by adjustments to current policies and a coherent medium-term strategy for increasing the workforce. Further shrinkage of the labour-market programmes and a significant downscaling of the Adult Education Initiative would be important elements.

More generally, the education system seems to have some weaknesses with repercussions on labour supply. Although difficult to measure, benchmaking studies suggest that the overall outcome is comparable with the average of other countries; in some respects performance is above average, while in others it is

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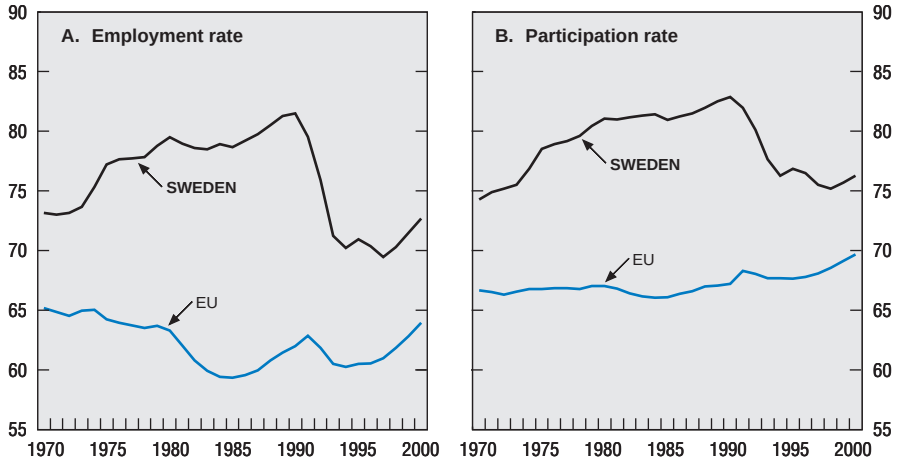
Figure 28. **Employment and participation rates**
Per cent of population aged 16 to 64



Source: OECD.

below (Lemne, 2000). However, Sweden spends more than 6 per cent of its GDP on education, which, next to the United States, is the highest in the OECD. Also, the average costs per student are high by international standards.

The high level of resources allocated to education partly occurs because some students take more than the prescribed time to pass through the system. A recent report from the Expert Group on Studies in Public Finance (ESO) suggests that there is considerable scope for improving efficiency and concludes that excessive time spent in the education system adds up to as much 130 000 full-year students, or the equivalent of 3 per cent of the labour force (Gustafsson *et al.*, 2000). The scope for improving inefficiency is mainly at the tertiary level:

– One out of three students does not complete upper secondary education within four years. An increasing share of those who graduate from upper secondary school need supplementary courses to qualify for enrolment in tertiary education. Moreover, admission criteria to tertiary education are often based on skills and knowledge of questionable relevance to the tertiary qualification sought, which boosts the number of students undertaking supplementary courses; in 1998/99 the number of students undertaking such courses represented 50 000 full-year equivalents.

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– The average age of completion of tertiary education is high, as small wage premiums and progressive income taxation combine to reduce the returns to education and individuals' incentives to complete their education quickly and move to the job market. As a result, 50 per cent of all students in tertiary education are 25 or older and 25 per cent are above 30 (Table 18).

By international standards the proportion of the population above 30 years (in particular in the age group 30 to 39 years) that undertake full-time education is very high (Table 18), which illustrates the great emphasis that has traditionally been put on adult education in Sweden. While this is a strength in a life-long-learning context (OECD, 2001), the accompanying costs – direct as well as indirect – need to be weighed against. Thus, even if it is not realistic to eliminate excess time spent in the education system altogether, a significant reduction should be a priority. This would serve the double purpose of lowering expenditures (including benefits to the participants) and stimulating labour supply. The budget for 2001 includes additional appropriations for primary schooling over the next five years and more generous student grants. These initiatives will further increase total spending on education, but may not do much to address the above problems. In sum, a reassessment of educational policies with a view to enhancing efficiency and strengthening individuals' incentives to acquire qualifications in a shorter period of time is warranted.

Working-time policies

Policies to boost labour supply serve the dual purpose of containing labour shortages and inflationary pressures in the near term and facilitating tax cuts (or avoiding tax increases) in the longer term. The link between labour supply

Table 18. **Share of population in selected age groups undertaking full-time education**
1998, per cent

	15-19 years	20-29 years	30-39 years	Above 40 years
Denmark	80.1	27.9	5.5	0.7
Finland	82.1	33.1	7.6	1.3
France	87.8	19.1	1.9	–
Germany	88.3	21.7	3.0	0.2
Ireland	80.7	15.5	2.2	–
Netherlands	86.0	22.0	3.5	1.4
Sweden	**86.1**	**30.4**	**13.3**	**2.7**
United Kingdom	69.5	18.1	8.8	3.2
United States	74.2	21.4	5.6	1.6
OECD	76.3	20.4	4.4	1.2

Source: *Education at a Glance*, OECD (2000).

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and the public finances is important, not least in countries like Sweden with its large public sector and a social safety net that provides broader coverage and higher benefits than in most other countries. Financing of welfare states such as in Sweden rests on the foundation of work. A broad tax base is not only a question of the participation rate, but also of the average number of hours worked. In contrast with the decline experienced in most other countries (with the United States as a notable exception), average working time edged up in Sweden during the 1990s. The tax reform in 1991, which broadened tax bases considerably and reduced marginal tax rates, is likely to have helped. Other contributing factors include a reversal of abnormally high rates of absenteeism in the late 1980s, and the fact that groups with below-average working hours may have been over-represented amongst those who were discouraged from the labour market in the early 1990s. However absenteeism is on the rise again, and the average working time has levelled off recently and may be starting to edge down.

Looking ahead, average working time is likely to decline over the medium term as the issue of cutting it has been high on the political agenda and a number of proposals have been advanced. While discussions about working time have often been linked to the persistence of high unemployment in many other countries, this has not been the case in Sweden due to the fact that full employment has been better maintained than elsewhere. Ironically, the issue of working time has surfaced in Sweden during the current upswing. A joint proposal from the main employee organisations presented in October 2000 suggested an extra week of vacation to be implemented over five years starting in 2002 and the introduction of *individual holiday accounts*. The latter would allow employees to pool hours from the recommended new holidays as well as from the existing fifth week and overtime exceeding 100 hours per year in a personal "account". Employees would, according to the proposal, have considerable flexibility as to how and when such hours can be "cashed in". The government has signalled its general support for this proposal, but the question of working time has temporarily been set aside while a commission reassesses the issue. Still, while the timing of its implementation is uncertain, the political commitment to reducing working time in some way is quite strong.

Reductions of average working hours have long-term implications for the public finances as a consequence of the induced fall in production (Box 7). While these may not be urgent concerns, given the current level of the budget surplus, the long-term trade-off should be recognised nevertheless; the tax-to-GDP ratio may have to be almost 1 percentage point higher (depending on the response of public consumption) if the aforementioned proposal were adopted than if working time is unchanged. In the nearer term, a concern related to working-time reductions is that they might add to labour-market pressures. As discussed in Chapter I, such pressures have been mounting recently and, with the prospects of continued declines in unemployment rates, they might grow further by the time working-time reductions might be implemented. In sum, even if the long-term costs for the public finances

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Box 7. Implications of reduced working hours on the public finances

The reduction of working hours in Sweden, in terms of the gradual introduction of an extra week of holidays, follows recent decisions in other countries, for example France, Italy and Denmark to reduce working hours. In Sweden, a central element in the political discussions has been whether shorter working hours should be imposed through legislation or bargained between the social partners. While these two different approaches are likely to have different repercussions on the bargaining system and thus wage formation more generally, they have the same adverse consequences for the public finances in the long term. These implications are quite similar to those of increasing dependency ratios. The main difference is that, while shorter working hours impinge only on tax bases, higher dependency ratios would normally be accompanied by increased expenditure pressures as well.

Assuming full carry-over on average working time, an extra week of vacation would imply a 2½ per cent reduction of labour supply. Production and income is likely to decline correspondingly in the long term relative to where it otherwise might have been, unless any efficiency gains are generated because of the change. The short-term effects may be different. To the extent that companies and public-sector workplaces respond by recruiting additional staff, production may not fall commensurately and unemployment may be reduced. However, unless additional labour supply is encouraged, this is likely to increase wage pressures, eventually bringing employment and unemployment back to their starting points, so that the working-time reduction will be fully reflected in the fall in production.

Therefore, total consumption (the sum of private and public consumption) will also have to be reduced by a similar amount. The reduction of private consumption will occur endogenously as households' disposable income decreases. By contrast, it may be a difficult political process to reduce public consumption or raise taxes, especially if the link between working time and the public finances is not widely understood. Therefore, the policy implications of reduced working hours are particularly pronounced in countries with a large public sector, especially where budgetary flexibility is limited.

are recognised and any shortening of working hours is phased in gradually, its timing is likely to be unhelpful in the short term, and it would certainly be at odds with the medium-term objective of increasing labour supply. Moreover, any compensation to employers in terms of reduced social-security contribution, as has been mooted, would prove problematic. It might leave the mistaken impression that reduced working time is a free lunch in terms of its consequences for the public finances. In addition, it might give rise to expectations that future reductions could be compensated as well. This could also result in a more permanent change of the long-standing practice of deciding on working-time issues through collective bargaining rather than through legislation.

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Transfer policies

Reducing transfers is crucial for increasing labour supply

Increasing labour supply obviously relies on appropriate tax and transfer policies. Most Swedes of working age who are not employed receive transfer income of some sort, and an increase in employment must therefore, almost by definition, be reflected in a decrease in the number of transfer recipients. In 1999, the non-employment rate was 28.5 per cent while the total of transfer recipients and ALMP participants amounted to 22.6 per cent of the working-age population (hereafter the "transfer rate") (Table 19). In a long-term perspective the transfer rate has been increasing in recent decades from some 8 per cent of the working-age population in 1965 to around 16 per cent in 1990 (Figure 27). This was mainly a reflection of more generous provisions for early-retirement pensions, sickness

Table 19. **Transfer recipients aged 16-64 years**

Thousand persons

	1965	1970	1975	1980	1985	1990	1995	1999
I: Transfers								
Early retirement pension	151	192	282	279	299	319	362	366
Sickness benefit	165	230	252	252	249	211	142	180
Parental leave	25	27	46	79	96	141	143	100
Temporary parental leave	0	0	0	0	0	0	20	16
Social security	0	0	0	0	31	72	118	115
Others	0	0	5	37	0	0	0	0
Unemployment	37	71	80	95	139	71	340	224
II: Labour market programmes (LMPs)[1]	25	35	50	69	105	61	198	284
III: Transfers and LMPs	403	555	715	811	919	875	1 323	1 285
IV: Public sector employment	577	806	1 044	1 300	1 413	1 437	1 348	1 328
V: Paid by the public, total	980	1 361	1 759	2 111	2 332	2 312	2 671	2 613
VI: Total employment	3 698	3 854	4 062	4 235	4 395	4 484	3 988	4 067
VII: Work force	3 742	3 913	4 129	4 320	4 529	4 559	4 320	4 308
VIII: Working age population	5 133	5 267	5 259	5 328	5 526	5 501	5 621	5 690
Dependency ratios								
A: III/VIII	7.9	10.5	13.6	15.2	18.3	15.9	23.5	22.6
B: V/VI	26.5	35.3	43.3	49.8	55.5	51.6	67.0	64.2
C: Non-employment rate	28.0	26.8	22.8	20.5	20.5	18.5	29.1	28.5
Tax-to-GDP ratio	35.1	41.7	45.0	48.2	49.5	54.0	48.1	53.6

Note: The number of transfer recipients reported are OECD estimates of full-year beneficiaries of the transfer schemes in question, based on available statistics.

1. Data from Public Employment Service including job-training and the Adult Education Initiative (AEI). Wage subsidy-programmes for persons with less than full working capacity are not included. Not all AEI participants receive public transfers, but are included here due to incomplete information on their sources of income. Consequently, the figure for 1999 is too high.

Source: Statistics Sweden; Olsson and Holmgren (1994); National Social Insurance Board; and OECD.

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benefits and parental leave – both in terms of easier accessibility and/or longer duration. Notwithstanding the preceding increase, the transfer rate rose much more rapidly in the early part of the 1990s. During the early-1990s recession it surged 10 percentage points and reached 25 per cent in 1994. Although it has fallen back since then, the transfer rate remains significantly higher than up to 1990. Comparing the variations in the transfer rate with those of the non-employment rate, the pattern is very different before and after 1990. Before the crisis the two series moved in opposite directions – the transfer rate increased when the non-employment rate declined. However, in the most recent decade they have moved in tandem.

While the transfer- and non-employment rates remain high in a historical perspective and are still causes for concern, their composition provides some comfort. Unlike in some other countries, Sweden has avoided an increase in the ratio of persons on early-retirement pension (disability) or other early-withdrawal schemes. As mentioned above, the employment rate of those aged 55 to 64 is among the highest in the OECD as a consequence. The majority of the increase in the transfer rate has been on unemployment and labour market programmes. Although significant reforms are likely to be required to permanently restore a significantly lower transfer rate, such reforms may be less painful than elsewhere.

However, sickness benefits are a notable exception. Sweden has a long record of strong cyclical fluctuations in absenteeism due to sickness. This has also been the case under the current upswing. Although still well below the peak levels seen in the previous expansion, sickness benefits have risen much more than expected by the authorities. From 1997 to 2000 the increase was some 50 per cent, and this has been very close to causing the expenditure ceilings to be breached (see Chapter II). Longer spells have particularly increased. A commission is currently analysing the possible causes for the recent surge. In a preliminary government report (SOU, 2000*b*), it was suggested that bad working environments played an important role, and an increase in the employer-paid period for sickness benefits from the first two weeks to two months was proposed. Although the analysis was widely accepted, the proposed remedy was massively rejected. Thus, no initiatives to deal with the problem were included in the budget for 2001. The highly cyclical pattern of absenteeism due to sickness would seem to put in doubt whether deterioration in the general work environment allegedly related to stress and excessive work loads is the main explanation for the surge in outlays. An alternative or at least supplementary explanation may be that higher sickness benefit outlays mirror lower inflows into the system of early retirement pensions following more stringent eligibility criteria for the latter as from the mid-1990s. In addition, the current system is fairly generous both in terms of the benefit level and – in the absence of a fixed time limit – potentially long-lasting benefit periods.

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Tax policies

Sweden faces a number of medium- and longer-term challenges in the domain of tax policies. The tax-to-GDP ratio of 52 per cent was the highest in the OECD in 1999, and high tax rates on labour income distort work incentives significantly. Against the background of collapsing labour force participation rates in the 1990s and the stated objective of a return towards earlier levels, it should be a high-priority medium-term objective to lower marginal income tax rates so as to make work relatively more attractive. At the same time, however, pressures on tax bases may emerge or intensify on some fronts in the future. Electronic commerce is likely to become gradually more important, putting some pressure on the indirect tax base, although its potential impact is difficult to assess. In addition, an alignment of taxes on alcohol and tobacco with the EU levels is inevitable, as the current restrictions on individuals' imports of such goods will be abolished in 2004. Finally, demographic developments will start restricting the size of the labour force and thus the tax base for personal income taxes. These will become an increasingly important factor after 2005. It marks a reversal of the trend of the recent decades where favourable demographic developments have helped to expand tax bases (see above). This might be reinforced if average working hours also decline. Establishing a coherent strategy for dealing with these challenges should have high priority, for example building on the work of the commission that is currently assessing the challenges caused by tax-base erosion, and which is to report in March 2002.

The comprehensive 1991 tax reform, which instituted a dual system of separate taxation of capital and labour income, was a major step in the right direction in terms of better neutrality of the system as a whole (OECD, 1999*a*). However, tax changes during the remainder of the decade were often less well designed and on some fronts could be seen as a retreat from the principles underlying the 1991 reform. VAT is a case in point: while the 1991 reform ensured a largely uniform rate, reductions on a range of goods and services were made over the intervening years. This has resulted in a less neutral system with several different rates. As a consequence, consumption behaviour is distorted, and pressures persist to extend the range of goods and services with reduced rates. The reduction of the VAT rate on public transport in 2001 serves as an illustration of further erosion, even if it is relatively inconsequential in itself. Another example is the indexing of the threshold values in the tax system; the general indexing principle stated in the legislation is that these should be inflated in line with the increase in consumer prices, but deviation from this occurred due to a predetermined temporary adjustment for the period 1995 to 1998 as a part of the consolidation programme.

To some extent, backing off from the principles of the 1991 reform could be justified by the sharp increase in the budget deficit in the years immediately after the reform, which made the restoration of sound public finances the

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paramount objective during the rest of the decade. However, in some cases, the deviations have worked to reduce tax revenues, thereby putting an additional burden on other instruments, most notably income taxes. These were increased during the period of fiscal consolidation, in particular by virtue of the introduction of employee-paid social-security contributions. Consequently, average marginal effective tax rates – excluding employers' social security contributions – were 4 percentage points higher in 2000 than in 1991 (Table 20). This is likely to have contributed to reducing labour supply.

Taxation of capital income (see Annex I) also presents a number of areas where changes have shifted away from the fundamental principles of the earlier tax reforms. Increased capital mobility continues to erode the tax base, since revenue requirements operate to keep tax levels relatively high. Nominal tax rates are difficult to compare reliably on an international basis, since rates may not bear a very close relationship to taxes actually paid. Some estimates of average effective tax rates have been made to provide an alternative basis for cross-country comparison (Carey and Tchilinguirian, 2000). These estimates, compiled from aggregate data, should necessarily be treated with considerable caution,[41] but they do indicate two features of Swedish capital income taxes that are important for understanding the broad international pressures at work. *First*, on these estimates, even after the reforms and with a relatively low nominal rate, in practice, Sweden's capital continues to be more highly taxed than around half of the OECD countries and clearly more than the EU average. *Second*, average effective rates on capital have risen in the last three years, and, calculated using the gross operating surplus method, they are back up to 1990 levels.[42]

The evidence suggests that Sweden has suffered from some mobility of the capital tax base. On a practical level, all countries have some degree of difficulty in applying the principle of residence that underlies taxation. And the higher the domestic tax rates are, the greater the incentive to avoid them altogether by

Table 20. **Average marginal effective tax rates, 1991 and 2000**[1]

	Unemployed and social benefit recipients	Employed		Others	All
		Part-time	Full-time		
1991	78	37	41	32	45
2000	78	41	45	40	49
Change	**0**	**4**	**4**	**8**	**4**

1. The reported numbers result from simulations on register data sets containing information on individuals' incomes as well as transfers received and taxes paid. An annual increase in gross incomes of SKr 12 000 has been assumed for all persons and the resulting change in taxes and transfers has been calculated under the tax- and transfer rules of 1991 and 2000 respectively.

Source: Ministry of Finance.

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investing abroad in instruments that have thus far been difficult for tax authorities to trace. The authorities have estimated that around SKr 332 billion of accumulated Swedish savings had been invested abroad by 1997, thereby largely escaping the domestic tax net (Ministry of Finance, 2000*c*). While diversification into foreign assets makes sense, in this case the overall allocation of savings is being distorted by high tax rates, instead of more closely reflecting the underlying economic merits – *i.e.* the gross relative rates of return – of different options. The consequences for public finances are exacerbated by the tax rebates on negative capital income, which are mostly mortgage interest payments and fully reported by households. Although the evidence would be difficult to pin down, there is a clear incentive for homeowners to borrow as much as possible and use the proceeds to purchase off-shore assets in order to exploit the tax arbitrage opportunity.

The combination of tax relief for interest costs and double taxation of dividends also influences the allocation of capital, by making it more attractive to issue debt than equity, as noted in the previous *Survey*. The treatment was significantly more neutral in 1994 when double taxation was temporarily abolished and capital gains tax was reduced to 12.5 per cent. But in 1995 taxation of new share issues and retained earnings both jumped, with new share issues faring worse. There is also some anecdotal evidence that differential taxation on dividends together with rules on interest deductions have played a part in a number of company decisions to relocate headquarters outside Sweden and have influenced the way relationships have been restructured between parent companies and subsidiaries, as well as affecting the way in which mergers have been implemented (Ministry of Finance, 2000*c*).

Sweden applies a dual-income system, where earned income is taxed progressively at higher rates than capital income. The lower rate on capital income is motivated by the need to have internationally competitive taxation of personal capital income. At the same time, rules are needed to prevent income-shifting especially for closely held companies, run by owner-managers. These enterprises face very complicated tax rules designed to avoid an incentive to reclassify all earnings as capital income so as to avoid the higher tax rates applying to wages. These so-called "3:12" rules particularly affect new and highly profitable operations, because their owners could end up paying significantly higher taxes than passive owners would, because a large part of the dividends and capital gains are treated as employment income. It is also claimed that these rules are particularly complex for small outfits, thereby involving relatively high compliance costs for the taxable income involved. The need to review these rules has been accepted and assigned to a commission of enquiry, which will report back in October 2001.

The late 1990s' trends in property taxation have probably been particularly unhelpful from a tax-efficiency perspective. Since real estate is immobile by nature, a gradual move towards higher property taxes, offset by lower taxes on

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mobile tax bases (including labour), would have been warranted. However, the opposite has been the case. The property tax on both owner-occupied houses and rentals was reduced from 1.7 to 1.5 per cent in 1998 and a temporary lower rate of 1.2 per cent applied to rentals in 1999 and 2000. Moreover, while the tax assessment values of the housing stock, which are the basis for the property tax, should reflect market values, these were effectively frozen from 1997 to 2000. As house prices have increased rapidly in recent years, a sizeable gap between the market values and the tax assessment values built up between 1997 and 2000. The gap has been estimated by the Swedish authorities to be some 35 per cent for rentals in 2000 and more than 40 per cent for owner-occupied houses, albeit with considerable regional disparities. As from 2001, the tax assessment values will be realigned with market values, but this will be largely offset by further reductions in property tax rates. As a result, the effective taxation of property has been reduced significantly during the second half of the 1990s, which is likely to have fuelled house-price increases.

Like many other European countries, Sweden has now embarked on a policy of reducing taxes. Against the background of comfortable budget surpluses, a multi-year tax-cut programme was announced in the budget for 2000. The main elements are individual tax discounts, which are designed to offset the employee-paid social-security contributions mentioned above, and gradual increases in the threshold for the 20 per cent state income tax that is levied in addition to municipal income taxes and social-security contributions. The aim is to reduce the proportion of tax payers facing the top marginal tax rate to 15 per cent, thereby bringing it back to the objective set in the 1991 tax reform. With the budgets for 2000 and 2001, half of this tax cut programme has now been implemented. This policy is laudable for its structural effects as it will reduce marginal tax rates for all taxpayers and thereby stimulate labour-force participation. However, larger effects on average marginal tax rates could have been achieved by further reducing the number of earners paying the state income tax. The share of full-time employees facing the state income tax is bound to be much higher, and a significant reduction could be obtained for relatively little revenue foregone by increasing the threshold further. In particular, this might encourage many part-time workers to increase working hours.

Competition and product markets

As noted in previous *Surveys*, Sweden has made considerable progress since the early 1990s in deregulating its product markets and improving competition, reflecting the strong political commitment to building a competitive market economy. Major initiatives have included strengthening competition law and aligning it with EU norms; deregulating important utilities markets such as telecommunications, electricity and postal services; promoting competition in the

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public sector; and reducing the burden of regulation. These reforms have generated more efficient sectors and lower prices and are often pointed to as a reason why Sweden's recent inflation performance has been so impressive. Nonetheless, the consumer price level was still more than 20 per cent above the EU average in 1998, and the Competition Authority has estimated that around half this difference can be attributed to a lack of competition in Sweden. Moreover, even in deregulated areas such as electricity and telecommunications, further improvements are still needed, especially where the former monopolist operator maintains a high degree of market power. Additional challenges persist in a range of sectors that continue to be under-exposed to competition, including retail distribution (food, pharmacies and petrol supplies), or excessively regulated (such as housing).

Electricity

Sweden has successfully applied market principles to the potentially competitive segments of the electricity supply industry. One key element of this approach was opening the market to international trade; another was the introduction of full end-user choice from the beginning of the reform. Some obstacles to effective choice have been recognised and recently removed. Competition is, thus, gradually increasing in the electricity sector, despite mergers and acquisitions in the market that have led to some increase in horizontal and vertical integration. However, while state-owned Vattenfall produces more than half of Sweden's electricity and has operations in other Nordic countries and elsewhere, its market dominance is being progressively diluted by the increasing integration of the Nordic electricity market, where it has a market share of only one-fifth (Table 21). The Nordic electricity exchange, NordPool, provides a 24-hour spot market and a weekly market for forward contracts up to three years. The market has continued to expand with some 61 Swedish participants (generators, distributors, industrial companies and traders) and 278 players in total at the beginning of 2000. Around 20 per cent of the area's trading now takes place through the exchange, which is gradually supplanting bilateral agreements. The market has clearly helped to smooth fluctuations in demand and supply and thereby lowered costs, although transmission capacity constraints can lead to local divergences in spot prices.

Households experienced a decline in power prices in 1999, by around 1 to 3 per cent, after several years of increases (Table 22). A major impediment to switching suppliers, the requirement for households to install hourly metering equipment, was abolished along with a shortening of the notification period from six months to one month. Switching fees are no longer permitted, and in the first two months after the new rules were implemented, some 50 000 to 60 000 consumers opted to change. By August 2000, almost 30 per cent of Swedish households had taken steps

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Table 21. **Market shares in Nordic electricity market**
1999

Generators	Energy generated TWh	Proportion of total Nordic countries, per cent
Vattenfall	79.6	21
Sydkraft	27.5	7
Total for Sweden	**150.5**	**40**
Statkfraft	33.5	9
Oslo Energi	8.1	2
Total for Norway	**122.9**	**33**
Fortum	35.0[1]	9
Pohjolan Voima Oy	20.3	5
Total for Finland	**66.8**	**18**
Elsam	22.8	6
Elkraft	14.2	4
Total for Denmark	**37.0**	**10**
Total for largest Nordic electricity generators	241.0	64
Total for the Nordic countries	**377.2**	**100**

1. Including 50 per cent of the energy generated by Birka Energi.
Source: Swedish National Energy Administration.

Table 22. **Electricity prices paid by households**
Average price including taxes as at 1 January, öre per kWh

	1995	1996	1997	1998	1999
Apartment	90.2	97.6	99.9	106.8	105.7
Single-family dwelling without electricity heating	82.7	90.0	92.4	97.8	95.4
Single-family dwelling with electricity heating	64.0	67.5	72.2	75.2	72.7

Source: Statistics Sweden.

to reduce their electricity costs, either by changing supplier or negotiating a new contract with their existing one.[43] More generally, the scope for active competition has also been constrained by long-term contracts between producers and distributors covering supply for some 80 per cent of the market: these were established before deregulation in 1996 and have only begun expiring in the last year or so.

Electricity distribution is an area where concerns about competition remain. The structure of the activity remains quite fragmented, with more than 200 distribution companies operating, mostly owned by municipalities. Around three-quarters of these have less than 20 000 subscribers. Most of these companies are

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also retailers of electricity. However, despite the legal restrictions separating distribution from other electricity supply activities, Vattenfall and Sydkraft between them have almost 2 million customers for distribution and even more retail supply customers. Furthermore a number of distributors have restructured their connection and transmission charges over the years to include a large fixed component, and seven of them apply only a fixed charge. The National Energy Administration, the supervisory authority for the sector was concerned that prices were not properly reflecting costs and therefore directed a number of companies to lower their delivery charges during 1999, effectively imposing a price cap. This is under appeal before the courts. In addition, distributors are now obliged by law to publish their tariffs, making it easier to compare rates and identify distortions.

Telecommunications

Some further progress has been achieved in strengthening competition in the telecommunications market since the last *Survey*, but a number of obstacles have been encountered, and technological developments are affecting the evolution of competition. Inter-modal competition and different options for narrow and broad-band Internet access are becoming more significant between fixed and mobile telephony, making the market somewhat more complex.

At the time of the last *Survey*, two developments introduced in late 1999 were expected to deliver greater competition in fixed telephony: pre-selection and number portability. Neither has lived up to its promise. Under pre-selection, customers should, in principle, find it easier to access their chosen operator's network. But due to technical constraints, pre-selection was not initially effective, because if the area code was not dialled, for communications within the local-calling area the calls were routed by default through the Telia network. This company, previously a monopoly, remains 70 per cent owned by the state and still has 98 per cent of all subscribers and owns the local loop.[44] This puts other operators at a disadvantage and has generated problems between operators, confusion for customers and conflicts with the regulator, the National Post and Telecom Agency (PTS). However, more recently technical and commercial developments have made it possible to dial local calls through the chosen operator. The effect of the reform can be seen in falling consumer prices due to growing competition. PTS is monitoring the pre-selection arrangements, and it has also ordered Telia to reduce its inter-connection charges for pre-selected calls by a factor of four. Telia has appealed. Number portability was also expected to make it less costly for customers to change operators. Again, this has produced little real change, in part because it concerns only those who can connect directly to an alternative to Telia. Only about 4 000 subscribers are currently directly accessing other operators, and, in the absence of unbundling of the local loop (*i.e.* allowing all operators to directly connect customers at a fair and reasonable price), this is unlikely to add much to effective competition.

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Unbundling the local loop is a high priority for the PTS, which presented draft legislation to this effect in autumn 1999. This is also in line with the recommendations of the European Union that Member states should move in this direction. However, this proposal may be in conflict with the Fundamental Law on Freedom of Expression, and this aspect is being investigated by the Media Constitutional Law Commission. If a constitutional change is deemed necessary, a decision must be taken by two separate Swedish Parliaments with an intervening election, meaning that reforms could not made earlier than late 2002. In the interim, Telia has offered access to the copper network to other operators since March 2000. But the prices – SKr 1 500 per year per customer – are around 50 per cent higher than Telia's subscription charges for its own customers, which are capped because of the company's position as the incumbent operator. Once unbundling is introduced, this price ceiling would need to be reviewed and the access fees charged between operators carefully monitored.

The mobile telephone market grew by 25 per cent in 1999, a slight slowdown from the previous year, and now comprises more than 5 million subscribers. A fourth operator entered in 1999, but its activities thus far remain marginal. Despite some restructuring of subscription packages, typical costs for a small user remained static between 1997 and 1999, before falling around 20 per cent in 2000. The PTS has twice obliged Telia to reduce its interconnection charges, in May 1999 and again in July 2000, to SKr 1.13 per minute, on average.[45] Telia has appealed the latter decision. Nonetheless, this has reduced prices for end-users calling from a fixed telephone to a mobile. But interconnection fees between mobile operators have remained more or less constant.

Several legislative changes have also been made in an effort to strengthen competition:

- Since May 2000, existing operators can be forced to sell excess capacity on "fair market" terms if another party wishes to purchase it. This does not seem to have made any significant difference, and some potential participants have criticised the change for its timidity.
- In June 2000, national roaming rights were imposed. This means that a new entrant can immediately offer the same geographical coverage as existing players, without having to install its own network infrastructure to provide it, for the first seven years after a licence is granted. Opinions in Sweden are divided on the likely impact of this measure on competition.
- Number portability will apply to mobiles from September 2001, which should, in principle, promote greater competition by making it easier for consumers to switch service providers.

The broader telecommunications infrastructure is developing at a rapid pace, with the advent of ADSL (Asymmetric Digital Subscriber Line) and cable for providing high-speed Internet connection and UMTS services in the future.

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Because ADSL relies on the existing copper network, it is hampered by Telia's control of the local loop. Effective competition from other ADSL operators is dependent on addressing this problem effectively. While ADSL and cable operations provide some competition for each other, cable network infrastructure has also raised local monopoly issues. The problem is that apartment buildings are being cabled under exclusive arrangements, which eliminates choice for their tenants. These contracts with building owners have covered both the infrastructure and content services and have in some cases involved long terms. However, this is a situation where the local-monopoly risks already seem to be receding. Proprietors surveyed in 2000 appeared less willing to accept long-term exclusive agreements than they were a year earlier, and operators have in some cases shortened their standard contract duration in response to negative customer feedback. PTS has stressed the importance of opening the local copper network to allow ADSL as an important measure to counterbalance cable.

Provision of broadband Internet access *via* third-generation mobile phone services (*i.e.* UMTS) will also create alternatives to physical networks. Sweden has decided to award four licences following a "beauty contest" based on four criteria: financial capacity, technical feasibility, commercial feasibility and the applicant's experience and expertise in telecommunications. Winners will pay an annual fee equal to 0.15 per cent of turnover. The motivation for choosing this allocation method rather than an auction was a belief that it would allow more rapid development of services, ensure wide geographical coverage and hold down user costs. This approach was criticised in Sweden in the wake of huge bids made in auctions held elsewhere: it is claimed that high auction prices provide a stronger incentive to bring services rapidly on-stream, in order to recoup heavy sunk costs, while providing a windfall gain to the government. Nevertheless, the government decided to press ahead, and the winners were announced in mid-December. Telia's bid for a licence was unsuccessful and the company has filed an appeal. Judicial proceedings may yet delay the process of getting the network up and running, although notwithstanding its court action, Telia has entered into a joint-venture arrangement with NetCom, which did obtain a licence.

Distribution

A number of concerns have been raised about weaknesses in competition in the distribution sector. Food and groceries distribution, pharmacies and petrol companies all manifest shortcomings. In the food retailing area, Sweden has some unusual features when compared with the rest of the OECD:

- There are three main retail chains, ICA (35 per cent of the national market for food retail distribution), Swedish Co-operative Union (KF, 20 per cent) and Axfood (20 per cent). Each of these comprises a block of semi-independent store-owners or co-operatives linked together horizontally and integrated vertically with a wholesaler.

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- The governance structures of these groups imply that they would be difficult to take over at the wholesale level. At the retail level, ICA and Axfood have first right to the business location if the owner wishes to sell, while KF already operates most of its supermarkets directly (the rest remaining in the hands of consumer co-operative associations).
- ICA and KF will not generally supply any retailers outside their respective groups.
- In a number of less populated areas, only one or two of these groups are present in the market.
- There are fewer hypermarkets per inhabitant than in many OECD countries, with less than 10 per cent of total sector turnover in 1995, but a higher proportion of supermarkets (Table 23).

It is widely accepted that competition is weak, although the high concentration ratio is in itself a relatively poor indicator of the degree of competition. Furthermore, economies of scale from larger operations can be significant, not only through hypermarkets but also *via* streamlined supply chain practices more generally.[46] However, empirical analysis of the Swedish market suggests that lack of competition at the wholesale level is the critical factor (Asplund and Friberg, 1999).[47]

Three barriers to entry would seem to be important. First, a new wholesale supplier entering the market would need to be able to attract existing independent retail outlets and/or buy up retailers. But there are few independents left, and the current chain agreements would seem to make it difficult for existing group members to break away. A block exemption from the relevant provisions of the Competition Act makes such agreements possible. This special arrangement will expire in June 2001, after having been extended beyond its original termination date. A new Swedish block exemption for vertical restraints in general came into force on 1 January 2001, modelled on the EU block exemption on vertical restraints. The only material difference is that in the Swedish block exemption the market share limit for allowed co-operation is 35 per cent, not 30 per cent. The new block exemption will terminate by the end of 2005 and the 35 per cent market share will then be revised. Second, the corporate governance structure chosen by the chains themselves makes it more difficult for an investor to acquire the integrated operation than if they were typical companies. It is not even clear exactly to whom a potential investor would make an offer and who within the target group would be empowered to accept a bid. However, these arrangements are not ones that the authorities can do anything to address. Third, building and planning restrictions may make it difficult for building new supermarkets and especially new hypermarkets. They are administered locally, and it is difficult to establish whether this is a critical factor. One investigation carried out by the County Administration Boards[48] was inconclusive and the Competition

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Table 23. **Supermarket and hypermarket shares**

	Number of supermarkets, 1995[1]	Supermarket density, 1995[1]	Share of food retailing accounted for by supermarkets, 1995	Number of hypermarkets, 1995[2]	Hypermarket density, 1995	Share of food retailing accounted for by hypermarkets, 1995	Share of food retailing in supermarkets and hypermarkets combined
Germany[3]	9 831	12.1	32	1 100	1.4	28	60
France	7 306	12.6	27	1 074	1.9	42	69
Italy	4 253	7.4	19	182	0.3	5	24
United Kingdom	n.a.	3.4	31	733	1.3	24	55
Austria	1 764	22.0	47	83	1.0	13	60
Belgium	2 003	19.8	58	98	1.0	18	76
Denmark	900	17.3	n.a.	18	0.3	0	n.a.
Finland	1 047	20.6	47	75	1.5	13	60
Greece	1 521	14.6	67	21	0.2	0	67
Ireland	550	15.4	n.a.	3	0.1	0	n.a.
Luxembourg	n.a.	13.5	n.a.	5	0.6	n.a.	n.a.
Netherlands	n.a.	13.8	54	40	0.3	4	58
Norway	1 343	31.0	63	30	0.7	n.a.	63
Portugal	613	6.2	25	31	0.3	40	65
Spain	7 478	19.1	44	221	0.6	32	76
Sweden	**2 063**	**23.5**	**59**	**74**	**0.8**	**8**	**65**
Switzerland	548	7.8	n.a.	55	0.8	n.a.	n.a.

1. Supermarkets are defined as covering between 400 and 2 499 m^2. Density is per 100 000 inhabitants.
2. Hypermarkets cover 2 500 m^2 and more.
3. German hypermarkets in 1995 consist of 554 superstores (over 5 000 m^2) and 2038 large stores (over 1 500 m^2). The number of hypermarkets as defined in the table (over 2 500 m^2) is around 1 100. The share of hypermarkets in food retailing covers superstores and large stores.

Source: Pilat (1997).

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Authority has been asked to draw attention to any obstacles to business establishment in this sector. Nevertheless, the widespread perception that permits would be difficult to obtain may, in itself, discourage new entrants, especially foreign companies. There has been some interest expressed by leading international chains in the Nordic market, but, more than anything, this seems to have sparked defensive cross-border agreements, between ICA and Norwegian and Dutch companies, and further consolidation of the alliance between KF and its Danish and Norwegian counterparts. Another possibility is competition from Internet retailers, especially for non-perishable items. However, these companies have so far gained little more than a foothold. In Sweden's case, Internet shopping accounts for just over 1½ per cent of total retail turnover (compared with less than 1 per cent in the United States), although prices on average are more than 10 per cent lower for goods and services purchased on-line (Ministry of Finance, 2000*d*).

Two other areas where retail trade faces weak competition have been under scrutiny. F*irst*, Sweden is unusual, if not unique, in operating retail pharmacies through a state-owned monopoly, with more than 9 000 employees and a turnover of SKr 26 billion in 1999, giving it 26th place among Sweden's top 500 companies. Apoteket is the sole outlet for both prescription and non-prescription medicines and supplies, although it is not permitted to sell beauty and other ancillary products often found in pharmacies in other countries. The strong political support for this approach has recently been reaffirmed, even though pharmaceutical prices were almost 30 per cent higher than the EU average, in 1997. Without a major restructuring of these arrangements, the scope for enhancing competition is very limited. Nonetheless facilitating parallel importing for pharmaceuticals is one option being studied, although the increase in the share of such imports from 2 to 9 per cent of the market between 1997 and 1999 is estimated to have delivered relatively small savings to consumers. The pricing system for medicines within the benefit system, the fixing of Apoteket's trade margins and the reference price system are also currently being reviewed.

S*econd*, the Competition Authority claims to have found evidence, during an investigation, for a cartel including five large petrol suppliers controlling 90 per cent of the market. The authority estimates that the cartel cost consumers an estimated SKr 500 million over the last five years, and it has asked the Courts to impose fines amounting to SKr 740 million on the companies concerned. This action has not only dealt with the cartel itself but has also built credibility for the Authority as a body with the willingness and expertise to mount such an operation.

Construction

Building and construction costs are very high relative to other European countries,[49] and the gap is widening: between 1995 and 1999, costs in Sweden rose

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by more than 10 per cent, compared with 6 per cent for the rest of the European Union. There are three problem areas:

- There are only three nation-wide construction and civil engineering companies, and they have around 60 per cent of the market between them. Foreign entry may be hampered by the set-up costs of mastering Swedish building regulations (including environmental standards) and adapting to local consumer preferences, which may be high relative to the size of the market.
- There are monopoly and duopoly situations in several sub-markets for building materials. Import competition here is hampered by slow progress within the European Union towards reaching agreement on the EU Construction Products directive.
- The distribution of building materials is characterised by high concentration. However, three of the four largest companies have become foreign-owned, which has resulted in greater price competition.

Public procurement

With a large public sector, public procurement and ensuring effective competition between public and private suppliers are particularly important. The public procurement rules are clearly laid out and are in line with the EU directives. But a review in 1998 indicated that only half of all entities that were studied complied with the 1994 Public Procurement Act. Only 15 per cent of the total value of calls for tender was put out to tender in official journals (Figure 29). The Act is also weak on enforcement, as only suppliers can submit complaints – and they are unlikely to do so. A large part of public procurement is undertaken by local government and even where the authorities can establish that a problem exists, they have relatively little ability to bring recalcitrants into line, in the absence of stronger legal enforcement provisions. The Government is currently examining the scope for allowing other authorities to submit complaints, and more effective supervisory arrangements. In addition, many education, health and welfare services are simply not open to outside competition, and private-sector shares of these publicly funded services remain small (Table 24). A government commission has been established to investigate the situation and propose measures to promote fair and equal treatment between the private and public sector, and will report its findings at the end of the March 2001.

Rental housing

The rental sector in Sweden makes up around 40 per cent of the housing stock, half of which is owned by public housing companies, established by the municipalities. The philosophy underlying public housing policy in Sweden is that

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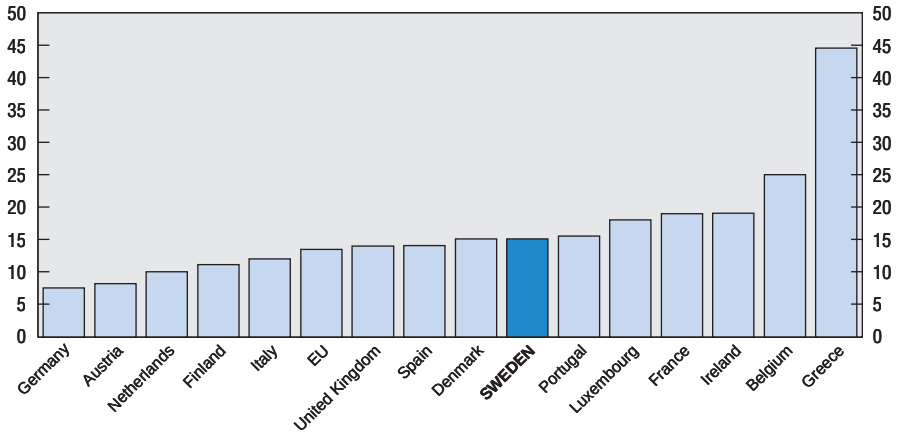
Figure 29. **Public procurement put to tender**

Value of calls for tenders published in official journals as a percentage of total, 1998



Source: European Commission, SEC(2000)1841.

Table 24. **Private-sector production share of social services**

1996, per cent

Services	Share
Childcare and schooling	
Institutional child care (pre-school)	12.5
Child care in the home (of the professional)	2.2
After-school care	4.5
Compulsory schooling	2.4
High school	1.9
Care of the elderly	
In nursing homes	8.3
In special apartments	5.1
In their own home	2.6
Medical	
Hospital care	4.3
Medical consultations	28.0
Share of doctors privately employed	10.0
Psychiatric wards	24.0
Children's dental care	5.0

Source: Werenfels Röttorp (1998).

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the state has a duty to ensure that all citizens have suitable accommodation, unlike some OECD countries, where public rental housing is provided to address concerns about affordability and equity of access to housing, based on an assessment of needs.[50] Thus, the stock of available state housing is allocated, by and large, though a queuing system. This allocation process immediately calls into question the efficiency with which the housing stock is utilised, since it is not even remotely based on tenants' willingness to pay true market rents. And once tenants have acquired a public apartment, they have little incentive to relinquish it, given the economic advantages that it confers.

A further concern is that the regulations concerning the rental housing market both reduce internal migration and also discourage building new private rental housing stock, even in areas where demand is very high, such as Stockholm, Gothenberg and Malmö:

- Although private-sector rents are officially negotiable, in practice they must be set in line with the "prevailing" normal rent level in the area, which has come to mean local public-housing rental rates. But rents for municipal properties are generally low because the stock is old, construction was heavily subsidised by the government, associated debts have been repaid and profits are not permitted. As a result, private-sector rents are suppressed and well below the construction costs of new rental housing, in particular.
- Existing tenants have strong and protected rights of possession relative to owners, and sub-let apartments are relatively common. Although the law limits the conditions and rents for sub-letting arrangements, in areas where rental housing is in short supply, the large gap between controlled rents and market rents provides a strong incentive for tenants to circumvent the rules, and profit on the deal, at the expense of the owners. Given the "grey market" nature of these transactions, it is difficult to ascertain their true extent, but recent media reports suggest rents on sub-let apartments may be more than 50 per cent above that paid by the original tenants.[51]

The framework governing the housing sector in general and rental housing in particular provides a very weak set of incentives for construction of new housing (or for that matter, renovation and upgrading quality). The currently low rate of housing construction despite apparent shortages (see Chapter I) has been attributed to a number of factors, such as planning failures by municipalities, unreasonable building code restrictions, high construction costs and a reluctance to use up green space. In addition, the tax equalisation scheme operating to redistribute revenue between local authorities means that a municipality reaps little benefit from expanding its population base. However, the lower returns to rental housing associated with the above characteristics do not seem to attract attention in the

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Swedish debate. Indeed, some municipalities have even started to sell their housing stock to existing tenants, and some 30 000 flats are expected to be sold in 2000-01, mostly in and around Stockholm.[52] The rules governing these sales allow significant windfall gains for the new owners at the expense of the municipality's other residents but should nevertheless provide some greater flexibility and reallocation. However, with the present limits on setting market rents, these apartments are unlikely to return to the rental sector.

Assessment and scope for further action

Sweden's long-term economic performance has finally picked up and structural improvements, combined with a buoyant cyclical position, provide it with an exceptional window of opportunity to make further reforms (Table 25). However, the pressure for reform seems to have dissipated now that sound public finances have been restored and activity is booming. Nevertheless, acting now on the outstanding areas where reforms are needed would ease the pain of adjustment, enhance medium-term growth prospects and attenuate the severity of the next cyclical downturn, by giving the economy greater capacity to absorb shocks and adjust. It would also give the economy a further edge in adapting to and embracing new technological opportunities as they emerge.

In essence, policies need to be oriented towards providing stronger incentives to work and to invest. Despite recent improvements, much more could be done here. Notwithstanding the recent changes, many aspects of the unemployment benefit system involve relatively weak incentives for job search. The decision to increase the effective replacement rate seems rather poorly timed since, given the current high vacancy rates, it may do more to finance leisure time between jobs than promote more efficient job-matching. The activity guarantee measure may prove to be a helpful initiative, but it could be improved by adding a time limit to participation. Otherwise, the incentive for individuals to exit from the scheme to normal employment is weakened: it amounts to an indefinite unemployment benefit. A finite period also puts greater onus on the administrators of the scheme to deliver effective programmes.

The situation for part-time workers is striking. The disparity between part-time unemployment rates and their full-time counterparts suggests that there is a serious malfunctioning of the system. It is hard to avoid the conclusion that some part-time workers are simply making the most of a generous government subsidy, which allows them to work shortened hours while maintaining higher incomes. The marginal benefits of working longer hours are very low. The generosity of these arrangements need to be pared back considerably: while several other countries allow the unemployed to work some part-time hours, none are as generous. The activity guarantee is not likely to change this situation fundamentally, as it will apply only after 27 months of unemployment. It would have been more efficient to

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Table 25. **Structural Policy Recommendations**

This table follows up on proposals from the 1999 *Survey* and summarises action taken since then, along with new recommendations arising in this *Survey*

Previous proposals	Action taken	Follow-up and new recommendations
Labour market		
Improve the wage bargaining system to increase wage flexibility.	This is considered a matter mainly for the social partners. The role of wage setting at the enterprise level has strengthened somewhat, and co-ordination seems to have improved. A New Mediation Authority was established in July 2 000. The Riksbank's inflation target has gained wider support and is likely to restrain wage increases at the aggregate level.	Continue to monitor closely the bargaining system's ability to provide aggregate wage increases in line with labour productivity trends and greater wage dispersion.
Review scope for secondary action and implement the recommendations of the Öberg report for proportionality in industrial action.	Recommendations were considered too difficult to implement and interfered excessively in the right to take industrial action.	Identify alternative ways of limiting the damage to third parties from industrial action.
Hiring and firing costs are still high and further efforts are needed to stimulate enterprises to take on new workers.	Some easing to EPL has been made. Employers can now exclude two persons from the strict LIFO order of priority, if employee retained is of significant importance to the undertaking.	Ease EPL further, for example by easing dismissal procedures and shortening notice periods. As a second-best approach, increase the number of employees exempted from LIFO.
Lower replacement rates in UI to allow the wage structure to adapt to differences in unemployment risk.	Effective replacement rates are likely to be raised, with a higher ceiling for the first 100 days.	Search for other ways of reducing effective replacement rates. Prevent any ceiling raise from being extended for a longer period or to other benefit recipients.
–	–	Reduce the generosity of part-time unemployment benefits by reducing effective benefit duration and raising the number of qualifying hours that must be worked.
Introduce an upper limit to the duration of UI benefits and ALMPs.	The activity guarantee removes the recycling element between UI and ALMPs, but the guarantee itself is open-ended.	Add a time limit to the activity guarantee, for example 1 or 2 years.
Reduce the scope and improve the quality of ALMPs.	The Adult Education Initiative has extended the scope of these programmes.	Significantly reduce the use of ALMPs outside the framework of the Activity Guarantee and ensure a sizeable overall downscaling. Enforce the Activity Guarantee rigorously, including firm sanctions for non-compliance. Evaluate programmes rigorously and eliminate poorly performing ones.

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Table 25. **Structural Policy Recommendations** (*cont.*)

This table follows up on proposals from the 1999 *Survey* and summarises action taken since then, along with new recommendations arising in this *Survey*

Previous proposals	Action taken	Follow-up and new recommendations
Strengthen basic-skills acquisition at compulsory school level.	Additional public funds have been allocated to primary schooling.	Put in place controls to ensure than funds are well-spent and make further efforts to stimulate quality improvements without increasing unit costs.
Stimulate educational innovation and ensure nation-wide education standards. Review the students' grants and loans scheme.	Grants have been made more generous. New programmes are underway for further education for teachers in science and technology and information technology.	Encourage quicker completion of tertiary education through less generous grants and promoting the acceptability of higher wage premiums for increased educational attainment.
–	–	Resist reductions in working time.
Tax and benefit policies		
Lower the top marginal income tax rate to 50 per cent when employees' social security contributions are included, while raising the threshold at which the 50 per cent rate applies and start lowering taxes on low and middle incomes.	A four-year tax-cut programme was announced in 1 999 and has been half implemented. As a consequence, marginal tax rates have been lowered for all wage earners, in particular for those with middle incomes, as the threshold for the state income tax has been increased.	Average marginal tax rates should continue being reduced, as part of a broad package of measures covering taxation of earned and capital income. In the current conjuncture, net reductions should also be matched by expenditure cuts. Broaden the tax base by restoring, and eventually increasing, the property tax level.
Introduce upper limits to the duration of sickness and unemployment insurance benefits.	Upper limits have been discussed but lack political support.	Follow through on earlier recommendation.
Develop a premium system for sickness, disability and occupational injury to reflect differences in risks between branches, enterprises and employees.	No action, although options for the addressing the surge in sickness benefits are being investigated.	Develop an experience-rating system for occupational injury and improve the incentive structure for sickness benefits by placing more responsibility on employers and employees.
Reconsider the norms embedded in the social assistance system.	Social assistance was changed in 1 998 and the government considers that no further review is necessary.	Re-examine the current norms.
Remove wealth taxation.	No action.	Follow through on earlier recommendation.
Alleviate the effects of double taxation of dividends and retained profits.	No action.	Examine ways to minimise potential distortions to the allocation of capital embodied in the tax system.

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Table 25. **Structural Policy Recommendations** (*cont.*)

This table follows up on proposals from the 1999 *Survey* and summarises action taken since then, along with new recommendations arising in this *Survey*

Previous proposals	Action taken	Follow-up and new recommendations
Product markets and competition		
Strengthen the scope of the Competition Act at the expense of the Municipal Act.	Under investigation.	Follow through on earlier recommendation, to ensure that effective competition applies to municipal entities.
Privatisation should continue with emphasis on separating natural monopolies from competitive activities.	There is no privatisation programme, *per se*, underway, although the government plans to sell a further share of Telia.	Follow through on earlier recommendation and develop a privatisation programme aimed at significantly downsizing state-ownership of market activity within a reasonable time-frame. Reconsider decision to keep Apoteket as a state monopoly.
–	–	Continue efforts to dilute incumbent power in electricity and telecommunications, and in particular, achieve "unbundling" of the local loop for telecommunications.
Remove government-induced obstacles to competition in distribution and construction	Continuing efforts are underway. The block exemption for retail trade will be allowed to expire in June 2 001.	Ensure that block exemption is abolished. Resolve the outstanding obstacles to implementing the EU directive for building materials, and encourage municipalities to facilitate competition in construction.
Review obstacles to entrepreneurship in labour-market and bankruptcy legislation. Proceed with efforts to simplify the tax code.	New legislation is being prepared to cover bankruptcy. Work to simplify rules and administrative burdens continues and the effects of new legislation of small businesses must be considered. Tax rules are under review.	Continue such efforts.
Remove the institutional blockages to more market-based rents for housing.	No action.	Implement earlier recommendations. As a first step, allow owners to charge market rents for new housing. Strengthen municipalities' incentives to supply suitable land sites for housing construction, for example through considering instituting a municipal property tax. Improve compliance with public procurement rules by strengthening sanctions.

Source: OECD.

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adapt the DELTA-commission's proposal of restricting the duration of supplementary benefits to 12 months. An easing of employment protection legislation might also help by reducing the incentive for firms to permanently scale back production by putting its workers onto shorter hours and shifting the costs onto the public.

Active labour market policies have long been a cornerstone of the Swedish strategy for addressing unemployment and improving skill levels. But the results have been disappointing given the sums involved. The present economic climate provides room for a reassessment of their purpose and effectiveness. Stricter rules and firmer sanctions in the case of non-compliance could do more to encourage individuals to search harder for work rather than enter a programme in the first place. Some valid role for ALMPs remains, but measures should be carefully targeted and evaluated rigorously: poorly performing programmes should be eliminated.

The welfare state requires a large tax base to finance it: the implication is a need for a high degree of participation and reasonably long working hours. But high average marginal tax rates discourage people from working and instead make leisure relatively more attractive. The public-finance consequences of reduced working hours, even where these are the result of voluntary agreements and not imposed by law, are often poorly understood, but negative, especially in the long run. Unless and until public expenditures can be reduced to make way for further reductions in taxes on labour, reductions in working hours should be resisted.

Taxes on capital income also need to be reviewed, especially in the light of evidence of eroding tax bases due to offshore portfolio investment. A comparison of the after-tax returns from investing in Sweden and elsewhere highlights the disincentive effects of its relatively high average effective tax rates. And if this offshore investment enables income to escape taxation altogether, it becomes even more attractive to investors. Redressing the situation would also require significant reductions in personal capital income tax rates, which by raising the after-tax return on assets would encourage saving, and also, to a lesser extent, lead to a better allocation of capital in productive investment. A sharp reduction here might not have a high fiscal cost, because it would simultaneously reduce the tax deductibility of mortgage payments as well as reducing rates on investment income from assets. However, timing would be important: since the household sector is a net debtor, it would be most appropriately managed in conjunction with reduced taxes on earned income. Furthermore, lowering capital tax rates in isolation would exacerbate the incentives to reclassify labour income as capital income in order to benefit from the lower rates. The potential distortions to the allocation of capital resulting from the double taxation of corporate profits also need to be addressed.

More efficient and responsive product markets are also important in improving Swedish economic performance. Significant progress has been made in recent years, but weak spots remain both in sectors that have been deregulated

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and in those where major changes have not yet occurred. Infrastructure services such as the supply of electricity and telecommunications are especially important because of their pervasive importance in economic activity.

The electricity generation market is set to become much more competitive, as a large proportion of the long-term contracts concluded before deregulation are due to expire soon and the Nordic electricity market is becoming increasingly integrated. Further improving cost-efficiency in local electricity distribution, however, is a difficult task, and while the increased transparency now required is a positive move, in itself it is unlikely to make a major difference. Capping tariffs is one way to constrain prices, but it has been criticised as second best as a strategy for dealing with local monopolies, because it provides insufficient incentive to companies to seek actively to maximise the efficiency gains. Another possible avenue might be to re-examine the ownership structure and consider whether moving from municipal ownership to consumer trusts might do more to ensure that any monopoly profits are returned to consumers as rebates.

The situation for telecommunications is evolving at a rapid pace. Unbundling the local loop as quickly as possible is key to adding effective competition, not only for voice transmission but also for allowing competition on broadband services between cable and copper networks. Third-generation mobile networks will add a further alternative mode of high-speed Internet access. Continued and constant monitoring of the state of competition will be necessary, particularly given Telia's persistent dominance. The sector is inevitably prone to disputes between parties over different aspects of inter-connection. These should be resolved as quickly as possible and appeals through the courts expedited in order to avoid delays to greater competition.

The distribution sector continues to display rather unusual features. Stimulating competition could be aided by more standard corporate governance arrangements in food and groceries wholesalers and distributors, although it is difficult to see what the authorities could do to facilitate this. The decision to allow the block special exemption to lapse at its current expiry date should also help. While the outcome would probably be much greater foreign presence in Sweden, the introduction of more efficient practices already in place elsewhere, together with exposure to the threat of take-over, should produce a one-off gain to consumers through lower prices for household consumables, as well as continuing advantages from greater competition and attention to the customer. Where municipalities stand in the way of allowing greater competition, their constituents will not reap the benefits. But once competition in some regions starts to generate lower prices, the real source of the lack of competition in others will become more evident.

Rental housing in Sweden is a highly distorted market that has generated serious adverse consequences for housing investment and has produced income redistribution effects that resemble those of a lottery. At the same time it is rather

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ironic that some types of "social needs" housing, usually a major focus of public housing in other OECD countries, are judged to be under-supplied. A thorough re-evaluation of rent-setting rules and their consequences needs to be carried out. A critical assessment would cover how public rental housing is allocated and what constitutes fairness, what would be the gains in allocative and dynamic efficiency by allowing market rents to be set for both public and private apartments, and whether the current favourable conditions for tenants, including sub-letting rights, strike the appropriate balance between landlords and tenants.

There are a number of other product markets where Sweden's performance is below par and further efforts are needed. In some cases central government faces strong resistance, which will require substantial efforts to overcome, for a number of different reasons. Although the state-owned monopoly, Apoteket, is a striking anomaly, political support for major change is not there, leaving scope for only minor improvements. Even so, one consequence of this seems to be significantly higher drugs prices than in other EU countries. More generally, state ownership of companies (including Vin and Sprit, and OM, owners of the Swedish stock exchange) should be reconsidered and the government's stake divested where the enterprise is already operating in a competitive market. In the building and construction area, greater efforts at the Europe-wide level to address the problems and difficulties in implementing the relevant EU directive would make it easier for foreign companies to enter the Swedish market. Municipalities could perhaps also do more to facilitate foreign companies' compliance with building regulations, but they face little incentive to do so. Public procurement is another area where the government has little leverage over the municipalities, despite their mixed record on compliance. Given the extent of publicly funded activities, strengthening the legislation to make it possible for government agencies and other parties to take recalcitrant municipalities to court and to provide stronger sanctions would both be useful modifications.

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IV. Enhancing environmentally sustainable growth

Background

Sweden's policies place a great deal of emphasis on sustainable development.[53] The environment forms one important part of this approach and is the focus of this chapter. Environmental concerns began in Sweden with nature protection in the first half of the twentieth century, and dealing with the local effects of industrial emissions became important already in the 1960s. Acidification problems were identified in the early 1970s and led to growing awareness of the international dimension of pollution. Sweden hosted the first UN Environment and Development Conference in 1972 and has been active in promoting international agreements for addressing cross-border environmental problems. These efforts have been extended to promoting international action to deal with climate change. Concern about the environmental effects of the production and use of chemicals, eutrophication and dealing with waste have also been major preoccupations of Swedish environmental policies. Its environmental performance is already good by OECD standards, and significant progress has been made in a number of dimensions (OECD, 1998*b*). Notwithstanding these achievements, the authorities remain dedicated to seeking further improvement in environmental outcomes but almost inevitably, the costs and challenges of achieving further environmental improvements rise with the progress already made.

Sweden has taken an integrated approach to environmental issues, and has recently adopted a new framework of environmental objectives. This approach has made more apparent some of the difficult choices and trade-offs that arise in reaching several different environmental outcomes – the most striking example perhaps being the difficulty of simultaneously reducing global greenhouse gas (GHG) emissions, closing nuclear power plants and renouncing further hydroelectric development. It has also helped to make the inter-linkages between policies more transparent. In light of these particular features, and the contrast with the more fragmented approach to these issues observed in some other OECD countries, the institutional settings for environmental policy-making will be discussed in some detail in the next section of this chapter.

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Although the Swedish framework for environmental policies is deliberately comprehensive, a set of specific environmental issues have been the main focus of attention and debate. This chapter will also concentrate on these concerns, organised according to desired environmental outcomes. While a sectoral approach is often adopted when discussing environmental issues in OECD Member countries, including Sweden, focusing on the goals to be achieved reinforces the importance of choosing policies that do so in the least-cost manner across different sectors. Thus, without implying that these are the only issues relevant to environmentally sustainable growth, the rest of this chapter focuses in turn on: reducing airborne emissions affecting acidification and climate change; addressing the damage from eutrophication; and dealing with waste. The final section will provide an overall assessment and policy recommendations.

Environmental policy making

The policy making process and its key players

The Swedish approach to policy-making in general could be characterised as a process of study, consultation and collective decision-making, followed by decentralised implementation, sectoral responsibility, and public information and education. There are many players in the policy-setting process (OECD, 1996), including a number of central government ministries and agencies, as well as local authorities and a range of non-governmental organisations. Public submissions and the co-opting of experts from outside the government are both ways in which a wider range of views are harnessed. The heavy emphasis on consultation and discussion means that the issues are thoroughly considered and results in relatively coherent actions rather than a fragmentation of effort. It also means that policies finally adopted usually enjoy a relatively high degree of support.

The Ministry of the Environment has responsibility for co-ordination of the government's environmental policy and the promotion of efforts to achieve sustainable development both within Sweden and internationally. Following the general structure for organising government responsibilities, the Ministry is small and deals only with policy issues, while the 13 agencies under its aegis are responsible for different aspects concerning the implementation of policies. In principle, these agencies are independent of the Ministry and are able to take decisions within the framework of the guidelines, laws and ordinances issued by the parliament and the government. The key central government agency is the Environmental Protection Agency, whose mandate is to:

- Promote ecologically sustainable development and contribute to achieving the objectives by taking on the role of co-ordinator and driving-force in environmental work both nationally and internationally.
- Compile and disseminate knowledge about the environment, *i.e.* the current situation and its development.

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- Promote sustainability in trade and industry, products, waste and infrastructure in co-operation with the sectoral authorities, as well as regional and local authorities.
- Contribute to the realisation of environmental policy and the achievement of established goals.
- Follow up and evaluate the condition of the environment and the work done to form the basis for further development of environmental policy.

Three other ministries are also heavily involved in setting environmental policies. The Ministry of Finance is especially concerned with environmental taxation and environmental programmes that involve public spending. The Ministry of Agriculture and the Ministry of Industry, Employment and Communications, have sectoral responsibilities with important environmental dimensions, in the latter case for energy and transport.

The Swedish Environmental Advisory Council, which advises the Government on environmental issues, has representatives of relevant sectors of the Swedish Cabinet Office and Ministries and provides advice to the government either on its own initiative or on issues assigned to it. It fulfils an important role in developing the government's position on the overall management of environmental issues. Most ministries have some environmental responsibilities, above and beyond the general requirement to take environmental dimensions into account in setting policies and managing their own affairs.

Local authorities also play an active part in environmental action, mostly using the framework of Agenda 21,[54] and, by the end of 1998, just over half the municipalities had adopted a local Agenda 21 plan. Initially plans focused primarily on waste and water management, together with "green" public procurement, but more recently the range of issues has expanded to include renewable energy, biological diversity, environmental management and auditing systems and sustainability indicators (Brundin and Eckerberg, 1999). Municipal boards for environmental protection and health also issue licences for small installations under the Environment Code and are charged with actively promoting nature conservation in their area. County administrative boards have major responsibility for the administration of environmental policies within their jurisdiction, including licences for medium-sized installations, and inspection also of those licences issued at the national level. They also manage national parks and nature reserves and co-ordinate policies within their areas.

Local authorities also assume a major role in Local Investment Programmes. These programmes were initiated in 1998 and have two key objectives: to reduce local environmental problems; and to create employment. They are based on the belief that the local authorities know most about the local environmental problems and conditions for sustainably-oriented development and are best suited to decide which measures are most important and useful in their local area. Project proposals

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from local authorities must show evidence of co-operation with the public, the business community and non-government organisations. In Sweden, local authorities play a major role in environmental protection and Local Investment Programmes constitute one way for the government to stimulate the shift towards an ecologically sustainable society. Parliament has now allocated SKr 7.2 billion in total to this programme, of which SKr 5 billion has already been disbursed, and the scheme has been extended to 2003. Both the environmental and economic value of these subsidies could be questioned, however, on three grounds. *First*, the employment-creation requirement may exclude some projects that would generate greater benefit for the environment. *Second*, the subsidies may be paid to projects that would have been privately funded anyway, and, third, an examination of the types of projects funded suggests that many of the environmental benefits are local and might be more appropriately funded through local revenue sources[55] (Table 26). However, the Local Investment Programmes are expected to have important effects on the environment, even at the national level, for example in reducing CO_2 emissions by almost 3 per cent. The first programmes were completed in 2000 and a broad review of the whole scheme is planned during 2001-04. Such a review should include a rigorous analysis of all the economic costs relative to the environmental and economic benefits obtained.

Environmental objectives

Environmental policies in Sweden are strongly oriented towards establishing goals and objectives. Recent efforts to identify what are the overall aims of the whole set of environmental policies are a clear strength of the institutional arrangements, particularly because they have made the potential tradeoffs between goals more transparent. In 1996 the Swedish Environmental Protection

Table 26. **Local Investment Programmes**
Funding by type of project, 1999

	Per cent
Conversion to renewable energy sources	20.5
Multi-dimensional projects	17.4
Waste management	10.2
Water and sewerage	9.6
Energy efficiency/energy saving projects	9.5
Remedial measures	9.5
Traffic	8.9
Nature conservation/biological diversity	6.3
Building projects	4.3
Other	3.8

Source: Ministry of the Environment.

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Agency identified some 170 environmental goals of various kinds that were embodied in current policy. The Agency pointed out that these goals did not fit together in a coherent fashion, and in many cases progress towards meeting them could not be monitored effectively, which made them unsuitable as a basis for developing sensible policy instruments. In response, the government developed the current framework of 15 key environmental objectives to steer policy: clean air; natural acidification only; no eutrophication; a non-toxic environment; a protective ozone layer; a safe radiation environment; high-quality groundwater; sustainable lakes and water courses; flourishing wetlands; a balanced marine environment, sustainable coastal areas and archipelagos; sustainable forests; a varied agricultural landscape; a magnificent mountain landscape; a good urban environment; and limited (influence on) climate change.

These were endorsed by the Parliament, and subsequent work has concentrated on developing proposed specific objectives, intermediate tasks, strategies and policy instruments. These were reported back to the Government in June 2000 (Committee on Environmental Objectives, 2000 and Klimatkommittén, 2000) which will soon deliver a bill to Parliament on these issues. It should be noted that these objectives are highly ambitious, and even if achieved in the most efficient manner, will inevitably be expensive. But this was seen as justified, given the parliamentary committee's valuation of the economic costs of ongoing environmental damage (excluding global warming) of SKr 20 billion per year (around 1 per cent of GDP) and their estimate of SKr 10 billion to achieve the goals, yielding a positive return to society of SKr 2 for each krona spent. Although the committee acknowledged the uncertainty of their estimates, they could in some sense be interpreted as an implicit, albeit crude, indication of the extent to which the Swedish public values the environment and is willing to pay for its improvement.

An alternative measure of the costs of sustainable development is the adjusted net national product, also known as "green" national accounts. This is obtained from the national accounts measure of net national product (*i.e.* the sum of consumption, net changes in real capital and the trade balance) plus increases in the natural capital stock, minus environmental damage. The National Institute of Economic Research has prepared estimates of adjusted net national product for 1993 and 1997 taking into account the change in natural resources, environmental damage, apart from global warming, depletion of the ozone layers, and reduced biological diversity. Their estimates indicate that Sweden's development is already very close to being environmentally sustainable, since the environmental damage and rundown of natural capital came to SKr 8.7 billion in 1997 or just over ½ per cent of net national product[56] (Table 27). While these are only estimates and preparing them has involved some difficult measurement and technical issues, they do provide a benchmark, which suggests that the policies designed to offset or avoid these environmental damages should not involve costs in excess of that amount or they would lead to an overall loss of welfare.

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Table 27. **Net national product (NNP) adjusted for the environment**
SKr million, 1997 prices

	1993	1997
Consumption	1 314.6	1 372.2
Net trade balance	78.0	161.8
Investment	209.9	244.7
Gross national product	1 602.5	1 778.7
Depreciation	–222.5	–212.6
Net national product	1 880.0	1 566.1
Increase in forestry stock	6.2	5.7
Mining	–1.2	–1.2
Natural capital erosion	–5.8	–5.4
Environmental damages	–3.4	–3.3
Adjusted national net product	1 375.7	1 561.8

Note: Natural capital degradation and environment damages are expressed in terms of real economic effects.
Source: Ministry of Finance.

Environmental policy design

Sweden has a wide range of specific environmental policies, as would be expected in a country that had many years' head start over other nations where environmental concerns have only more recently come to the fore. There are two key features of its specific policies to address environmental concerns: the sectoral approach and the emphasis on economic instruments. Despite the orientation given to setting environmental outcomes in terms of reducing environmental damage, specific policies tend to have a distinctly sectoral flavour and aim at reducing damage, sector by sector. To a large extent this reflects the logical organisation of government work and has been reinforced by Parliament's adoption of the principle of sectoral responsibility for the environment. While such an approach is both logical and efficient for most policies, it can present significant drawbacks for sound environmental policies, where several sectors are producing the same type of environmental damage because it becomes harder to ensure that marginal abatement costs be equalised across all economic actors. Although the extensive discussions that typically take place before any policy is adopted in Sweden should minimise this risk, there is no built-in mechanism that explicitly or implicitly "tests" sectoral policies to check that they deliver equalised marginal abatement costs.

A particular issue arises over exemptions or other special treatment for specific sectors, in particular the special rules for energy-intensive enterprises and the lower rates of CO_2 taxes for industry and agriculture. These are used in Sweden and elsewhere, in relation to products that are, or may be, internationally traded. The argument, put forward in support of special treatment, is that without

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exemptions, unilateral efforts to internalise environmental costs will penalise exporters and favour imports from less stringent countries relative to domestic producers, both hurting local producers and leading to an increase in global environmental damage by shifting production to enterprises and countries with lower environmental standards. But allowing tax breaks for some producers means that others must face higher taxes and the approach encourages rent-seeking behaviour. Moreover, whether leakage of harmful emissions would actually occur is essentially an empirical matter and depends on the responses across all sectors of the economy to the removal of exemptions and the production functions of alternative foreign producers given the emission reduction policies that apply in their countries.

Another relevant issue, particularly in Sweden, is switching taxation from labour to environmental taxes, especially on energy. This change is often assumed to provide a "double dividend" by simultaneously generating both environmental and economic benefits. However, this is an empirical question that depends on the industrial structure of the economy, its mix of exports and imports, and the elasticities of demand and supply for labour and energy and the existing tax structure. Results from computable general equilibrium models for Sweden suggest that making such a shift might leave the country economically worse off, even if the switch were revenue-neutral and involved lowering labour taxes, which are among the most distortionary (Swedish Green Tax Commission, 1997). For example, a doubling of the carbon tax (approximately SKr 1 000 per household) offset by lower labour taxes would incur a "welfare" loss of SKr 4 billion and reduce carbon dioxide emissions by around 0.1 per cent.[57] Moreover, subsequent work indicates that with the current set of exemptions and reductions for some sectors, such a tax swap would provide limited environmental benefits, because emissions could actually increase in exempted sectors (Bränlund and Kriström, 1997). Depending on trade elasticities, it is possible that global emissions would actually rise, as a result. Notwithstanding these analyses, the Swedish Government has decided to go ahead with such a tax switch from labour to CO_2 taxes, amounting to around SKr 30 billion (1½ per cent of GDP in 2000) over 10 years; the first SKr 3.35 billion of which is incorporated in the 2001 budget (see Chapter II). While this approach may satisfy political commitments, there is a danger that it proceeds in a piecemeal fashion that over time drifts away from sound tax principles and it would be better instead to evaluate each tax on its own merits and only raise CO_2 taxes after the exemptions had been removed.

Policy instruments

Along with its Nordic neighbours, Sweden was one of the first countries to develop and implement economic instruments. Indeed, the OECD's judgement in 1996 was that Sweden probably had more economic instruments in place than

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any other Member country (OECD, 1996). Economic instruments currently in place cover a range of incentives, including taxes and charges on emissions (*e.g.* CO_2, nitrogen oxide and sulphur), pollutants (*e.g.* pesticides tax), environmentally differentiated charges (*e.g.* fairway dues for shipping), and subsidies (*e.g.* energy technology funds and investments). Notably absent are tradable emissions permit schemes, which are more extensively used in North America. However, last Spring an inquiry into the feasibility of using flexible mechanisms recommended that Sweden immediately begin developing a trading system with other EU members and applicants, and EEA countries. A well-designed set of environmental taxes or tradable permits are both cost-efficient ways of achieving desired environmental outcomes.

Sweden also relies quite extensively on regulation and control in certain areas. This is perhaps most evident in controls over agricultural farming practices, land use more generally, hazardous chemicals, licences for certain emissions from stationary sources and waste management. The legislation governing these and other aspects of environment regulation is the Environmental Code that came into force on 1 January 1999, which replaced some 15 separate pieces of legislation. The purpose of implementing the Code was threefold: to put more emphasis on goals and results management; to provide a more coherent and integrated treatment across the environment; and to integrate more recently developed instruments, such as eco-labelling and environmental management systems, into environmental legislation. However, although the Environmental Protection Agency is charged with undertaking cost-benefit analysis of central government regulations, mechanisms to ensure that regulations applied by regional and local authorities are based on rigorous cost-benefit analysis remain lacking.

Public funding of research into environmental matters is also a high policy priority. Together with an early preoccupation with environmental issues, this has meant both ample financing and significant results. However, most of this funding has come from the taxpayer. It is generally accepted that because of the externalities involved, governments have a clear role and responsibility for funding basic scientific research, and much of the research into environmental damage and its causes clearly falls into this category and makes an important contribution to international scientific understanding. However, it is less clear whether the government should also be financing projects with commercial potential. Access to quality research about environmental damage and its causes would certainly make it easier for policy-makers to design sensible policies. However, there is also a risk that the present broader research programme might lead to an over-emphasis on technical solutions without sufficient analysis of the economic consequences and costs. Developing some best-practice guidelines on project evaluation that addressed economic considerations in a more rigorous fashion might help.

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Air quality, climate change and emissions

Sweden's policies on air-borne emissions are governed by two main concerns: acidification and global warming. These two problems are quite distinct in that while acidification affects Sweden directly, climate change affects the planet as a whole, albeit with significantly different regional effects.[58] In each case, the environmental consequences of harmful emissions on Sweden are the same regardless of where and how they originated, so the key social objective ought to be to reduce the relevant emissions in the least-cost way or to maximise the emissions reductions that can be achieved for a given cost. Choosing policies to achieve this may require stepping back from a sectoral approach and instead focusing on economic incentives to encourage those who can reduce emissions most cheaply to do so. Sweden has already achieved some notable successes in using economic instruments as well as regulation to reduce emissions, which have fallen in a number of areas. It should also be noted that since the beginning of the 1990s efforts have been made to achieve cost-efficient reductions in global and regional emissions by directing a major share of development aid to countries in the Baltic region and eastern Europe with large emissions. The emphasis on environment and enlargement as key priorities during Sweden's presidency of the EU is part of the same strategy. Table 28 shows the targets together with the reductions in emissions achieved thus far.

Acidification

Acidification of soils and waterways is caused by emissions of sulphur, nitrogen oxide (NO_x), ammonia and volatile organic compounds. It is a particular problem in Sweden because of a combination of abundant precipitation and soil composition. Sweden's low average critical load (*i.e.* where the level of deposition exceeds nature's ability to compensate, and environmental damage to soil and water ensues) has required an extensive programme of liming to offset the damage, at an annual cost of more than SKr 200 million, around 85 per cent of which is publicly funded. The most difficult policy challenge for Sweden in reducing damaging emissions arises from the essentially transboundary nature of the problem. In 1997, for example, more than 123 000 tonnes of sulphur was deposited on Swedish soil, of which only 12 000 tonnes originated in Sweden itself (see Figure 30). Meanwhile, total Swedish emissions amounted to only 34 000 tonnes, one-third of which was deposited at sea. Clearly then, effective action requires international co-operation.

Sweden has been working actively since the first UN Environmental and Development Conference in 1972 towards international action to deal with the problem. The Convention on Long-Range Transboundary Air Pollution was agreed 20 years ago, and protocols have contained quantitative reduction targets since 1985 (Ministry of the Environment, 1999). The most recent achievement has

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Table 28. **Emissions: achievements and targets**

	1980	1990	1995	Latest	Targets
	1 000 tonnes				
Emissions of SO_x	508.0	136.0	94.0	91.0	Previous target:
of which:					– Emissions in 2000 equal to only one-fifth of the level prevailing in 1980 (already achieved)
Mobile sources	44.0	37.0	24.0	24.0	
Stationary sources	464.0	99.0	70.0	67.0	Proposed new target:
					– By 2010, decreased by at least 25 per cent from 1 995 levels, to 72 000 tons
	1 000 tonnes				
Emissions of NO_x	448.0	388.0	354.0	337.0	Previous target:
of which:					– Emissions in 1995 to be 30 per cent below 1980 levels (achieved in 1997)
Mobile sources	313.0	315.0	286.0	274.0	
Stationary sources	135.0	74.0	63.0	63.0	Proposed new target:
					– By 2010, decreased by at least 55 per cent from 1995 levels to 155 000 tons
	Million tonnes				
Emissions of CO_2	77.1	55.4	58.1	57.0	Previous target:
					– Emissions stabilised at 1990 levels by 2000 (not achieved)
Emissions of GHG[1]	–	69.4	68.4	73.8	Kyoto/EU burden-sharing target:
					– Emissions for greenhouse gases shall be 4 per cent higher than 1990 levels on average from 2008-12
					Proposed new national target:
					– Emissions for greenhouse gases shall be 2 per cent lower than 1 990 levels on average from 2 008-12

1. Greenhouse gases in CO_2 equivalents.
Source: Swedish authorities.

been the Göteborg Protocol, signed in December 1999, which deals with four types of pollutant and specifies more stringent targets. If the agreed targets for emissions reduction under this agreement are met, only 4 per cent of Sweden's surface area will receive more than the critical load by 2010, compared with 17 per cent in 1990.[59] In principle, the Protocol is designed so that those parties whose emissions have a more severe environmental or health impact and whose emissions are relatively cheap to reduce will have to make the biggest cuts. While it remains up to each country to determine its own set of commitments, parties are encouraged to use economic instruments in order to promote the search for least-cost abatement strategies.

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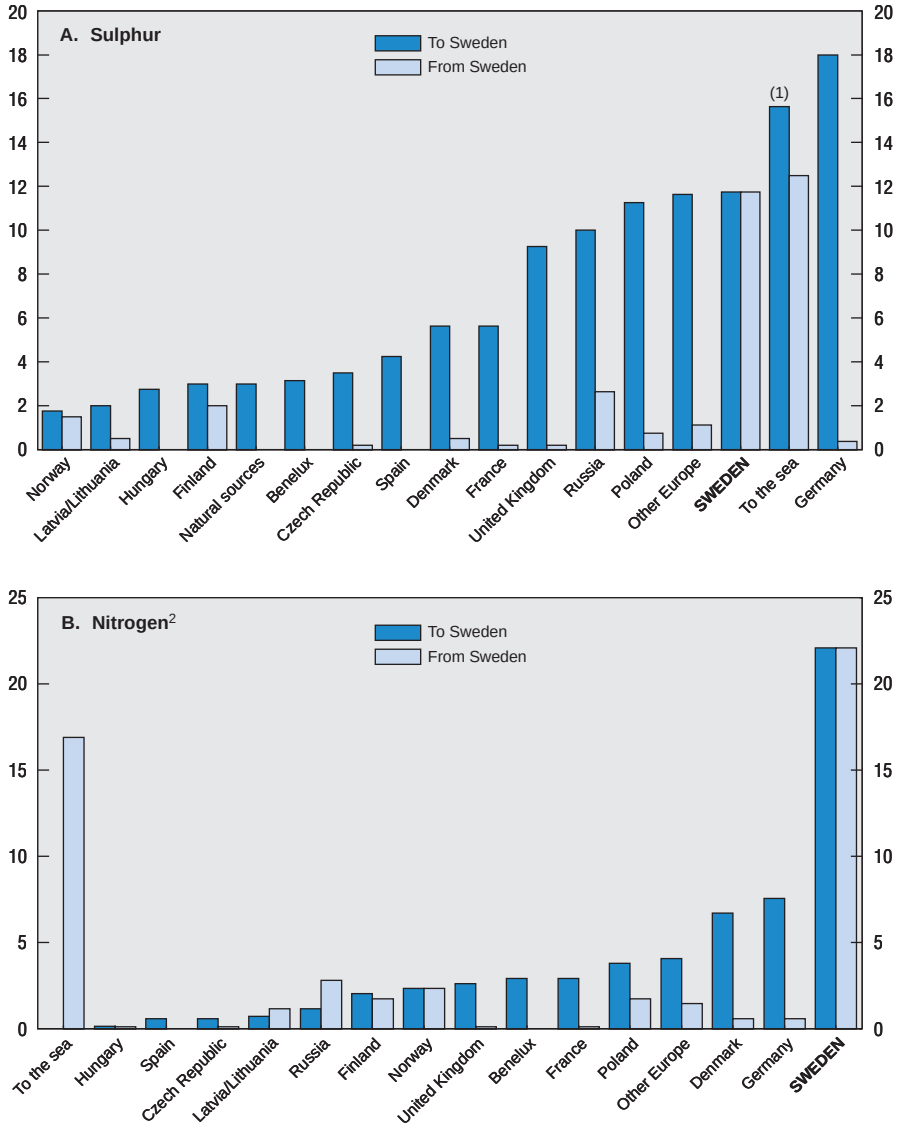
Figure 30. **Exports and imports of air-borne pollutants, 1997**
Thousand tonnes



1. Emissions from shipping.
2. Reduced.

Source: Environmental Protection Agency.

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Swedish emissions of sulphur have dropped dramatically since 1970, in large part due to the shift away from fossil fuels, and emissions have fallen further since the sulphur tax was introduced in 1991. NO_x emissions have also fallen, but more slowly, and the NO_x charge, imposed since 1992 on large point-sources, seems to have played an important role (Box 8). It should be noted that the variety of sources of emissions makes it impossible to link all emissions to a single stable tax base and thereby design a single economic instrument that in itself will equalise abatement costs (Figure 31). Instead, distinctions must be made between large stationary sources (for whom cap-and-trade systems or taxes can work equally well) and mobile emitters. Indeed, the guidance document on

Box 8. Sweden's NO_x charge: an effective economic instrument

Sweden's NO_x charge on large stationary plants – SKr 40 per kg of NO_x emitted – is an example of an effective economic instrument. Steps to reduce emissions started once the bill was passed by Parliament in June 1990, even though the charges applied only from 1992. Between 1990 and 1995, emissions from plants in the scheme dropped on average by 60 per cent per megajoule of energy generated. Total emissions, however, showed a smaller decline, because the energy generated had increased by 25 per cent over the same period.

In 1996 and 1997 the coverage was expanded to include all installations producing more than 25 GWh of useful energy per year, and many of these newly-covered plants had higher emission rates, which has meant that average emissions have remained stable over more recent years. Currently some 400 units are covered by the charge.

The charge is not a tax, since the total paid is returned to the payers in proportion to their share of the total of useful energy produced. It is administered by the Environmental Protection Agency, whose costs of less than 0.5 per cent of revenue are deducted from the pay-out. The structure of the charge means that any producer with emissions lower than the industry average will receive net benefits, while those with higher emissions face a net cost.

However, there are drawbacks. Measures to reduce NO_x emissions can result in higher emissions of other air pollutants: increased ammonia in the case of flue gas cleaning, and increased emissions of nitrous oxide and carbon monoxide, where improvements are made in the combustion process. An assessment made by the EPA indicated that the cost of reducing one kilogram of NO_x amounted to SKr 19, of which more than half was the cost of damage through increases in other harmful emissions. Nevertheless, the same assessment still valued the gross socio-economic benefit per kilogram at SKr 40, implying a significant net benefit from the measures.

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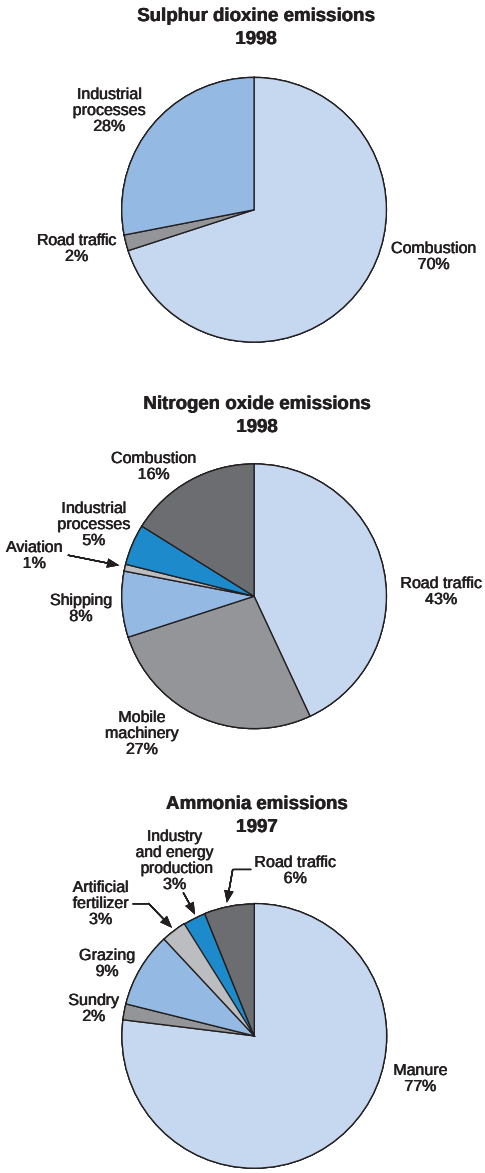
Figure 31. **Sources of emissions for sulphur, NO_x and ammonia by sector**
Per cent



Source: Environment Protection Agency.

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economic instruments associated with the Göteborg Protocol analyses the applicability of different economic instruments to different types of emitters and indicates their current use in various countries (Table 29). In the case of small point-source emitters, no country has yet put in place a workable economic instrument, and regulation may be the only option, although even here it would have to be balanced against compliance costs.

The first-best option at this point would be to persuade other countries to do more to reduce the damage they inflict on Sweden. Given the domestic reductions already achieved and the high proportion of imported depositions, the abatement costs for reducing emissions in other countries, especially in Eastern Europe, would almost certainly be lower than domestically. However, although Sweden has continued its efforts at the international level since the Göteborg protocol was signed, the key policy challenges now involve reducing its own emissions in three main areas: Baltic shipping, off-road vehicles and machinery, and farm manure.

Baltic shipping now contributes more than half of the sulphur depositions in Sweden. The sulphur content of shipping fuel in the region remains around 4.5 per cent, despite the 1997 International Maritime Organisation's agreement to reduce it to 1.5 per cent.[60] Shipping also contributes to high emissions of NO_x. Since these vessels are operating in international waters, achieving reductions can be difficult and require co-operative efforts, and Sweden is actively pursuing the issue with other Baltic states and the EU. It also has instituted a system of differentiated fairway dues (*i.e.* charges imposed to cover the costs of services such as navigational aids and ice-breaking) in order to provide economic incentives for reduced emissions from 1999 (Box 9). A number of Swedish ports also use differentiated harbour fees, although the incentives are somewhat weaker, since ports are required to cover costs and are competing with each other. These economic instruments do seem to be attractive to shipping companies, since within one year of implementation 65 per cent of annual ferry tonnage and 30 per cent of cargo tonnage were using low-sulphur fuel[61] (although some, especially coastal, vessels had already been using low sulphur fuel). The NO_x-related part of the discount has taken longer to produce results, since it requires significant capital investment. But Sweden provides a subsidy to any ships calling at its ports that install NO_x abatement equipment, by reimbursing the fairway dues paid during the first five years after 1 January 1998. Since fairway dues in aggregate are currently set to cover full fairway costs, this reimbursement approach may prove to be costly.

Off-road vehicles and machinery (including in agriculture) currently account for around 20 per cent of NO_x emissions. The main approach being taken here is regulatory, particularly through EU directives, in line with the government's objective of further reducing emissions. Further measures could include even stricter

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Table 29. **Economic instruments for sources of air-borne pollution**

	Tradable permits and quotas	Emission and process taxes/charges[1]	Product taxes and tax differentiation	Subsidies and fiscal inducements
NO_x: (large) point sources	– Emissions trading: United States	– Incentive charge on emissions: **Sweden**		– Emission related: Netherlands[2] – Energy related: Austria, Canada, Denmark, Germany, Netherlands, Norway, Poland, **Sweden**, United Kingdom – Industry related: Canada, France, Germany, Greece, Netherlands, Poland, Portugal
	– "Internal bubbles": Denmark, emissions: France, Italy, Poland, Netherlands	– Financing charges/taxes on emissions: France, Italy, Poland, Slovakia, Switzerland		
SO_2: large point sources	– Emissions trading: Poland, United States – "Internal bubbles": Netherlands, United Kingdom	– Financing charges/taxes on emissions: Czech Republic, France, Italy, Poland, Slovakia, Spain	– Taxation of fuels differentiated according to sulphur content: Belgium, Denmark, Finland, France, Norway, Portugal, **Sweden**, Switzerland, United Kingdom	– Energy related: Austria, Canada, Denmark, Germany, Netherlands, Norway, Poland, **Sweden**, United Kingdom – Industry related: Canada, France, Germany, Greece, Netherlands, Poland, Portugal
VOCs: large point sources	– Emissions trading: United States	– Financing charges/taxes on emissions: Czech Republic, France, Poland	– VAT reduction for low-solvent paint: Czech Republic, Slovakia	– Energy related: Austria, Canada, Denmark, Germany, Netherlands, Norway, Poland, **Sweden**, United Kingdom
		– Financing charge on processes: Poland	– Tax on solvents (as from 2000): Switzerland	– Industry related: Canada, France, Germany, Greece, Netherlands, Poland, Portugal

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Table 29. **Economic instruments for sources of air-borne pollution** (*cont.*)

	Tradable permits and quotas	Emission and process taxes/charges[1]	Product taxes and tax differentiation	Subsidies and fiscal inducements
Ammonia: large point sources		– Financing charges/taxes: Czech Republic, Poland		– Industry related: Canada, France, Germany, Greece, Netherlands, Poland, Portugal
NO_x, SO_2, and VOCs: mobile sources	– Only used on the urban scale: Cracow (Poland), Singapore	– Environmentally motivated road pricing: Austria	– Lower taxes on "cleaner" vehicles and/or fuels: Austria, Belgium, Canada, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Netherlands, Norway, Portugal, Slovakia, Spain, **Sweden**, Switzerland, Turkey, United Kingdom, United States	– Investments: Hungary, Poland – Car scrapping schemes: Ireland, **Sweden**, Norway, France, United Kingdom
Ammonia: agriculture	– Emissions trading ("offsets"): Netherlands[3]	– Charge on surplus manure: Belgium, Netherlands	– Charge/tax on N-fertiliser: Austria,[2] Finland,[2] Norway, **Sweden**, United States	
SO_2: small point sources	– None	– None	– None	– None
VOCS: small point sources and products	– None	– None	– None	– None

1. Excluding non-compliance fees.
2. Abolished.
3. Sub-national level.

Source: United Nations/ECE (2000).

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Box 9. Economic incentives to reduce emissions from shipping

Shipping produces two types of emissions: sulphur, from burning high-sulphur fuel, and nitrogen oxides, from any form of combustion. Sweden has introduced a system of differentiated fairway dues to provide economic incentives to reduce them.

Sulphur

- Sulphur emissions can be reduced by switching to low-sulphur fuel, without any modification to equipment. However, such fuel is more expensive, and the marginal cost increases as the sulphur content falls. In late 1999, for example, the price of high-sulphur bunker fuel was around $130 per tonne; for 1 per cent sulphur content the price was about $10 higher, and with 0.5 per cent sulphur content an additional $20 was demanded.
- For international shipping, imposing a unilateral tax on sulphur content would lead ships to refuel elsewhere, making it ineffective as an economic instrument. Such an approach would only work if it were imposed on a joint basis, and even then leakage may occur outside the region.
- In a number of northern European countries, including Sweden, Norway, Finland and Denmark, there exist some forms of fairway dues, designed to cover the costs of providing safety-related services such as navigational aids and ice-breaking. However, only Sweden comes close to cost-recovery for the services provided.
- Sweden's fairway dues are based on the ships' gross registered tonnage (GRT) and on the amount of cargo, and the environmental differential applies to the GRT component. Standard fees since the beginning of 1998 have been SKr 5.30 per GRT for oil tankers and SKr 5.00 per GRT for ferries and other ships. Ship-owners who continuously operate ferries on fuels using sulphur content less than 0.5 per cent and other vessels on less than 1.0 per cent receive a discount of SKr 0.90 per GRT.

Nitrogen oxides

- Since nitrogen oxides (NO_x) are a by-product of combustion in ships' engines, the options for reducing emissions are to modify the motor or to treat the exhaust gas. There are a number of technical solutions, all involving some capital investment.
- A discount is also given on fairway dues for low-NO_x emissions, gradually increasing for emissions below 12 grams/kWh to a maximum discount of SKr 1.60 per GRT for emissions of less than 2 grams per kWh.
- To speed investment, the Swedish Maritime Administration is also reimbursing a portion of the fairways dues paid by ships installing the necessary equipment, over the five years to 2002.

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Box 9. **Economic incentives to reduce emissions from shipping** (*cont.*)

Differentiated fairway dues as an economic instrument

- The Swedish system is not perfect, since it operates only on ships that actually call regularly at its ports. Baltic shipping outside of Sweden inflicts considerable amounts of domestic damage, but the present system provides little incentive for ships that call rarely at Swedish ports to modify their behaviour. International co-operation, at least within the Baltic region, where short-sea shipping accounts for almost 90 per cent of all port calls, would clearly be more effective, and Sweden is actively working towards this.
- A further limitation of the Swedish system is that fairway dues are not linked to the distance travelled, although the environmental damage clearly is. However, distance-based charges would make the system much more complicated to administer.
- Given these constraints, the Swedish system does seem to be effective, and it has been successful in encouraging shippers to modify their operations in order to reduce environmental damage.

Source: Kågeson (1999).

exhaust standards, incentives to accelerate the introduction of cleaner equipment and encouraging the installation of abatement equipment, such as catalytic converters. According to one study, these measures would involve lower abatement costs than some of the other measures already in place for emissions reduction (Ågren, 2000). They would help to reduce greenhouse gas emissions as well.

Farm manure is the main source of ammonia, with 90 per cent of all ammonia emissions coming from agriculture. Farming techniques play a significant role here. Practices such as turning manure into the ground within four hours can significantly reduce air emissions, but many add to risks of nitrates leaking into aquifers. Application times and corresponding storage capacities are also relevant. These complex relationships may make it difficult to design an effective and reliable economic instrument based on manure volumes or herd numbers. It would also be difficult in practice to measure emissions at the farm level, making an emissions-based instrument impractical. Given these constraints, regulation and education may be the only effective way of changing farmers' behaviour, although realistically, without economic incentives, they are unlikely to voluntarily make more than minor changes to production structures and practices.

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Regulations on the storage of manure were introduced in 1995 and have contributed to reductions in ammonia emissions. As it happens, Sweden's adoption of the EU Common Agricultural Policy (CAP) in 1995 led to structural changes that have reduced cattle numbers in any case. This has eased this particular source of environmental stress (Swedish Board of Agriculture, 1999), although the overall effect of the CAP on the environment is far more complex and beyond the scope of this *Survey*.

Greenhouse gases and global warming

Sweden's emissions of CO_2 are insignificant on a global scale, amounting to just 0.4 per cent of OECD emissions and a mere 0.24 per cent of the world total, while its emissions of methane and nitrous oxide are also minuscule by global standards. On a per capita basis, Sweden releases around 6 tonnes of CO_2 per year, compared with around 9 tonnes for the European Union as a whole and more than 20 tonnes for the United States. It is therefore clear that, acting on its own, Sweden can make very little difference to global outcomes. However, Sweden has been an active leader in seeking collective action to confront and address climate change and strategies to deal with it at an international level, through international agreements on reducing greenhouse gas (GHG) emissions and then sharing the burden of achieving the targets laid down in them. Under the Kyoto Protocol Sweden, along with all other EU member states, agreed to reduce GHG emissions (expressed as CO_2 equivalents) to 8 per cent below 1990 levels by 2008-12. In the burden-sharing arrangements agreed within the European Union since the Protocol was signed, Sweden negotiated an increase in emissions of 4 per cent above 1990 levels, arguing that it had less scope to reduce these emissions than other EU countries, given the already exceptionally low reliance on fossil fuels for electricity generation, leaving it with a higher proportion of other fossil-fuel uses, such as transport, that are more difficult to cut (Figure 32). The decision to phase out nuclear power plants by 2010 was another factor taken into account.

Most OECD countries have found that GHG emissions have continued to grow since 1990, and that, to achieve their targets, significant reductions from present levels will be required. In Sweden's case, CO_2 emissions also fluctuate significantly with the weather, since reserve generating capacity and marginal electricity supply mainly use fossil fuels, while oil is an important fuel for heating. For example, the cold and dry winter in 1996 drove emissions up to 11 per cent above 1990 levels whereas in 1998 they were only 6 per cent higher (Figure 33). Recent high economic growth will have almost certainly spurred an increase in emissions. Two government agencies have recently made projections of emissions in 2010 based on present policies: the National Institute of Economic Research estimates emissions will be 15 per cent higher than 1990 levels, while the National Energy Administration predicts only a 5 per cent rise (Klimatkommittén, 2000).

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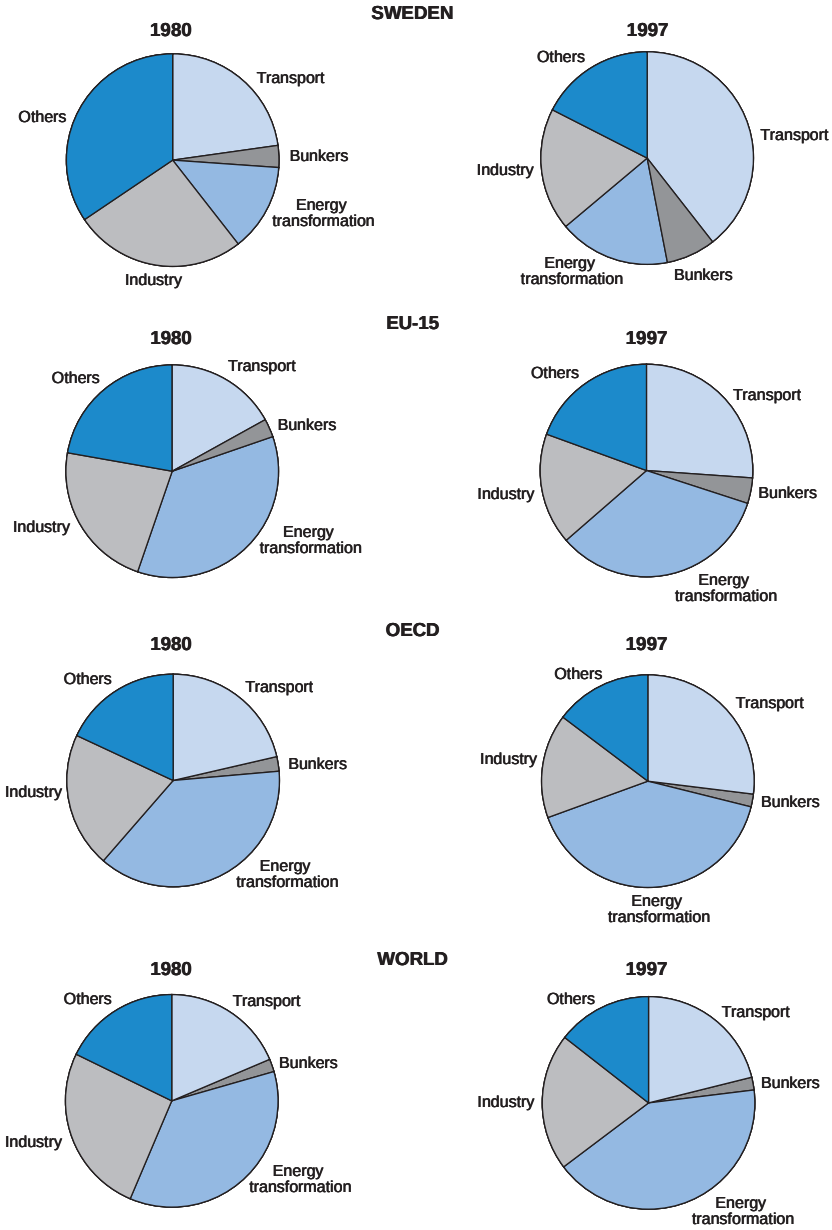
Figure 32. **CO_2 emissions by source**



Source: OECD Environmental Data 1999.

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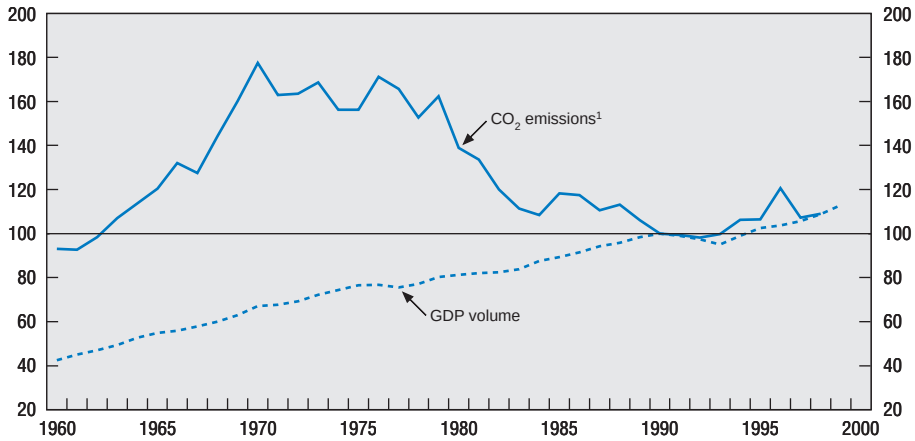
Figure 33. **CO_2 emissions and GDP growth**
1990 = 100



1. CO_2 emissions from combustion of total fossil fuels.
Source: Environmental Protection Agency; International Energy Agency (1999); OECD.

Policies to reduce CO_2 emissions

Sweden already has in place a set of policies designed to reduce GHG emissions. The main emphasis is on economic instruments (*i.e.* taxes and subsidies), although Swedish research has also highlighted the influence on motor vehicle emissions of such factors as road characteristics, engine temperature and driving practices. Energy taxes have a long tradition in Sweden. Petrol and alcohol-based motor fuels were first taxed in 1929, and the general energy tax was established in 1957 for purely revenue-raising reasons. It was only in the 1970s that energy-policy objectives became a supplementary justification for the taxes. Sweden first introduced a CO_2 tax in 1991 in conjunction with a reform to energy taxes. Joining the EU necessitated some further changes, and the present tax structure was put in place in 1995.

The present tax structure comprises three elements: an energy tax, a CO_2 tax and a sulphur tax (Table 30). The energy tax is levied on fuel oil, coal and natural gas, but electricity generation and industry are exempted. Electricity consumption is also taxed, but again industry is not charged. This tax is also differentiated, with lower rates in North Sweden than in the rest of the country. Specific taxes also apply to nuclear and hydro-electric generation.[62] The CO_2 tax is based on the carbon content of the fuel, and industry and agriculture pay only one-third

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Table 30. **Energy and CO_2 tax rates**[1]

SKr, 1 January 2000

Product	Energy tax	Carbon dioxide tax		Total tax	
		Industry	Others	Industry	Others
Petrol *per litre*					
Environmental Class 1	3.61	0.86	0.86	4.47	4.47
Environmental Class 2	3.64	0.86	0.86	4.50	4.50
Other	4.27	0.86	0.86	0.59	0.59
Gas oil, kerosene, heavy fuel oil[2] *per* m^3	743	529	1 058	1 272	1 801
Liquefied Petroleum Gas (LPG) *per tonne*	145	556	912	701	1 257
Natural gas and methane *per* 1 000 m^3	241	396	792	637	1 033
Coal and petroleum coke *per tonne*	316	460	920	776	1 236
Crude tallolja[3] *per* m^3	1 801	529	–	2 330	1 801
Electricity[4] *per kWh*					
Manufacturing industry and commercial greenhouse cultivation	0	0	–	0	–
Other consumption in certain areas, mainly in northern Sweden	0.106	–	0.0121	–	0.0121
Electricity, gas, heating or water supplies in areas other than above	0.139	–	0.0159	–	0.0159
Other consumption	0.162	–	0.0185	–	0.0185

1. Value Added Tax and sulphur tax may also apply.
2. Marked oil. Marked oil is used in stationary motors and ships for heating purposes. Higher rates are paid on consumption of "unmarked" oil used for the propulsion of motor-driven vehicles and boats.
3. Tallolja ("pine oil") is a byproduct from the pulp and paper industry that can be used as a fuel.
4. Electricity consumed in big electrically heated boilers (> 2 MW) is taxed at higher rates during November-March. Relief from energy tax is granted if the electricity is produced in a certain manner or used for certain specific purposes, *viz.* produced in a wind power station, produced and consumed on board a craft or other means of transportation; used in connection with the production of electricity; produced in a reserve power station; if electricity is produced with other fuels than taxed fuels and the electricity is used by the producer himself for the supplies of electricity, gas, heat or fresh water. Energy and carbon taxes on fuels are exempted when the fuels are used for the production of electricity.

Source: Ministry of Finance.

of the rate paid by other consumers.[63] The sulphur tax is applied to emissions where these are measured, or the sulphur content of oil used. In both cases, the tax is uniform across all users. Together all these taxes generated revenue equivalent to 2.7 per cent of GDP in 1999, half of which from the energy tax.

Sweden's tax regime raises three key policy issues: the interaction between energy and emissions-based taxes raises questions about the appropriate level and mix of environmental taxes on energy; the variation in taxes for different users of energy means marginal abatement costs are not equalised across different economic activities, implying that emissions may not be reduced in the most cost-efficient way; and the desire to encourage a particular mix of energy supply, within certain constraints, may render superfluous the signals produced by economic instruments.

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Appropriate level and mix of taxes

As noted above, energy taxation was originally introduced to raise government revenue. It gradually acquired "green" credentials for encouraging energy conservation. CO_2 and sulphur taxes are rather more directly oriented towards reducing environmental damage. However, what matters for influencing behaviour are the overall prices for different kinds of energy. Notwithstanding the various taxes applying to electricity, prices are amongst the lowest in the OECD, although motor fuel prices are well above average (Figure 34). But the impact of these prices on the choices that consumers make relative to the environmental externalities involved are quite different. Taxes based on energy content discourage all energy use, relative to other inputs, while the emissions-linked taxes affect the relative prices of different forms of energy, reflecting not only their production cost, but also their relative environmental impact.

The effects of energy taxes on overall energy use have been mixed. Significant efficiency gains have been achieved in housing, cars, appliances, and industrial processes, but energy intensity has fallen more slowly in Sweden than elsewhere in Europe (Figure 35), suggesting that some of the improvements are being offset by higher energy consumption. For example, improvements in fuel economy of cars have been offset by longer distances travelled and larger vehicles. Also households have a widening range of appliances. (These results are entirely consistent with the income and substitution effects arising from any improvements in technology. However, the level of energy intensity in different countries will also affect the scope for changes.) While higher energy prices have probably spurred the search for greater efficiency, it is hard to identify how much difference it has made, and whether the social benefits (*i.e.* over and above the private benefits that individuals reap from more energy-efficient machines) justify the dead-weight losses associated with the tax. In any event, Sweden is able to generate electricity at low average variable cost and with very low emissions because of the high proportion of hydro-electricity and nuclear energy.[64] In light of this, it is not clear that the energy tax plays a useful role in a climate change strategy, since it serves to make energy in general more expensive, regardless of the environmental damage it generates. In any case, the energy tax system has evolved over the years into a complicated and uneven set of arrangements and in its 2001 budget, the government announced reforms, clarifying its objectives as being to promote energy efficiency and raise government revenue. In the long run, the energy tax should be simplified and the Government's ambition is to move in the direction of aligning taxes to energy content, including that of biofuels.

CO_2 taxes are more tightly focused on environmental damage, and there seems little doubt that Swedish CO_2 taxes have reduced emissions. According to estimates carried out for the Swedish EPA, emissions in 1994 were 5 million tonnes (9 per cent) lower than they would have been in the absence of the tax.

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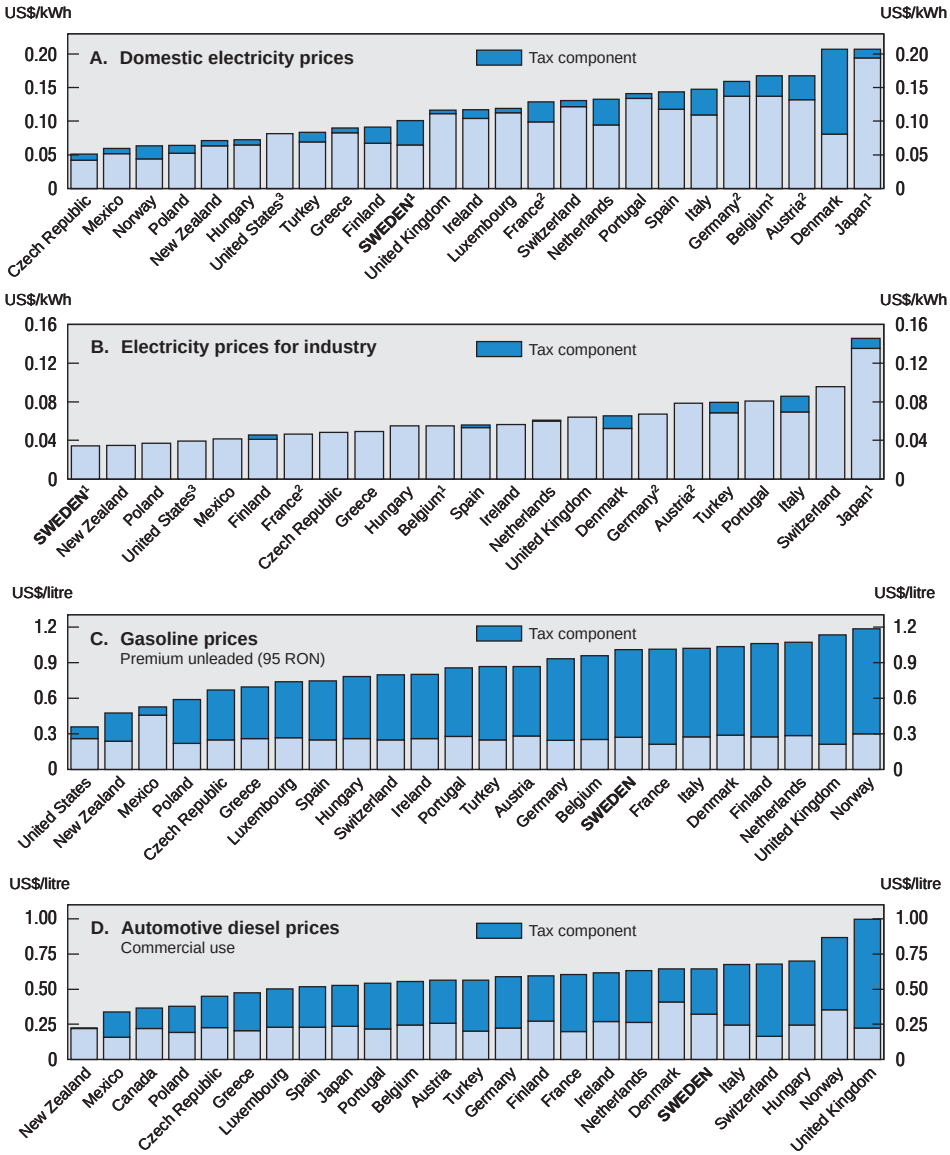
Figure 34. **Energy and fuel prices in OECD countries, 1999**



1. 1997.
2. 1998.
3. Ex-tax price for the United States.

Source: International Energy Agency.

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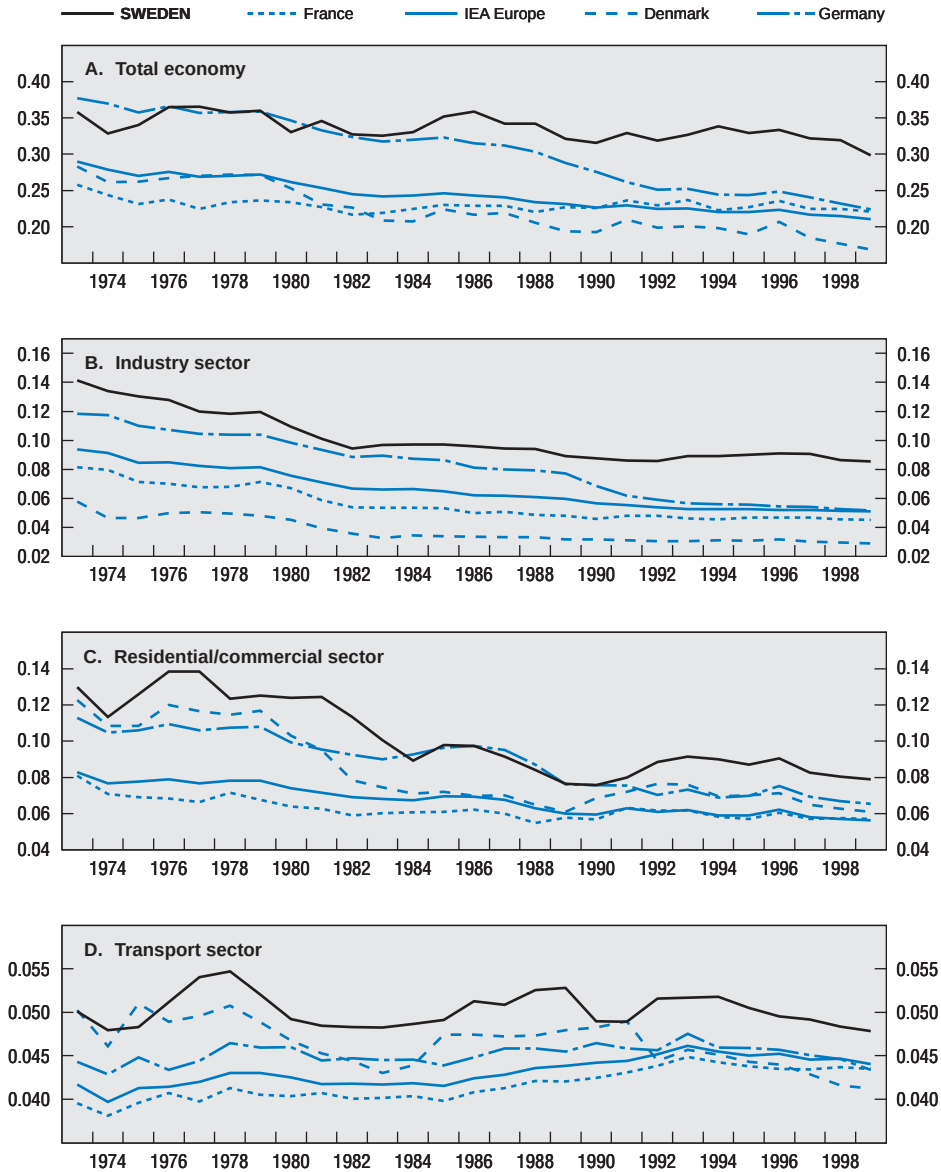
Figure 35. **Energy intensity in Sweden and other IEA countries**

Toe per thousand US$ at 1990 prices
and purchasing power parities



Source: International Energy Agency (2000).

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The district heating sector underwent the greatest adjustment to the tax. The resulting substantial price differences between heavy oil and biomass fuels[65] produced a significant switch towards biomass, which rose from 25 to 42 per cent of all district heating supplied between 1991 and 1995.

The issue at stake for CO_2 taxes is whether they are set at the right level. The basic principle for any environmental tax is that is it should internalise the negative externalities. But in this context of climate change, there is a great deal of uncertainty about what the "right" global CO_2 tax would be. Nevertheless, Sweden is committed to the Kyoto Protocol, which provides an international framework with agreed quantitative emissions targets: in this context, the appropriate CO_2 tax for Sweden is simply the one that achieves the necessary national reductions (although the Kyoto Protocol does not specifically require domestic reduction). Given the projected evolution for Swedish emissions, it is clear that CO_2 taxes would need to rise from present levels, and simulations suggest that CO_2 taxes would need to be 2½ times higher than 1998 levels (*i.e.* rising to SKr 0.91 per kg CO_2 emitted) in order to lower emissions to the 4 per cent increase agreed under the burden sharing arrangements (Nilsson and Huhtala, 2000): the 2001 budget incorporates a rise in the carbon tax to SKr 0.53 per kilogram of CO_2 gas emitted. However, how high taxes will need to rise in order to meet Kyoto commitments will depend on the extent to which "joint implementation" and emissions trading are to take place within the Kyoto framework. OECD modelling work suggests that if international emissions trading succeeded in equalising marginal abatement costs across the Annex B countries[66] it would reduce the economic costs of meeting the targets by one-third (OECD, 1999*b*). It should be noted, however, that reducing CO_2 emissions may not be as cost-effective for Sweden as reducing other GHG emissions or increasing forests as sinks. It is important, therefore, that the strategy on climate change takes all these into account. A clear and positive outcome of the negotiations within the Conference of the Parties to the UN Framework Convention on Climate Change, and full ratification of the Kyoto Protocol would give Sweden a much clearer basis on which to establish the appropriate tax levels or move to domestic, and eventually international, emissions-trading permits. A further complication arises to the extent that an increase in CO_2 taxes leads to a rise in all electricity prices, including those for hydro-electricity and nuclear energy. This would generate an economic rent for these two groups since their supply capacity has been fixed by law. It would be appropriate to apply a resource tax to such economic rents.

Equalising abatement costs

The disparity of tax rates among users is a significant feature of the Swedish taxation regime for both the energy and the CO_2 tax. Energy products used in industry and agriculture are both exempt from energy tax and pay only one-third

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of the general CO_2 rate. In addition, special rules apply to energy-intensive industries, which play a particularly strong role in the Swedish economy. The pulp and paper industry alone consumes one-third of all electricity used by industry, and energy typically makes up almost 15 per cent of marginal costs in newsprint production from virgin pulp. Altogether, some 60 enterprises, mainly producing iron and steel, chemicals or pulp and paper, are able to benefit from preferential arrangements for energy-intensive operations, which apply if the CO_2 tax paid at the normal industry rate would be greater than 0.8 per cent of sales. The marginal tax burden for these firms is reduced to 12 per cent of the rate paid by industry in general.

In Sweden, low electricity prices are a key source of competitive advantage, but the country has chosen to impose higher environmental taxes than most other countries and in advance of concerted international action to address global warming. Thus, the Swedish authorities have chosen the present arrangements to deal with the consequences of acting alone. The concern is that if industry and agriculture faced the full tax rates, they would rendered less competitive against production from less environmentally-conscious countries, even if they would be the most efficient producers in a situation where all countries were making similar efforts to reduce emissions. At the same time, if Swedish production was displaced by other countries, total harmful emissions might even rise, although the extent to which this would occur is an empirical matter.[67] The Swedish authorities argue that if a uniform rate was applied, this competitiveness consideration places a low ceiling on it. However, such a low rate would in turn involve only a weak incentive for other users to reduce emissions. But from an economic perspective, the current differential arrangements effectively provide subsidies to particular users, with the most intensive energy users benefiting most, which leads to an inefficient allocation of resources. In addition, the emissions reductions that are achieved carry a higher cost than otherwise, because the marginal costs of abatement are not equalised across the economy. While these problems should largely dissipate once the Kyoto Protocol is implemented, in the interim, better arrangements would promote the least cost reduction in emissions while compensating industry for the temporary economic losses that would result. Within a tax-based system, one solution would be to impose the same tax rates on all users, but grant an ongoing tax credit equivalent to, say, the additional environmental taxes paid in the first year that businesses faced the full rates. Under a tradable emissions permits scheme, existing emitters could be allocated permits equivalent to their current emissions, a so-called grandfathering arrangement.

A further issue is whether marginal abatement costs are the same for the two main sources of CO_2 emissions in Sweden: energy and transport. Sweden's CO_2 tax per kg of emissions is the same for transport as for elsewhere, except for the reductions and ceilings that apply in industry. Energy tax has long been applied to motor fuels for fiscal reasons and 80 per cent of taxes on motor fuels are energy

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taxes. It could be argued that these are more closely related to CO_2 emissions than energy taxes elsewhere, and on those grounds the *de facto* carbon taxes could be considered as higher on transport. But the analysis is complicated by the other externalities associated with transport, including congestion, local pollution and accidents, and the degree of spillover between policies to address each of these issues. A comprehensive cost-benefit approach taking into account all the externalities associated with transport might justify higher abatement costs in this sector than would be rational on the grounds of global warming alone, because of those ancillary benefits (OECD, 2000*d*). Indeed, a comparison of the energy taxes on petrol and diesel with external costs, excluding CO_2 emissions, showed that the energy tax was significantly higher than externalities for petrol cars but neutral or lower for diesel, depending on traffic (Swedish Green Tax Commission, 1997). Fuel taxes might not be the ideal way of internalising some of these costs; the roles played by other related policies such as the shift to unleaded petrol, catalytic converters and car-scrapping schemes are important. However, the role of these taxes as instruments for raising revenue also needs to be taken into account. The government has recently announced a major review of the total effects of all traffic-related taxes on the environment as a whole as well as their role in raising revenue to cover the economic costs of the road network, which should bring all these elements together in a coherent framework. Taxes related to these ancillary benefits will be separated from energy taxes on fuel.

Energy supply

A major feature of Sweden environment policy is the strategy concerning energy supply. A state-owned enterprise, Vattenfall AB, owns half of the country's electricity-generating capacity, while municipalities own almost a quarter. As part of its environmental objectives Sweden has renounced further hydroelectric development and has also committed itself to phasing out nuclear energy, despite the potentially high costs associated with this decision.[68] All methods for electricity generation have advantages and disadvantages from an environmental point of view. In terms of low operating costs and GHG emissions, both existing hydro-electricity and nuclear energy have clear advantages.

Official policy is that nuclear energy can be replaced by greater use of renewable energy sources together with gains in energy efficiency that translate into reductions in energy intensity. Given the weight of nuclear energy in current electricity production, this seems rather optimistic, especially if such a switch is to be realised over a relatively short period of time. In fact, Sweden's nuclear policy has evolved significantly since the original referendum in 1980, which led to the sequence of decisions culminating in the 1997 change in legislation that led to the closure of one reactor (Barsebäck 1 in November 1999). The Swedish parliament has now agreed on stringent conditions to be met before future closures are

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accepted, including that of Barsebäck 2, which was due for closure on 1 July 2001. In essence, the requirements to be met before Barsebäck 2 can be closed are that it must be possible to compensate the loss of electricity production from the reactor by an increase in electricity supply from other energy sources, and a reduction in electricity use. As a result, it seems unlikely that this plant will close in the immediate future. The government has announced that, after the closure of both Barsebäck reactors, the developments in, *inter alia*, electricity prices, investments, environmental impact, employment, functioning of the electricity market and R&D, are to be followed up. The conclusions should then act as a guide to future changes.

The Swedish authorities have placed a lot of emphasis on renewable sources of energy, especially biofuels, wind power and solar energy. A government-funded energy research, development and demonstration programme to promote an ecologically-sustainable energy system was established at the beginning of 1998 with funding of SKr 5 billion over seven years. Biofuels (*i.e.* wood and forest-product wastes) already provide around 15 per cent of Sweden's total primary energy supply, but most of it is used within the forestry sector. However, its use in district heating has been vigorously promoted by the government, and combined heat and power plants (CHPs) using biofuels have benefited from government-funded research programmes and investment subsidies. But although district heating has the advantage of reaping economies of scale, its promotion may have regulatory implications, by establishing *de facto* local monopolies (IEA, 2000). Biomass is also being developed as an alternative source of motor fuel, through production of ethanol, biogas-based methane, dimethyl ether and additives for petrol and diesel oil. Total government funding into research in these areas amounts to around SKr 250 million spread over several years.

While wind power has been judged to offer a potential for supplying up to 10tWh, current production is negligible, and it seems unlikely that significant expansion would take place without heavy subsidisation. The Danish experience with promoting the development of wind energy (OECD, 2000*e*) suggests that the cost could be high, even though technological improvements are reducing unit costs. Average production cost on turbines installed in Denmark in 1998 was around SKr 0.36 per kWh compared with a spot price in the Nord Pool electricity market that year of only SKr 0.120/kWh and a hydro generation cost of 0.01 kWh (Figure 36). Furthermore, transmission constraints limit the siting of wind farms to within around 150 km of users, unless they are large enough to justify investment in high-voltage transmission capacity (Cassedy, 2000). Although considerable government-funded research has gone into developing Swedish-designed large-scale wind-power technology, as the IEA has noted, it has not yet led to commercial output. Research into solar energy, through photovoltaics and artificial photosynthesis, is at an earlier stage and is not likely to provide any generation capacity within the next 10 years.

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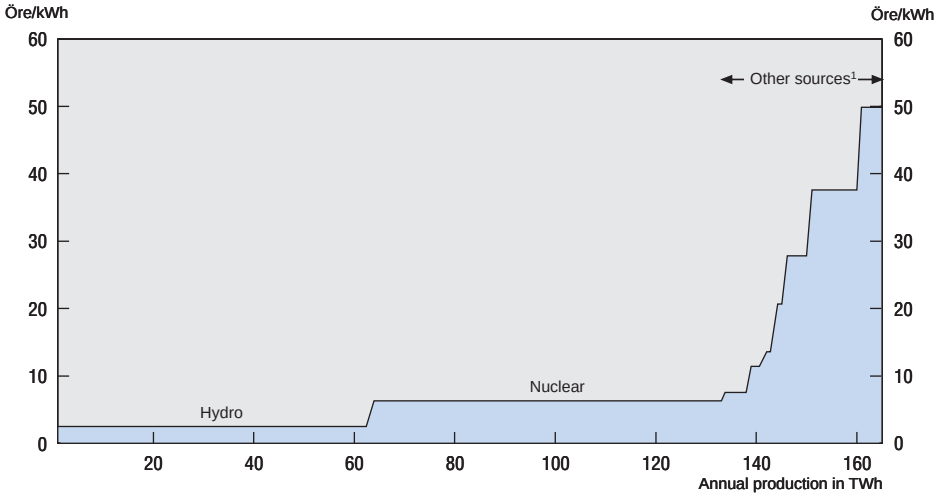
Figure 36. **Electricity generation capacity and variable costs in Sweden**



1. Combined heat and power, oil condensing and gas turbine sources.
Source: OECD (1999).

Strategy for addressing global warming in the future

Notwithstanding all the efforts already made towards reducing global emissions, both through promoting international action and by adopting economic instruments to curb domestic emissions, the parliamentary commission charged with examining the environmental objective of "limited climatic impact" has recently recommended that Sweden should adopt a long-term objective of a 50 per cent reduction in greenhouse gases from 1990 levels by 2050, with further reductions beyond that point. As an intermediate target, it proposed that emissions should on average be 2 per cent lower over the period 2008-12 than in 1990,[69] *i.e.* 6 per cent lower emissions than agreed within the European Union, in order to meet its targets under the Kyoto Protocol. Despite cross-party support within the Parliament, this recommendation has nonetheless drawn criticism from many economists. It should be recognised that this report reflects a laudable desire to do more to reduce global warming. But, by concentrating on reducing Swedish emissions further, it would reduce the global environmental gains that could be achieved for a given domestic economic sacrifice through greater efforts to reduce emissions in other countries where the marginal abatement costs would be lower, either through flexible mechanisms provided in the Kyoto Protocol, or through separate aid and assistance outside that framework.

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The commission proposed a number of measures. It stressed the importance of economic instruments, CO_2 taxes and/or emissions trading systems, as essential instruments if the targets for 2008-12 are to be met, but argued that given the uncertainties at an international level, it may not be possible to take clear decisions before 2003-04. In light of this, it has drawn up a set of base measures that could be implemented immediately. In fact, however, most of the almost 100 measures, are either statements of intent or reflect a continuation of existing policy rather than specific policy changes. The measures to finance these proposals are perhaps more likely to affect the environment, since the commission proposes to increase the electricity consumption tax, index the CO_2 tax increases to GDP growth, impose differentiated sales tax rates on new cars based on their CO_2 emissions, and increase the vehicle tax. In any case, experience with economic instruments does suggest that using price signals alone might achieve many of the changes that the commission wants to see, in a rather more direct manner and at a lower cost than the myriad of other proposals put forward. Alternatively, a domestic cap-and-trade system for GHG emissions could be instituted immediately, and integrated into international-trading schemes as soon as these become operational. And while in principle cap-and-trade systems or correctly set taxes will produce the same environmental results for the same economic cost, there may be practical advantages to permit trading systems, not least because emissions reductions are fixed and then the appropriate "price" determined, whereas getting taxes right depends on trial and error. International trading of emission permits would in turn make it possible to achieve reductions in emissions at a lower cost than if only national schemes are instituted.

Water quality

Sweden has abundant water supply: it has low annual per capita abstractions of around 300 m^3 (compared with the US level of 1 870 m^3) and one of the lowest intensity of abstractions among OECD countries. The main emphasis is therefore on preserving its quality and avoiding pollution of waterways. The problem of acidification has already been discussed above, but eutrophication[70] is also a major concern. Wastewater has progressively been subject to increasingly extensive chemical and biological treatment to reduce damage from organic material and from phosphorus. Now, the main source of nutrients is nitrogen that comes from artificial fertiliser and farm manure (although air-borne emissions also contribute). These have not only affected Sweden's waterways, they have also added to eutrophication in the Baltic sea. Furthermore, nitrogen has found its way into drinking water, with around 100 000 people now dependent on drinking water which contains higher nitrates than the Swedish health limit, while several municipal waterworks have had to be closed for the same reason.

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Policies to reduce nitrogen

Agriculture is the main source of excess nutrients, with emissions from crop production and livestock management either through nitrate leaching[71] or through ammonium evaporation. Structural shifts towards larger farms, greater specialisation and more intensive agriculture over the years have exacerbated the environmental problems. Ironically, these trends have been encouraged by agricultural support policies, as the OECD had pointed out to Sweden already in the early 1990s (OECD, 1994) even before Sweden adopted the EU CAP. The significance of artificially high agricultural output prices for environmental outcomes can be seen in modelling work for Denmark, which estimates that a 10 per cent reduction in EU cereal prices would lead to changes in both output and input mix, resulting in a 2.4 per cent fall in nitrogen loading, without any change in environmental programmes or incentives (Wier *et al.*, 1999). A sensible starting point for reform would, therefore, be market-based prices, but although Sweden has strongly advocated CAP reform, it is dependent on progress made at the EU level.

In the meantime, policies to address the environmental damage from agriculture have centred on trying to engineer a switch to less damaging farming methods, through legislation, information and subsidies. For example, Sweden has tightened its regulations limiting the proportion of arable land that can be left fallow during autumn and winter, since newly ploughed fields release more nutrients, whereas crops can absorb nitrogen. Organic farming is also actively encouraged, with a sharp increase since 1995 when less than 4 per cent of farm outputs were organic: the goal of cultivating 10 per cent of total acreage organically by 2000 is likely to be met. Funding for this switch is coming from the EU's agri-environmental programme of compensation, which in total costs SKr 2.8 billion per year (half of which comes from the Swedish budget). Organic livestock farming by adapting the animal density to the farm's capacity to produce fodder, was expected to reduce large animal concentrations and thereby lower nutrient leaching. But this amounts to saying that it only works by reducing inter-farm trade. In any case, recent evidence suggests that organic farming has not reduced the nutrient leaching load.

Fertiliser taxes have been in place since 1984, but seem to play a relatively small role in the strategy for dealing with nitrogen leaching from agriculture. Current tax rates are SKr 1.80 per kg of nitrogen and SKr 30 per gram of cadmium, for cadmium exceeding 5 grams per tonne of phosphorus. (There is also a charge on pesticides of SKr 20 per kg of active ingredients.) The relationship between fertiliser application and nitrogen leeching is in any case complex and depends both on crop mix and farming methods. The first-best approach, taxing the nitrogen leaching directly, would involve high measurement costs. A "next-best" approach could involve taxing on estimated mineral losses (residual balances) for

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each farm. Both Denmark and the Netherlands already require farmers to keep detailed records of nitrogen application and absorption, a key requirement for developing such a taxation system. The Dutch authorities have noted that minerals accounting does not involve much extra paperwork for farmers, as the relevant input and output values are already closely monitored for financial reasons. (It does, however, involve significant implementation costs for the government, and monitoring and control costs would rise if the accounts were used as a tax base.) In the Netherlands, however, a standard allowable loss rate per hectare is applied to each farm and a tax applied to any excess loss. From an economic point of view this would not necessarily lead to the most efficient minerals use. Either taxing all nitrogen losses or allocating loss permits to each farmer and allowing them to trade would be more likely to result in an equalisation of marginal costs of nitrogen reduction.

Another approach that has been suggested recently in Sweden and may have some merit would be to shift from the present regulatory and input-oriented system to an incentive system based on farm performance against a set of environmental outcomes. This would provide a means for farmers to optimise their activities with more freedom. An environmental charge system which redistributed income from those with poor environmental performances to above-average ones, as operates with the NO_x charge, bears further investigation, although defining a workable aggregate indicator of environment performance in agriculture may be far from straightforward.

Solid waste management

There are basically three types of waste generated in Sweden: production waste, hazardous waste and general waste from households and businesses. Production waste is generally homogenous and produced in Sweden by a few large producers in the mining sector, iron and steel industry, forestry and food production. The waste generated is generally managed at source. In the case of the mining sector, which generates the bulk of this waste, the Swedish Foundation for Strategic Environmental Research (MISTRA) is currently funding a major project to find ways of mitigating the damage associated with mining waste. A large proportion of hazardous waste also results from industrial processes in a relatively small number of enterprises. Some 30 companies have licences for on-site treatment and management. All municipalities are responsible for collecting and handling household hazardous waste, such as batteries and paint cans, while around half of Sweden's municipalities have voluntarily established facilities for handling industrial hazardous waste. Some hazardous waste is transported directly to the national centre for hazardous waste treatment at Kumla in central Sweden, while some hazardous waste is treated in specially dedicated plants. For example, the SAFT recycling plant deals with 97 per cent of the industrial

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nickel-cadmium batteries disposed of in Sweden, as well as imported batteries for recycling. Costs are covered by a specific charge levied on environmentally hazardous batteries.

To facilitate the separation of hazardous waste from general household waste, municipalities often have agreements with petrol stations, pharmacies and paint dealers to provide easy collection points for the public. They also run special campaigns to collect hazardous household waste to help keep it separate from the general selection system. The importance of these measures is reinforced by the leaching of toxic chemicals from landfills, which has occurred in several places, because earlier standards were not stringent enough. However, non-hazardous waste, generated by households and businesses, is the main focus of the efforts to reduce waste, where the strategy involves encouraging recycling through extended producer responsibility and minimising the use of landfill.

Extended producer responsibility

Extended Producer Responsibility, which is part of a broader Integrated Product Policy (IPP),[72] is a cornerstone of Swedish environmental policy. Originating with the Agenda 21 Action Plan for Sustainable Development, adopted in 1992, it has been put into place more extensively in Sweden than any other OECD country. Producer responsibility was introduced for packaging, waste paper and tyres in 1994 and for cars in 1997. It will apply to electrical and electronic products from July 2001. Sweden is also actively promoting IPP within the European Union, arguing that this approach is underpinned by the basic principles laid down in the Amsterdam Treaty and would "support an efficient single market that provides effective safeguards for human health and environment" (Government Communication to Parliament, 2000).

The key principle underlying producer responsibility is that resources should be conserved and recycled. Regulations oblige each producer to take back their products and recycle a certain proportion of them. They are required to set up collection systems to manage the process and inform and encourage the public to co-operate, although legislation also forces households to separate waste and take it to producers' collection points. For products such as tyres and cars where there are few producers and the items are large and infrequently disposed of, sectoral responsibility could work well as a means of internalising costs. However, for packaging and electrical and electronic products, there are many more producers and diverse products. Given the cost of establishing collection systems, the producer responsibility principle provides a deliberately strong incentive to co-ordinate efforts within the sector, as has happened in Sweden.[73] The costs of collection are then shared among producers on the basis of units of packaging

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type produced. It should be noted that in the Swedish case importers of products are classed as producers for the purposes of this legislation. In some countries this approach might raise concerns about the wisdom of encouraging firms within a sector to co-operate so closely, in view of the risks to competition more generally. It may also make it more difficult for new firms or importers to enter the market (regardless of their overall environmental credentials). These potential drawbacks do not seem to be considered important in Sweden and would anyway need to be balanced against the economies of scale and density that favour unified collection systems.

Another key feature of the Swedish approach is its emphasis on products rather than materials. This approach might seem to suggest that plastics used in packaging, for example, are considered more harmful to the environment than plastic in other items such as toys. In fact, the priority has been to deal first with products where producer responsibility can most easily be instituted, before moving on to deal with those for which it is harder to set up collection systems. In any case, if the concern were about harmful health effects of the products, then it would be more logical to tax the harmful agent, regardless of the products it is embodied in, while concerns about resource depletion would also argue for an explicitly materials-based approach, such as could be provided by environmental taxes. But these have both advantages and drawbacks (see Annex II). If instead the intention is to reduce landfill volume *per se*, then an incentive structure designed to address that more directly would make more sense, while encouraging least-cost reductions of waste to be adopted.

Of even greater consequence however is the strong assumption underlying this policy that recycling is always worth undertaking, regardless of the economic cost. No cost-benefit analysis was undertaken before these policies were adopted in 1994, and only one assessment has been undertaken since then (Radetzki, 1999). That analysis notes that the producer responsibility strategy has been successful according to its own objectives, namely raising the level of recycling, but judged that it has been achieved at a high cost. For packaging waste, the marginal cost of recycling packaging waste is estimated at SKr 2 220 per ton, which is paid by producers and presumably passed back to consumers via higher prices. This contrasts with SKr 1 500 per ton for incineration and SKr 1 200 for landfill. In addition, there is a significant additional cost to households and firms in sorting, cleaning and transporting packaging waste to the collection point. On the assumption that each household spends an extra half-hour per week, these costs could add up to SKr 1 660 per household over a year, bringing the total costs to society of recycled packaging to an estimated SKr 34 000 per ton and to SKr 6 400 for recycled newsprint. The critical conclusions of this analysis have spurred the Swedish Environmental Protection Agency into carrying out its own economic evaluation, the results of which should be available in 2001.

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The aforementioned study highlights the degree of complexity embodied in the Swedish approach. In several OECD countries recycling is carried out as part of the municipal rubbish collection system and requires relatively little sorting effort by households. Allowing co-mingling of recyclable materials has also been shown to increase overall recycling rates (Judge and Becker, 1993). Additional sorting and sale of recyclable material takes place centrally, in response to economic signals – *i.e.* the waste handling company can easily assess whether the extra costs of recuperating specific materials can be justified by their sale. Economic instruments to promote recycling in general or of particular materials could easily be incorporated into this system by shifting relative prices, and recourse to targets and regulations would not be necessary.[74] Another study found that packaging consumed by households fell by only 4 per cent after the introduction of the green dot scheme in Germany (Rousso and Shah, 1994). A further aspect called into question is whether recycling can be taken too far. For Finland, one study showed that the optimal recycling rate was between 30 and 50 per cent, while only cardboard and metal demonstrated net positive social benefits from recycling: glass and paper did not (Huhtala, 1997). Another analysis using US data found an optimal recycling rate of only 7 per cent for solid waste (Palmer and Walls, 1997). While the conclusions of these studies cannot be automatically applied to the situation in Sweden, they do illustrate the importance for policy of a thorough assessment of the producer-responsibility rules already in place. The government has initiated an evaluation of producer responsibility.

Landfill minimisation

Sweden's goal is that only a small fraction of waste will be landfilled over the long term and has set itself the intermediate goal that landfilled waste (excluding mining waste) should be 50 per cent lower than 1994 levels by 2005. To help achieve this objective, no combustible waste should be sent to landfill after 2002 and no organic waste after 2005. Sweden is not alone in adopting this waste hierarchy approach, which treats landfill as inherently the least desirable option from the environmental and public health points of view. But, as pointed out by the European Commission (European Commission, 1999) landfill, composting and incineration all have environmental draw-backs, and careful assessments are required to establish which disposal method is optimal and under what local and technological conditions. One US analysis for example, indicates that, with the upgraded federal regulations on landfill since 1976 and modern land-fill management techniques, the external costs can and have been largely internalised, as reflected in a doubling of average disposal costs (Beede and Bloom, 1995). A further complication in setting a policy of incinerating all possible materials is that it requires installing more sophisticated equipment at incineration plants than would need be necessary if certain of them, particularly PVC products, were sent to landfill instead. This significantly increases either the economic cost or the

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environmental damage of incineration if the equipment is not installed (Brown *et al.*, 2000), although Swedish incineration plants already comply with strict emissions standards. More generally, the relative economic cost of alternative disposal methods does not feature anyway in the waste hierarchy approach, which takes into account only environmental considerations.

A more promising development is the consideration of differentiated waste collection rates and treatment charges and a tax on landfilling waste of SKr 250 per tonne, which has now been implemented. These economic instruments would provide an incentive to waste generators to modify their behaviour, while ensuring that those who can cut down on waste for the least cost do so ahead of others, thus achieving an efficient economic outcome. It should be noted that the largest benefit of waste reduction is actually economic, through the savings in resources used in collection and disposal, but this only becomes fully apparent if landfill is subject to a tipping fee that covers full costs (Dewees, 1998).

Unit-based pricing for household waste collection has been implemented in a number of OECD countries, either charging on a volume basis (*e.g.* by requiring that pre-paid bags of different sizes be used) or on a subscription basis, where households pre-pay for the right to have a certain volume collected each week, whether or not this volume is generated. Increasingly, albeit more rarely, weight-based charging systems have been implemented by local authorities in Sweden as well as in Australia, Canada, Denmark and the United States. These come closest to marginal cost pricing, but they also involve significantly higher collection costs and are more difficult to adapt to multi-household dwellings. However, various empirical studies in the United States show that demand for solid waste disposal is relatively price inelastic, so that fee increases have done relatively little to reduce the weight of garbage, though volume-based systems have induced greater home compaction of rubbish (Kinnaman and Fullerton, 1999). It could also be expected that recycling rates would rise, but the empirical evidence is inconclusive. However, studies have shown that illegal dumping and burning have risen where unit pricing schemes have been introduced.

Conclusions and policy recommendations

Taken overall, Sweden's approach to environmental policies has a number of obvious strengths, but there are also some notable weaknesses and areas where policies could be improved. Among its strengths are the clarity of its objectives and the degree of discussion and debate that has taken place in establishing them. And there is no doubt that environmental issues play a central role in the policy choices made, unlike in some OECD countries, where environmental policies are not really integrated into the rest of the policy framework. Another feature is the extent to which the authorities have given economic instruments a role to play in achieving environmental outcomes and are willing to consider extending their use. However, there is also a tendency in Swedish policy to emphasise

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sectoral responses to produce particular results, for example with agriculture, producer responsibility, energy and transport. While the motivation behind this approach is understandable, the wish to steer a sector towards developing in a particular way may actually compromise the search for the best environmental results as well as incurring higher economic costs than necessary.

The energy sector provides a case in point. Here, as well as the use of economic instruments such as the energy, CO_2 and sulphur taxes, the authorities are actively promoting renewable energy development and energy efficiency, while banning further expansion of emissions-free capacity in hydroelectricity and reducing nuclear generation. However, the effective deregulation of the Nordic electricity market has reduced electricity prices (see Chapter III), making efficiency measures and alternative generation technology relatively less attractive. The economic costs associated with reaping the environmental benefits from these policies has risen, and they have thus become harder to justify. Against this background, policies to address climate change should focus directly on emissions, since they cause the environmental damage. Taxes on electricity generated from emissions-free sources and promotion of energy efficiency do not seem to have an obvious role in a climate-change strategy, especially where capacity constraints are self-imposed. Indeed the environmental damage associated with electricity consumption from low-cost renewable sources is relatively hard to value and local in nature.

Sweden is committed to meeting its Kyoto targets, although on current projections of emissions it is clear that significant further efforts will be needed. Essentially the options are much higher CO_2 taxes or shifting to a cap-and-trade system of permits. Without one or other of these, the targets look clearly unattainable, while either of them should provide the right signals to general economic efficiency improvements in the use of fossil fuels and should encourage the development of alternative energy sources. The extent to which Sweden has to reduce its own emissions in order to reach its commitments under the EU burden-sharing arrangements depends on the final form of the ratified Kyoto Protocol. The greater the scope for using international emissions-trading, Joint Implementation and the Clean Development Mechanism in meeting Sweden's targets, the lower the overall costs are likely to be. If Sweden wished to reduce emissions even further than required under Kyoto in the interests of making a greater contribution to reducing global warming, using these same instruments would enable Sweden to make a larger environmental contribution than it could by cutting its domestic emissions even further, as was suggested by the Climate Change Commission.

Sweden faces all the challenges of dealing effectively with transboundary pollution in its efforts to deal with acidification and its associated damage: the recently signed Göteborg Protocol represents a significant milestone in international co-operation. While it is hoped that this agreement will be fulfilled and countries deliver the reductions promised, it will be necessary to monitor progress carefully and maintain the pressure on them to comply with their

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commitments. In the meantime, Sweden has been doing its best to address those factors where it can directly influence the amount of acid rain that falls on its territory. The recent introduction of economic instruments to reduce emissions from Baltic shipping is already showing good results and would be even more effective if coverage could be widened. However, regulations may be the only option for off-road machinery and farm-based emissions, where the challenge is rather to ensure that these regulations are well designed, efficient and cost-minimising.

Dealing with the fall-off in water quality due to eutrophication is a home-grown problem, with nitrogen the main pollutant and farming the culprit. Here, despite the use of an economic instrument, namely the fertiliser tax, the main policy efforts seem to be misplaced. Regulations on farming methods and, more recently, significant subsidies to promote organic farming have been the main tools. These approaches are not only costly, they also overlook the alternative of designing and implementing a more effective economic instrument. Following the example of some other OECD countries, a tax on the nitrogen outflow from farms could provide a much stronger incentive to farmers to tackle and reduce their net nitrogen use. A shift to more organic methods would be one possible outcome, but farmers themselves could judge whether this was the most cost-effective route, given all factors, including the demand for goods qualifying as organically produced. The alternative approach mooted recently of shifting to a more hands-off approach to agriculture with incentives for overall environmental outcomes merits further study.

Sweden's policies concerning solid waste management are an area where economic analysis seems to have been rather overlooked and cost-benefit assessments of policies and alternatives have rarely been carried out. Instead, the policies are based on an environmental impact studies and a "waste hierarchy" approach that forms the cornerstone of the Swedish Government's waste policy. Nevertheless, the decision to ban landfill of combustibles and organic waste should be reconsidered, and a full analysis, taking into account the economic cost as well as the environmental damage of different waste management strategies should be carried out. Meanwhile for recycling and extended producer responsibility, the Swedish government has already initiated an evaluation which should include a thorough assessment of the overall costs and benefits, drawing not only on Swedish experience but also comparing it with alternative approaches to recycling used elsewhere, especially those which involve less effort on the part of households. A report is expected by July 2001.

Looking across the whole range of policies and specific measures examined in this chapter, the nation would benefit from closer scrutiny of the economic costs associated with achieving its environmental objectives. This would enable policies to be more clearly and constructively assessed, both in relation to the net environmental benefits expected to accrue, and in comparison with alternative policies intended to achieve the same outcome. Specific recommendations for action are presented in Box 10.

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Box 10. Recommendations for further action

Environmental policy-making

- Ensure that rigorous cost-benefit analysis is systematically used to evaluate environmental policy options.

Acidification

- Reinforce the economic incentive to install NO_x abatement equipment on Baltic shipping by raising fairway dues on those ships operating without the equipment, to cover some of the cost of the subsidy to convert (with the revenue raised used to offset the reimbursement of fairway dues over five years for those ships that invest in abatement technology).
- Ensure that for those areas where regulation is the only practical solution (off-road machinery and vehicles and farming), the approach adopted is subject to rigorous cost-benefit analysis and can demonstrably generate net benefits compared with the absence of regulations.

Climate change

- Restructure energy and carbon taxes so that marginal abatement costs are equalised across the economy while taking appropriate measures to address competitiveness concerns, such as tax credits or grandfathered tradable emission rights.
- Consider the possibility of using a domestic cap-and-trade approach to meeting Kyoto targets, instead of, or as well as, using environmental taxes. Use emissions-trading, Joint Implementation and the Clean Development Mechanism provisions under the Kyoto Protocol to meet targets in the most cost-effective manner.
- If Sweden wishes to make a larger contribution than required under the EU burden-sharing arrangements, then more extensive use of flexible mechanisms (including international trading) should be considered, in order to achieve the most cost-efficient reductions.
- Re-examine energy efficiency promotion to ensure that measures generate net social benefits and do not simply strive to achieve maximum engineering efficiency.
- Examine carefully the policies to promote renewable energy sources to ensure that they do not overstep the mark and become, *de facto*, industrial policy. Allow market signals, adjusted for environmental externalities, to play a greater role in determining the appropriate mix of energy supply. Build sunset clauses into research and development projects.
- Ensure that the competition drawbacks of encouraging a switch from individual to district heating are fully addressed.
- Align energy taxes with energy content and apply uniformly across all energy consumers.

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Box 10. **Recommendations for further action** (*cont.*)

Water quality

- Examine the merits of either adopting a minerals accounting approach and taxing nitrogen losses (*i.e.* residual balances) to provide an economic incentive to reduce the damage from agriculture or shifting to an environmental-outcomes-incentive charge for farmers. Evaluate costs and benefits compared with present arrangements.

Solid waste management

- Ensure that the economic cost is taken properly into consideration as well as the environmental consequences of different waste management strategies.
- Reconsider the emphasis placed on recycling and identify which items are worthwhile recycling on the basis of cost-benefit analysis. Evaluate the relative merits of extended producer responsibility in the light of international experience with alternative approaches to recycling.
- Consider moving to a weight-based general waste collection system to provide a stronger economic incentive to reduce waste.
- Undertake cost-benefit analysis before proceeding with the planned bans on landfill of combustibles and organic waste.

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Notes

1. Swedish estimates of the growth rate of potential output range between 2 and 2½ per cent.
2. This projection was first published in *Economic Outlook* 68, which only takes into account information up till early November.
3. This excludes the statistical effect of the reclassification of the activities of the Church of Sweden in 2000 from the general government sector to households, which technically increased private consumption by another 1.0 per cent.
4. The Riksbank assumes an elasticity of 0.02 within the first year.
5. Rents are negotiable but must be set in line with the normal prevailing rent level in the community. This has been interpreted as the rent in municipally-owned housing companies which thereby set a ceiling to local rents.
6. The equalisation scheme effectively implies that individual municipalities have no economic incentives to attract new inhabitants inasmuch as the income earned by the latter has no bearing on municipalities' tax base. Rather, the tax base of all municipalities is adjusted in accordance with the average aggregate increase.
7. The short-term employment statistics indicate a year-on-year increase in the number of public-sector employees of 1.1 per cent in the third quarter of 2000. The preliminary national accounts, on the other hand, show a decline of 1 per cent in the total number of hours worked even after adjustment for the reclassification of the Church of Sweden.
8. Interest-rate fluctuations affect the CPI directly, most notably through their impact on housing costs. Thus, an increase in the interest rate causes the CPI to increase in the short term.
9. The calculation of the housing component is very complex and in some respects difficult to interpret. For example, the method used for assessing the rate of depreciation of the housing stock was changed recently, thereby contributing significantly to reducing the measured increase in housing costs. Also, a number of taxes, as well as the indexing of the tax assessment values of houses affect the calculation. It should be noted that heating and electricity, which are included in the housing component, have had only modest effects.
10. Falling interest rates may also have factored in. Lower capital costs in the municipally-owned housing companies may have been passed on to rents, which are typically cost-based. To the extent this is the case, it could be argued that such effects should be ignored by the monetary authorities, as is the case with interest payments in the owner-occupied sector.

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11. The measured growth rate in productivity is affected, amongst other things, by the methods underlying constant-price calculations in the national accounts. In particular, the quality adjustments of ICT equipment may potentially be important in this respect. As for production, investment and import of hardware the Swedish national accounts rely partly on hedonic price indices for their constant price calculations. However, unlike in, for example, the United States, such indices are not used for the deflation of software investments. According to Statistics Sweden this is of second-order importance only; assuming the same methods for quality adjustments of software as in the United States had been applied in Sweden, the statistical office has calculated that real GDP growth would have averaged 0.1 percentage point higher per year. This rather small difference is in large part due to the fact that a very high proportion of software is imported (SCB, 2000).
12. The official measure of unemployment is based on the labour force survey. However, in contrast with international conventions, the official unemployment measure does not include full-time students even if they comply with the ILO criteria. Therefore, unlike in other countries, the standardised unemployment rate is somewhat higher in Sweden than the commonly used measure: in 2000 the standardised unemployment rate was 5.9 per cent, while the official measure was 4.7 per cent. The gap between the two, which was non-existing at the outset of the 1990s, reached a peak of 1.9 percentage points in 1997.
13. The projection is identical to that presented in *Economic Outlook* 68, finalised in early November. Information published since then includes revised national accounts data for 1997 to 1999 and preliminary data for the third quarter 2000 as well as fourth quarter indicators of private consumption, consumer and business confidence and employment and unemployment. In combination with indicators of a sharper than expected deceleration in the United States, this information may suggest slower growth in demand in the fourth quarter than expected in EO68.
14. The Government has the power to decide, after consultations with the Bank, the exchange-rate regime, while it is the responsibility of the Bank to implement the regime. This means, for example, that the Bank decides on the practical application of policies in a floating-rate system and would choose the central rate and the band width in a fixed-exchange-rate system (Berg and Lindberg, 2000; Ministry of Finance, 2000).
15. While there is widespread consensus that the operational definition of price stability currently applied is appropriate, some have advocated the adoption of a stricter target, arguing that the ongoing deregulation of various utilities sectors as well as recent improvements of the framework for competition policies (see Chapter III) is temporarily imparting downward pressure on inflation and that the application of a 2 per cent inflation target, therefore, implicitly allows for a higher average increase in other prices. This critique was repudiated by the Bank and others, who emphasised, among other things, the importance of adhering consistently to one well-known measure of inflation.
16. For example, the Bank was criticised by a majority in the Parliament's Standing Finance Committee in mid-1999. Against the background of undershooting the target in each of the years 1996 to 1998, the majority expressed the view that it had lowered interest rates too little and too late (Berg and Lindberg, 2000).
17. In a clarification, in February 1999, the Executive Board identified two situations in which the direction of monetary policy could depart from the inflation target. The first would be if the CPI were to be influenced by factors that are not considered to have a permanent effect on inflation, such as changes in household interest expenditure or in indirect taxes and subsidies. The second situation, which has so far not occurred, would

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be if a quick return to the targeted rate would be costly for the real economy, in which case there may be grounds for aiming to bring inflation back to the target more gradually. Any deviation for these reasons from the inflation target, defined in terms of the CPI 1 to 2 years ahead, would be announced by the Bank in advance.

18. In its third-quarter 2000 Inflation Report, for example, the Bank projects the increase in both the CPI and UND1X to be 1.9 per cent two years ahead.

19. An increase in the TCW index expresses a weakening of the nominal effective exchange rate.

20. The changes in the placement provisions, which were effective from the autumn of 2000, follow from the comprehensive reform of the pension system in 1999. An element in this reform was the division of the State Pension Fund (AP-*fonden*) into seven smaller funds and the ability for individuals to choose between these for part of their accumulated pension savings. At the same time the new pension funds are allowed to invest a larger proportion of their portfolios in foreign securities than was the State Pension Fund previously.

21. The cyclically-adjusted government balance required to meet the 3 per cent deficit criterion is higher for Sweden than for any other EU country. OECD estimates suggest that to meet that criterion with 90 per cent confidence over a three-year time horizon, a cyclically-adjusted surplus of 1 per cent of GDP is required (Dalsgaard and De Serres, 2000). Over a five-year time horizon, the required surplus is 5 per cent of GDP.

22. According to OECD estimates, a one per cent increase in GDP improves the fiscal balance by 0.7 percentage point (Van den Noord, 2000). Other studies have even found the cyclical sensitivity to be slightly higher: Braconier and Holden (1999), for example, estimate that figure to be 0.9 percentage point.

23. In its budget for 2001, any surplus above 2 per cent of GDP is from 2002 onwards technically transferred to households, and the medium-term projection of the public debt to GDP ratio is based on a 2 per cent surplus.

24. This, in turn, reflects the distinctive path of unemployment in Sweden. Unemployment remained low until around 1990, then rose very rapidly for five years and has subsequently come down significantly again in only a few years, although not quite to the level before 1990. The fact that Sweden has experienced only one "unemployment cycle" makes the assessment of the NAIRU particularly difficult.

25. The cap is SKr 1 150 per month for the first child; SKr 767 for the second, and SKr 383 for the third. For further children there is no charge. Households with an income below that for which the cap becomes effective will pay a maximum of 3 per cent of their income for the first child, 2 per cent for the second and 1 per cent for additional children. The original proposal from the government was a flat-rate charge. This would have been much simpler and, by avoiding the income-dependency of the charges now adopted, it would have reduced average marginal tax rates more.

26. A proper decomposition of the expenditures inside and outside the ceiling into different types of outlays would require detailed information on the type of internal transactions between different levels of governments. Such internal transactions amounted to SKr 171 billion in 1999, equivalent to 16 per cent of total expenditures. Gross central-government outlays to other government sectors, including transfers, are included under the ceiling – the single most important is the block grant to municipalities, which amounts to SKr 80 billion.

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27. When the introduction of the expenditure ceilings was first announced in 1995, it was with the intention that they should cover all public expenditures. However, it was subsequently assessed that including local government's expenditures was irreconcilable with the principle of municipal self-government (SOU, 2000*a*).

28. This refers to cyclical savings that were not envisaged when the ceiling was fixed.

29. For example, as from 1 October 1999 ALMP-related wage subsidies have been technically posted on the revenue side of the budget as tax credits to employers, while they had previously been paid directly to employers and were booked on the expenditure side of the budget.

30. The municipalities hold a constitutional right to impose income taxes, and the average municipal tax rate of 30.4 per cent is the highest in the OECD area. A 50 per cent tax on municipal tax increases was decided in 1996, *i.e.* individual municipalities were to pay half the revenue raised from increases in their income tax rates to the central government. However, in 1998 the Supreme Court ruled that this violated the constitution.

31. The Entrepreneurial Research Consortium project asks a sample of adults whether they are currently trying to start a business, alone or with others (Davidson and Henrekson, 2000). The relative degree of nascent entrepreneurship between countries is particularly revealing because the study is co-ordinated and deliberately designed to facilitate international comparison.

32. OECD estimates of the structural unemployment rate (NAIRU) have been recently updated using a revised estimation technique (see Richardson *et al.*, 2000). In the case of Sweden the NAIRU estimate for 1999 was virtually unaffected by the change in methodology. The estimated unemployment gap in 1999 is robust to using the standardised unemployment rate, which is higher than the official figures.

33. At first glance, this may seem inconsistent with the current low inflation, and, indeed, the latter suggests that the estimate for Sweden be interpreted with caution. On the other hand, however, the estimation procedure deliberately implies a somewhat gradual adjustment of the NAIRU in part to avoid end-point problems. The NAIRU estimate should be interpreted as an indicator of the rate of unemployment that is sustainable over the medium term. Moreover, declining profit shares in the business sector and abnormally low increases in housing rents, partly due to rent controls, have recently acted to keep inflation below what could be expected over the longer term.

34. For example, if an individual's weekly working time is 30 hours, supplementary benefits for 10 hours per week may be received for approximately six years.

35. This figure is based on estimates from the DELTA Commission (SOU, 1999*b*). The majority of non-PES-registered part-time unemployed were either students or recipients of partial pensions.

36. The EPL rules may also have been important for the marked increase in the share of workers on temporary contracts during the 1990s. While in 2000 the number of persons employed on permanent contracts was 15 per cent below the 1990 level, the number of workers on temporary contracts was some 25 per cent higher.

37. The AEI was introduced in 1997 and allows – subject to some preconditions – employed and unemployed people to undertake education while receiving a government-paid allowance comparable to the unemployment benefit level. Some 135 000 persons participated in such activities in 2000 of which 35 000 was financed by local governments. The AEI was extended to 2003 in the 2000 Spring Budget.

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38. To the extent that ALMPs take the form of education, the participants will not be counted in the workforce and a decline in the latter may, therefore, be observed even if the scope of ALMPs is boosted.
39. The effect of "taxing leisure" by requiring participation in labour-market programmes is likely to manifest itself as enhanced search behaviour even before entering programmes (*i.e.* to avoid entering). The importance of this effect is not captured by studies of changes in employment probability for programme participants.
40. Those who were born before 1937 will follow the old rules, while the new pension system will apply fully to everybody born in 1954 or later. The cohorts in between will be entitled to pension according to a mix of the old and the new system.
41. Because these estimates are drawn from macroeconomic data, they are subject to a number of limitations, and may result in some overstatement of the average taxes paid on capital income; the authors discuss several significant areas of difficulty and some ways of dealing with them. However, the magnitude of any bias may vary considerably between countries, depending on the structure of taxation systems. For example, the methodology does not adjust for relief from the double taxation of dividends, which leads to an overstatement of average effective capital tax rates for those countries with such relief, but does not affect the calculation for countries like Sweden where double taxation applies. In addition, the estimates assume that a standard taxation regime applies, rather than the dual-income system that applies in Sweden. Adapting the analysis to reflect more closely the actual system would produce a more robust result. The best approach to measuring effective capital tax rates is a matter for ongoing debate and further work is underway (OECD, 2000*c*).
42. Two sets of estimates have been prepared, one using a concept of gross and the others net operating surplus. Consumption of fixed capital is the difference between the two measures, which is important for tax purposes, but the procedures for calculating it vary significantly across countries.
43. Of the 10 per cent of households that switched supplier, 9 out of 10 state that they are pleased with their new one.
44. Parliament has already agreed that the government's stake in Telia should be reduced to 49 per cent, but the timing of any sale will depend on stock-market conditions.
45. Even at the reduced rate, these charges remain ten times as high as those between Telia and other fixed operators.
46. Indeed even in the United States, in 18 out of the 100 largest urban areas, the largest company has more than a 40 per cent market share.
47. Their analysis indicated that if the market were to expand from three equal wholesalers to four, this would lower prices by around 4 or 5 per cent.
48. These are local representatives of the central government.
49. According to Eurostat figures, the gap was some 24 per cent in 1996.
50. In contrast, needs assessment surveys indicate that in 60 per cent of municipalities there is a shortfall of housing suitable for the elderly and disabled, while 40 per cent report a lack of accommodation for students (*www.boverket.se/novo/filelib/personal/hangus/euaccesstilleu*000613*.doc*).
51. Such income may also escape the tax net.
52. The recent clarification of the legal status of owner-occupied apartments facilitates this process.

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53. See Sustainable Sweden website at *www.hallbarasverige.gov.se* for further discussion of sustainable development concepts.
54. The global plan of action for sustainable development, agreed at the UN Conference on Environment and Development in 1992 considered the participation of local authorities as a determining factor in achieving successful implementation.
55. Some examples of projects funded are: a campaign promoting energy-efficient windows and investment grants towards purchase; a grant to restore marshland and overgrown pastureland; a grant to a company manufacturing new detergent tabs for washing machines and dishwashers; a package of measures for reducing transport-related CO_2 emissions; a project to strengthen biological diversity in deciduous forests and promote nature tourism; digging new ponds and restoring wetlands; measures to reduce fertiliser run-off and water purification units; and the construction of a district central heating plant that burns bio-fuel and industrial sludge.
56. This represents the immediate and probable future environmental impacts of activity in 1997. An alternative measure would show the environmental damages reflected in the year they impact rather than when they occur (Ministry of Finance, 2000*c*).
57. The value of these damages avoided are estimated to be between SKr 5 to 130 million, using the cost data published by the UN Panel on Climate Change. The model results do not take account of any long-run impacts from innovation, which the oil-shocks experience suggest could be large.
58. Latest estimates suggest that global temperatures would be anywhere between 1.5° and 6° Celsius higher by 2100, and rises are projected to be higher towards the poles than average (Watson, 2000). Regional effects remain even more uncertain. Sweden's EPA notes that the effects on agriculture and forestry yields would be positive, but biodiversity would suffer and some coastal flooding would occur.
59. However, recent simulations have illustrated that these estimates are subject to a high degree of uncertainty, and the assumed reductions in critical loads implicit in the Protocol may be rather optimistic (Alveteg *et al.*, 2000).
60. This agreement forms part of Annex VI of the International Convention for the Prevention of Pollution from Ships, MARPOL 73/78.
61. However, random spot checks carried out in 1999 by the authorities on 150 vessels that were certified as running on low-sulphur fuel, found that three vessels were in fact cheating, including one large ferry. Their certificates were withdrawn, and they were required to re-apply, but no fines or other penalties were imposed.
62. Nuclear energy also pays a rent tax, based on generating capacity.
63. Until 1 January 2001, industry and agriculture paid half the standard rate. But their rate was held constant while the standard rate was raised in the 2001 Budget.
64. Furthermore, if desired, hydro capacity could be significantly expanded but such development has been ruled out, in the interests of protecting the natural environment.
65. In 1994, heavy oil for large heating plants cost SKr 22.9/100 kWh, while that of biomass fuels was only SKr 10.9/kWh.
66. Annex B countries under the Kyoto Protocol are all OECD countries except Korea, Mexico and Turkey and a number of countries in transition, including Russia.
67. One case study of the effect of revoking the special rules on the Swedish pulp and paper industry shows plausible outcomes ranging from CO_2 emissions falling slightly while those of SO_2 and NO_x rise, to emissions of all three falling significantly (OECD,

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1999*c*). Whether emissions would go up or down depends crucially on several key assumptions. *First* is that although the average emissions from electricity generation are low, marginal emissions would be quite high, because they draw on Swedish fossil-fuel reserve capacity or on imported electricity from Nordic neighbours, especially Denmark. *Second*, pulp from virgin wood is significantly more energy-intensive than pulp from recycled paper, so that a shift away from virgin pulp, in which Sweden has a comparative advantage, could imply less energy use for the same world output of paper.

68. Several estimates of the possible costs have been made, (for example, Andersson and Hådén, 1996 and Nordhaus, 1997). But these calculations depend critically on assumptions about the time profile of phase out, the costs of alternative technologies and the degree of competition in the electricity market. The last two factors have evolved significantly in recent years.

69. It should be noted that adopting a new even tougher national target is not without precedent. When the Swedish Parliament ratified the UN Climate Convention, it adopted an interim target that emissions should be stabilised at 1990 levels by the year 2000 and thereafter decrease. This national target has remained, despite the subsequent Kyoto-linked burden-sharing agreement.

70. This is the process whereby excess nutrients in waterways leads to denser vegetation and abundant plankton algae, until waterways become choked with dead flora and fauna.

71. Crop production can also involve denitrification which contributes to global warming and ozone depletion but does not contribute to eutrophication, acidification or contamination of ground water.

72. IPP in Sweden also covers eco-labelling, life-cycle analyses, environmental management systems, etc.

73. Even greater co-ordination has occurred in Germany, where producers and retailers have set up the *Duales System von Deutschland* (DSD), which operates a national recycling system on behalf of all participants, whose products are marked with a green dot.

74. In any case, one empirical analysis of targets and regulations for recycling in the United States shows that they have had little effect on either the amount of recycling or the amount of garbage generated, even in those municipalities where mandatory recycling is reinforced by fines (Duggal *et al.*, 1991).

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Annex I

Capital taxes

Several taxes apply to capital in Sweden and together accounted for just over SKr 100 billion (5½ per cent of GDP in 1997) (Table A1). The main taxes are:

– Capital taxes paid at a flat rate of 30 per cent on income earned by individuals on dividends, interest, rental income from housing, and capital gains on sales of real and financial assets.* Unlike most countries, there is no income threshold below which no capital income tax is payable. However, if capital income is negative, because of mortgage interest payments or capital losses, 30 per cent of the losses up to SKr 100 000 and 21 per cent of losses above this, may be used to offset state and local income tax due on earned income from employment.

Table A1. **Taxes on capital, 1997**

SKr million

	Paid by individuals	Paid by companies	Total
Income tax on capital income	11 313		11 313
Income tax on company profits		46 509	46 509
Real estate tax	14 870	12 210	27 080
Wealth tax	5 329	124	5 453
Inheritance tax	1 573		1 573
Gift tax	222		222
Tax on dividends to non-residents	1 663		1 663
Tax on pension fund earnings		12 293	12 293
Sub-total	**34 970**	**71 136**	**106 106**
Stamp duty			2 624
Total			**108 730**

Source: National Tax Administration.

* On housing only 50 per cent of the capital gains are taxable, and this can be deferred if an equivalent property is purchased within 12 months. Sale of other private real estate is taxable only if the profit is more than SKr 50 000.

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- Corporate tax of 28 per cent is also levied on company profits before dividends are paid out, resulting in double taxation of dividends. Double taxation was temporarily abolished in 1994 and then reinstated in 1995.
- Real estate taxes are currently levied at 1.5 per cent of the tax value of detached housing, 1.2 per cent for apartments, 1.0 per cent for business premises and 0.5 per cent on industrial property. These rates were adjusted down to these levels in 1998. In principle, the tax value should correspond to 75 per cent of the market value, but tax values of private homes have been held frozen at 1997 levels. The 2001 Swedish budget restores the link between tax assessment values and market rates and will offset this with further reductions in the tax rates, by 0.3 percentage point for houses and 0.7 percentage point for apartments.
- Wealth taxes are levied on the net wealth of the household (and some co-operatives and foundations). It is calculated at 1.5 per cent on net worth above SKr 900 000, although the 2001 budget proposes to raise the thresholds to SKr 1 million for those living alone and SKr 1.5 million for jointly taxed couples. The taxable base contains the tax value of housing, but does not include assets employed in businesses of the self-employed or shares in most non-listed companies. Shares in listed companies are assessed at 80 per cent of market value except for shareholders holding more than 25 per cent of voting rights.

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Annex II

Extended producer responsibility versus environmental taxes

Extended producer responsibility (EPR) and environmental taxes are two alternative ways of applying the polluter pays principle to waste products (*e.g.* packaging). There are two objectives to be met: to discourage waste and to pay for its recycling/treatment. It should be noted that implicit in both approaches is the assumption that these particular products do more harm to the environment than general waste.

Extended Producer Responsibility

EPR is based on the explicit assumption that producers should be responsible for the environmental damage caused by their products. In addition, at least implicitly, there is an assumption that recycling these products generates more net economic benefit than using other methods of disposal such as incineration or landfill. EPR involves a voluntary agreement by producers within a sector to manage the collection and recycling of their products. Its strengths and weakness are the following:

Strengths

- The aggregate cost is the actual cost of collection and recycling, so that in a direct sense, the polluter pays.
- Fees paid by each producer can be calculated on a per unit basis to provide an incentive to reduce production. In this way the relative prices of the products are shifted to encourage more economic use of the product.
- The scheme is designed by the sector itself, so that a high degree of compliance is to be expected.
- It promotes recycling efforts, in line with agreed targets for total recycling goals by product.

Weaknesses

- It emphasises products rather than the materials; this may compromise the principle of equalising marginal abatement costs across the economy. It also may provide only very indirect encouragement to switching towards producing products that use less harmful materials.
- Producers outside the voluntary agreement can free ride and have no incentive to modify their behaviour. It may also difficult to integrate imports into the system.
- Voluntary agreements may risk encouraging overly co-operative (even collusive) behaviour among firms.

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- It may encourage over-recycling, if it may be more cost-effective and environmentally appropriate to use general collection and disposal.
- It promotes multiple waste-collection systems, which may be less efficient than a single integrated system.

Environmental taxes

An alternative approach would be to tax the materials in line with the additional disposal costs that they present, over and above the general cost of waste disposal, by landfill or incineration, to internalise the environmental externality they generate. The strengths and weaknesses in this approach are summarised below:

Strengths

- The taxes can be set to cover the costs of environmental damage regardless of who takes care of the disposal and the method chosen. Recycling will only take place if it is cost-effective in economic terms.
- By altering relative prices, they provide an incentive to economise on the use of the taxed material, in whatever way it is used. It also provides a mechanism for equalising abatement costs across the economy.
- It is compulsory and does not rely on industry co-operation.

Weaknesses

- Tax collection costs may be high and a set of materials-based taxes that provide the environmentally correct signals may become extremely complex.
- A great deal of cost-benefit information is required in order to be able to set the taxes at the right level.
- It can be difficult to integrate imports into the tax base.

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Annex III

Calendar of main economic events

1999

September

The budget for 2000 is presented, following a political agreement between the government and its coalition partners, the Green Party and the Left Party. Actual outlays for the ceiling-restricted expenditure areas are estimated at SKr 763.5 billion, or SKr 1.5 billion below the expenditure ceiling. A tax-cut programme is announced in the budget, consisting of two main elements:

- compensation, in terms of individual tax discounts, for the employee pension contribution of 7 per cent of gross earnings; and
- an increase in the threshold for the state income tax so as to reduce the proportion of taxpayers facing it to 15 per cent, which was the intended coverage in the 1991 tax reform.

The tax programme is to be introduced over four years with decisions being taken for one year at a time, depending on the state of the economy. The first step in 2000 involves a reduction of tax revenues by SKr 12½ billion (0.7 per cent of GDP).

November

The Riksbank raises the repo rate by 0.35 basis points to 3.25 per cent.

Sweden presents its updated Convergence Programme to the European Union. It foresees net government lending showing a surplus of 2 per cent of GDP in each of the years 2000 to 2002 and consolidated gross debt declining from 66 per cent of GDP in 1999 to 52 per cent in 2002.

2000

February

The Riksbank raises the repo rate to 3.75 per cent.

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March

Parliament passes into law a number of changes in the existing labour market code, including:

- A new Mediation Authority is established as from 1 July 2000 to help underpin an improvement of the wage bargaining process. The new authority has slightly greater powers than its predecessor. Its role is to mediate in labour disputes and call parties to discussions, or keep informed in other ways about current or future wage negotiations.
- The set-up of active labour market policies is reformed by the introduction of an activity guarantee, which is to come into force by August 2000. The activity guarantee, which is open-ended, requires long-term unemployed to participate continuously in full-time activation measures after approximately two years of unemployment.

April

The spring budget for 2000 is presented following a political agreement between the government, the Green Party and the Left Party. The ceiling on central government expenditures (excluding interest payments) for 2003 is set at SKr 847 billion, an increase of 3.7 per over the previously set ceiling for 2002. Higher spending in a number of areas is announced, amounting to SKr 4½ billion in 2001 and increasing to SKr 20 billion in 2003. Additional resources are reserved for more generous educational grants and child benefits. Moreover, a cap on parents' payment for childcare services is instituted as from 2002, thereby reducing average user charges some 40 to 50 per cent but leaving them means tested. Municipalities are compensated for the loss of revenue, although any increase in demand will be financed fully by the municipalities.

The Adult Education Initiative is prolonged to 2003.

May

Sweden submits its 2000 Action Plan for Employment to the European Union, reiterating the objectives of reducing the unemployment rate to 4 per cent by 2000 and increasing the proportion of persons aged 20 to 64 years in unsubsidised employment to 80 per cent by 2004. The main focus is on the adjustments of labour market policies endorsed by the parliament in March.

The Government presented a Competition Policy Bill, to implement articles 81 (on cartels) and 82 (on abuse of dominant position) of the EU Treaty.

July

The Öresund-bridge between Malmö and Copenhagen is opened, thereby linking the southern part of Sweden closer together economically with the Copenhagen region. At the same time, more liberal provisions for individuals' cross-border imports of alcohol and tobacco come into force, following an agreement earlier in the year between the Swedish government and the EU Commission. Restrictions on such imports must be abolished altogether by 2004.

September

The budget for 2001 is presented, following a political agreement between the government, the Green Party and the Left Party. The budget contains:

- Projected spending in 2001 at just below the expenditure ceiling and no major expenditure initiatives.

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- The second step of the four-year tax-cut programme announced in the budget for 2000. Thus, an additional quarter of employees' pension contributions is offset through individual tax discounts as from 2001, and the threshold for the 20 per cent state income tax is raised, thereby reducing the proportion of taxpayers facing it. The tax concessions announced in the budget amount to SKr 13½ billion, and in combination with earlier decisions, total tax cuts of SKr 27 billion come into force in 2001 (1.3 per cent of GDP).
- A reduction in the property tax from 1.5 to 1.2 per cent of tax assessment values for owner-occupied housing, and from 1.2 to 0.7 per cent on rental housing. These reductions offset the realignment of the tax assessment values to market values after the former had been nominally frozen since 1997.
- A first step in the ten-year "green tax-swap programme" announced in the spring budget is implemented from 2001; environmental taxes, mainly on diesel and CO_2, are raised by SKr 3.4 billion, but offset by an increase in the basic personal tax allowance and lower employer social security contributions.

Legislation on reduced working hours was discussed in the budget negotiations but decision of this is postponed until a newly established commission has reported.

November

Sweden submits its updated Convergence Programme to the European Union. It foresees a budget surplus in 2001 of 3½ per cent of GDP and 3.3 and 3.6 per cent of GDP in the two following years, respectively. Gross consolidated debt is expected to decline from 58.9 per cent in 2000 to 48.2 per cent in 2003.

December

The Riksbank raises the repo rate to 4.0 per cent. The deposit rate and the lending rate were both raised by ½ percentage point to 3.25 and 4.75 per cent, respectively, thereby establishing a symmetric interest rate corridor around the repo rate. The Bank states that the deposit and lending rates will no longer signal monetary policy, and that they will henceforth be adjusted in line with changes of the repo rate.

UMTS mobile telephone licences are awarded to four operators after a "beauty contest".

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